

MATH 647 Random Matrix Theory in Machine Learning

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Abstract

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Many aspects of random matrix theory Random Matrix Theory (RMT) lies at the intersection of probability, linear algebra, and mathematical physics, and has become a central tool in modern data science and machine learning. Despite its breadth, many of its most powerful phenomena are governed by a small number of unifying principles.

The goal of this course is to highlight these core ideas and develop the key techniques that repeatedly appear across the theory. Rather than attempting a comprehensive survey, we will focus on conceptual understanding and transferable methods, and then illustrate how they arise naturally in contemporary machine learning problems.

The following references provide complementary perspectives on RMT, depending on your interests:

- For connections with mathematical physics and spectral universality, see [AGZ10, Tao23], as well as [BGK16] for local spectral statistics.
- For a non-asymptotic, high-dimensional probability viewpoint with emphasis on data science, see [Ver18, Ver10, VH14].
- For applications in high-dimensional statistics, machine learning, and optimization, see [BS10, CL22, Paq].

1 Asymptotic theory for random matrices

1.1 Wigner’s semicircle law

Wigner’s semicircle law can be considered the starting point of RMT.

Definition 1 (Wigner matrix). Let $W_n = (w_{ij})_{1 \leq i, j \leq n}$ be a real symmetric *Wigner matrix*:

- $\{w_{ij} : 1 \leq i \leq j \leq n\}$ are independent;
- for $i < j$, $\mathbb{E}[w_{ij}] = 0$, $\mathbb{E}[w_{ij}^2] = 1$, and $\mathbb{E}[|w_{ij}|^k] < \infty$ for all fixed k ;
- for $i = j$, $\mathbb{E}[w_{ii}] = 0$ and $\sup_n \mathbb{E}[w_{ii}^2] < \infty$.

Define the normalized matrix

$$H_n := \frac{1}{\sqrt{n}} W_n.$$

Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of H_n , and define the empirical spectral distribution (ESD)

$$\mu_{H_n} := \frac{1}{n} \sum_{i=1}^n \delta_{\lambda_i}.$$

Let μ_{sc} be the semicircle law on $[-2, 2]$ with density

$$\rho_{\text{sc}}(x) := \frac{1}{2\pi} \sqrt{4 - x^2} \mathbf{1}_{\{|x| \leq 2\}}.$$

Theorem 1 (Wigner semicircle law). *Let W_n be a Wigner matrix. Then $\mu_{H_n} \Rightarrow \mu_{\text{sc}}$ weakly in probability as $n \rightarrow \infty$.*

Method of moments To show the semicircle law, we rely on the following well-known theorem:

Lemma 1 (Carleman condition). *Let μ be a Borel probability measure on $\mathbb{R}$ with finite moments of all orders*

$$m_k := \int_{\mathbb{R}} |x|^k d\mu(x) < \infty \quad \text{for all } k \geq 0.$$

If

$$\sum_{k=1}^{\infty} m_{2k}^{-\frac{1}{2k}} = \infty,$$

then μ is uniquely determined by its moments. That is, if ν is another Borel probability measure on $\mathbb{R}$ satisfying

$$\int_{\mathbb{R}} x^k d\nu(x) = \int_{\mathbb{R}} x^k d\mu(x) \quad \text{for all } k \geq 0,$$

then $\nu = \mu$.

Lemma 2 (Semicircle moments). *For $k \geq 0$,*

$$\int x^{2k+1} \rho_{\text{sc}}(x) dx = 0, \quad \int x^{2k} \rho_{\text{sc}}(x) dx = C_k := \frac{1}{k+1} \binom{2k}{k}.$$

Here C_k is the k -th Catalan number, which counts different types of combinatorial structure, including non-crossing partitions, rooted planar trees, and Dyck paths [Sta15].

One can check μ_{sc} satisfies the Carleman condition. Therefore, to show μ_{H_n} converges weakly in probability to μ_{sc} , we will show that $m_k(\mu_{H_n}) = \frac{1}{n} \text{Tr}(H_n^k)$ converges to $\int x^k d\mu_{\text{sc}}$ in probability¹. Note that μ_{H_n} is a *random measure*. We can first evaluate its expected moments $\mathbb{E}m_k(\mu_{H_n})$, then upgrade it to convergence in probability by a variance calculation.

¹a more detailed discussion on moment convergence theorem can be found in [BS10, Appendix B] and [AGZ10, Section 2.1.2]

Proof of Theorem 1. Step 1: Expected trace as closed-walk counts

Fix $k \in \mathbb{N}$. We write

$$m_k(\mu_{H_n}) = \frac{1}{n} \text{Tr}(H_n^k) = \frac{1}{n} \sum_{i_1, \dots, i_k=1}^n (H_n)_{i_1 i_2} (H_n)_{i_2 i_3} \cdots (H_n)_{i_k i_1}.$$

Since $H_n = \frac{1}{\sqrt{n}} W_n$, this becomes

$$\mathbb{E} m_k(\mu_{H_n}) = \frac{1}{n} \cdot \frac{1}{n^{k/2}} \sum_{i_1, \dots, i_k=1}^n \mathbb{E} \left[\prod_{t=1}^k w_{i_t i_{t+1}} \right], \quad (i_{k+1} := i_1). \quad (1)$$

We interpret $(i_1, \dots, i_k)$ as a closed walk of length k on the complete graph with vertex set $[n]$.

Let $G(I)$ be the labelled multi-graph (loops are allowed) spanned by the closed walk $(i_1, \dots, i_k)$. Since w_{ij} are centered, the only index sequences $I = (i_1, \dots, i_k)$ that contribute are given by the graphs $G(I)$ where each edge is visited at *least twice*.

Step 2: Non-leading terms

Let $G(I) = (V(I), E(I))$. Let $v(I)$ be the number of distinct vertices visited by the walk $(i_1, \dots, i_k)$, and let $e(I)$ be the number of distinct edges used by the walk.

For a contributing closed walk I , each distinct edge is used at least twice, so

$$2e(I) \leq k \quad \Rightarrow \quad e(I) \leq \lfloor k/2 \rfloor.$$

If $e(I) \leq k/2 - 1$, then the walk uses at most $k/2 - 1$ distinct edges. The number of choices of the vertex labels for such a walk is at most $O(n^{v(I)})$, and one always has the graph-theoretic inequality $v(I) \leq e(I) + 1$ for a connected graph traced by a closed walk. Hence here

$$v(I) \leq e(I) + 1 \leq \frac{k}{2}.$$

Therefore the total number of such index sequences is $O(n^{k/2})$. Using that all moments of w_{ij} are finite, the absolute value of each expectation in (1) is $O_k(1)$ (a constant depending only on k). Plugging into (1), the contribution of all closed walks with $e(I) \leq k/2 - 1$ is bounded by

$$\frac{1}{n^{1+k/2}} \cdot O(n^{k/2}) \cdot O_k(1) = O_k(n^{-1}).$$

So only the “maximal edge” case matters asymptotically: $e(I) = k/2$. This forces k to be even.

Step 3: the leading contribution Assume $k = 2\ell$ is even. From a contributing closed walk, $e(I) = \ell$ and $v(I) = \ell + 1$.

These are exactly the closed walks I whose associated multigraph $G(I)$ is a tree on $\ell + 1$ vertices, with each of its ℓ edges traversed exactly twice. Such walks are in bijection with rooted planar trees on $\ell + 1$ vertices.

So we restrict to $G(I)$ with $v(I) = \ell + 1$ and each of the ℓ edges used exactly twice. For any fixed rooted planar tree and a breadth-first search walk on $\ell + 1$ vertices, the number of labelings by distinct elements of $[n]$ is

$$n(n-1) \cdots (n-\ell) = n^{\ell+1} + O(n^\ell).$$

For each labeling, independence and $\mathbb{E}[w_{ij}^2] = 1$ imply

$$\mathbb{E}\left[\prod_{t=1}^{2\ell} w_{i_t i_{t+1}}\right] = 1.$$

Hence, the total contribution from one unlabelled tree is

$$\frac{1}{n^{1+\ell}}(n^{\ell+1} + O(n^\ell)) \cdot 1 = 1 + O(n^{-1}).$$

Since there are Catalan number C_ℓ many rooted planar trees,

$$\mathbb{E} m_{2\ell}(\mu_{H_n}) = C_\ell + o(1).$$

Now we have obtain the convergence of expected moments for all $\ell \geq 1$:

$$\lim_{n \rightarrow \infty} \mathbb{E} m_{2\ell}(\mu_{H_n}) = C_\ell, \quad \lim_{n \rightarrow \infty} \mathbb{E} m_{2\ell+1}(\mu_{H_n}) = 0.$$

Step 4: Variance bound To upgrade convergence of expectations to convergence in probability, we show that for each fixed k ,

$$m_k(\mu_{H_n}) - \mathbb{E} m_k(\mu_{H_n}) \xrightarrow{\mathbb{P}} 0.$$

It suffices to prove $\text{Var}(m_k(\mu_{H_n})) \rightarrow 0$.

Write $m_k = \frac{1}{n} \text{Tr}(H_n^k)$ and expand as in (1). Then

$$\text{Var}(m_k) = \mathbb{E}[m_k^2] - (\mathbb{E}[m_k])^2.$$

Expanding m_k^2 yields a double sum over two closed walks of length k , say $(i_1, \dots, i_k)$ and $(j_1, \dots, j_k)$:

$$\mathbb{E}[m_k^2] = \frac{1}{n^2} \cdot \frac{1}{n^k} \sum_{I, J} \mathbb{E}\left[\prod_{t=1}^k w_{i_t i_{t+1}} \prod_{s=1}^k w_{j_s j_{s+1}}\right].$$

Similarly, $(\mathbb{E}[m_k])^2$ is the same sum but with the expectation factorized:

$$(\mathbb{E}[m_k])^2 = \frac{1}{n^2} \cdot \frac{1}{n^k} \sum_{I, J} \mathbb{E}\left[\prod_{t=1}^k w_{i_t i_{t+1}}\right] \mathbb{E}\left[\prod_{s=1}^k w_{j_s j_{s+1}}\right].$$

Subtracting, the only pairs of walks that contribute are those for which the two products are *not independent*, i.e. the two walks share at least one common edge variable w_{ab} . Indeed, if the sets of involved edges are disjoint, independence implies the expectations factor, and the term cancels exactly.

Let $G(I \cup J)$ be the graph spanned by the closed walks I and J . Then $G(I \cup J)$ is a connected graph. Moreover, each edge in $G(I \cup J)$ is visited at least twice by the closed walks I, J , otherwise the expectation contribution is zero. This gives $e(I \cup J) \leq k$ and $v(I \cup J) \leq e(I \cup J) + 1 \leq k + 1$.

Next, we will show that $v(I \cup J) \leq k$. Suppose not, then there exists some I, J such that $v(I \cup J) = k + 1$. Then $G(I \cup J)$ must be a tree with $e(I \cup J) = k$. Since both $G(I), G(J)$ are trees, they must overlap on at least one edge. Then the overlapped edge is visited 4 times, enforcing $e(I \cup J) = k - 1$, a contradiction.

Therefore, the total number of pairs I, J with nonzero expectation is $O(n^k)$. Each mixed moment is bounded by a constant $O_k(1)$ using the finiteness of moments. Therefore

$$\text{Var}(m_k) \leq \frac{1}{n^2} \cdot \frac{1}{n^k} \cdot O(n^k) \cdot O_k(1) = O_k(n^{-2}),$$

Therefore $m_k(\mu_{H_n}) \rightarrow m_k(\mu_{sc})$ in probability. This finishes the proof of Theorem 1. In fact, for each fixed k , $\sum_n \text{Var}(m_k) < \infty$. So by Borel-Cantelli, $m_k(\mu_{H_n})$ also converges almost surely. $\square$

Weaker conditions and extension Our definition of the Wigner matrix in Definition 1 requires all moments to be finite. In fact, this is not necessary. If the upper triangular part of W_n is i.i.d. with variance 1, the semicircle law holds. It also holds for the non-i.i.d. case, under a Linderberg's condition, see [BS10, Section 2]. If the variance of H_n satisfies $\sum_i \text{Var}(H_{ij}) = 1$ for all $j \in [n]$, this is called a generalized Wigner matrix, and a semicircle law also holds [Zhu20]. However, if the row sum of the variance is not constant, the limiting law is no longer semicircular, and it is called a Wigner-type inhomogeneous random matrix, see [AEK17, Zhu20].

The same proof also works for the adjacency matrix of Erdős-Rényi graphs $G(n, p)$. Let A_n be the adjacency matrix of $G(n, p)$. One can show that the ESD of $\frac{A_n}{\sqrt{np}}$ converges in probability to the semicircle law as long as $np \rightarrow \infty$ [TVW13].

1.2 Marchenko-Pastur law

Another class of random matrices widely used in high-dimensional statistics is the sample covariance matrix.

Definition 2 (Sample covariance matrix and ESD). Let $X = X_{p,n} = (x_{ij}) \in \mathbb{R}^{p \times n}$ have independent entries with

$$\mathbb{E}x_{11} = 0, \quad \mathbb{E}x_{11}^2 = 1,$$

and assume that for every fixed $k \geq 1$ we have $\mathbb{E}|x_{11}|^k < \infty$. Define the sample covariance matrix

$$S := \frac{1}{n} X X^\top \in \mathbb{R}^{p \times p}.$$

Let $\lambda_1, \dots, \lambda_p$ be the eigenvalues of S . The *empirical spectral distribution* (ESD) of S is

$$\mu_S := \frac{1}{p} \sum_{i=1}^p \delta_{\lambda_i}.$$

Throughout, we consider the high-dimensional regime

$$\frac{p}{n} \rightarrow y \in (0, \infty) \quad (p, n \rightarrow \infty).$$

y is called the aspect ratio.

Definition 3 (Marchenko-Pastur distribution). Fix $y \in (0, \infty)$ and set

$$a := (1 - \sqrt{y})^2, \quad b := (1 + \sqrt{y})^2.$$

The *Marchenko–Pastur distribution* with parameter y , denoted $\mu_{\text{MP},y}$, is the probability measure on $[0, \infty)$ defined by

$$\mu_{\text{MP},y} := \left(1 - \frac{1}{y}\right)_+ \delta_0 + f_y(x) dx,$$

where

$$f_y(x) := \frac{1}{2\pi y x} \sqrt{(b-x)(x-a)} \mathbf{1}_{[a,b]}(x).$$

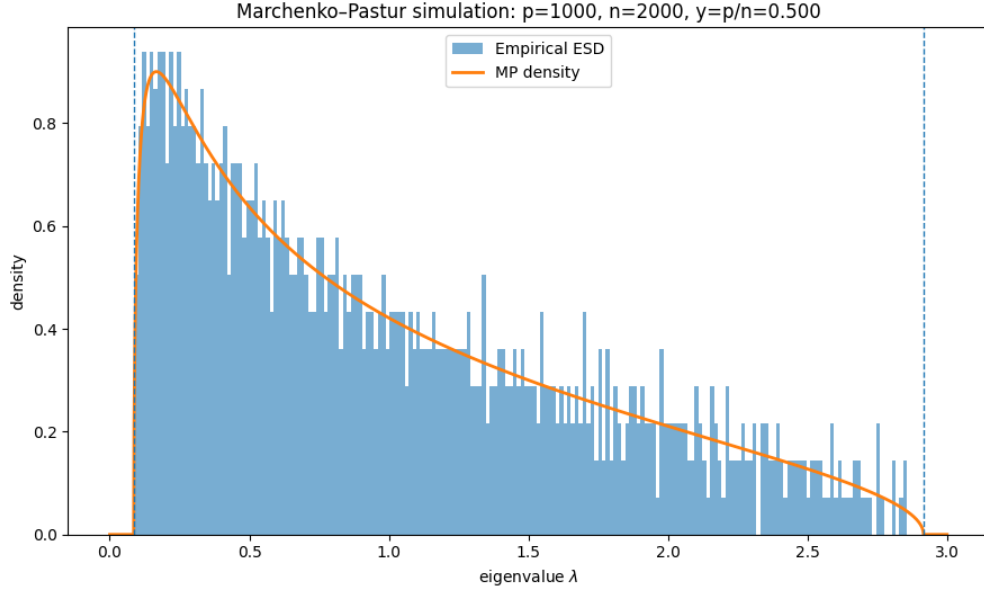


Figure 1: MP law for a sample covariance matrix

Definition 4 (Narayana numbers and MP moments). For integers $k \geq 1$ and $1 \leq r \leq k$, define the Narayana numbers

$$N(k, r) := \frac{1}{k} \binom{k}{r} \binom{k}{r-1}.$$

Define the sequence $(m_k(y))_{k \geq 0}$ by $m_0(y) := 1$ and, for $k \geq 1$,

$$m_k(y) := \sum_{r=1}^k N(k, r) y^{r-1}. \quad (2)$$

Lemma 3 (Moments of the MP law). For every $y \in (0, \infty)$ and every integer $k \geq 0$,

$$\int_0^\infty x^k \mu_{\text{MP},y}(dx) = m_k(y),$$

where $m_k(y)$ is given by (2).

Theorem 2 (Marchenko–Pastur law). Let $S = \frac{1}{n} X X^\top$ be as above and assume $p/n \rightarrow y \in (0, \infty)$. Then the ESD μ_S converges weakly to $\mu_{\text{MP},y}$ in probability:

$$\mu_S \Rightarrow \mu_{\text{MP},y} \quad \text{in probability as } p, n \rightarrow \infty.$$

Remark 1 (Atom at zero from rank deficiency). If $y > 1$, then $\mu_{\text{MP},y}$ has an atom at 0 of mass $1 - 1/y$ because $\text{rank}(S) = \text{rank}(X) \leq \min\{n, p\} = n$. The number of zero eigenvalues is at least $p - n$. When $y > 1$, the MP-law implies $\frac{\text{rank} X}{p} \rightarrow y^{-1}$ in probability.

The Marchenko–Pastur (MP) law is one of the central limit theorems in modern random matrix theory and high-dimensional statistics: it describes the *typical spectrum* of a sample covariance matrix when the dimension is comparable to the sample size ($p/n \rightarrow y \in (0, \infty)$). In this regime, many classical “fixed p , large n ” intuitions break. For instance, even when the population covariance is the identity, the eigenvalues of the sample covariance do *not* concentrate at 1; instead they spread over the interval $[(1 - \sqrt{y})^2, (1 + \sqrt{y})^2]$ (and, when $y > 1$, a positive fraction of eigenvalues are exactly 0 due to rank deficiency). This spectral spreading is the quantitative origin of a wide range of high-dimensional phenomena. For instance, it underlies the double-descent behavior observed in high-dimensional ridge regression [HMRT22]. We will discuss this connection later.

Moment method proof For each integer $k \geq 1$, define the k -th empirical moment of μ_S :

$$M_{p,k} := \int x^k \mu_S(dx) = \frac{1}{p} \text{Tr}(S^k).$$

Theorem 2 will follow once we show $M_{p,k} \rightarrow m_k(y)$ in probability for every fixed k as $p \rightarrow \infty$.

Let $k \geq 1$ be fixed and write $i_{k+1} := i_1$. Use $S = \frac{1}{n} X X^\top$ and expand $\text{Tr}(S^k)$ as a cyclic product of entries of $X X^\top$, then expand each $(X X^\top)_{ab} = \sum_{j=1}^n x_{aj} x_{bj}$. We have

$$\text{Tr}(S^k) = \frac{1}{n^k} \sum_{i_1, \dots, i_k=1}^p \sum_{j_1, \dots, j_k=1}^n \prod_{t=1}^k x_{i_t j_t} x_{i_{t+1} j_t}.$$

Consequently,

$$\mathbb{E} M_{p,k} = \frac{1}{p n^k} \sum_{i_1, \dots, i_k=1}^p \sum_{j_1, \dots, j_k=1}^n \mathbb{E} \left[\prod_{t=1}^k x_{i_t j_t} x_{i_{t+1} j_t} \right].$$

The sequence pair $I = (i_1, \dots, i_k), J = (j_1, \dots, j_k)$ forms a closed walk of length $2k$. Fix $(i_1, \dots, i_k) \in [p]^k$ and $(j_1, \dots, j_k) \in [n]^k$. Let

$$r := |\{i_1, \dots, i_k\}|, \quad s := |\{j_1, \dots, j_k\}|.$$

be the number of distinct vertices. Let e be the number of distinct pairs (“edges”)

$$(i_t, j_t) \quad \text{and} \quad (i_{t+1}, j_t) \quad (t = 1, \dots, k)$$

appearing among the $2k$ -index sequence. Let $G(I, J)$ be its associated bipartite graph.

If $\mathbb{E} \left[\prod_{t=1}^k x_{i_t j_t} x_{i_{t+1} j_t} \right] \neq 0$, then every distinct edge (i, j) appearing in the product appears at least twice. In particular, $e \leq k$.

Since $G(I, J)$ is a connected bipartite graph on $v = r + s$ distinct vertices and e distinct edges. Then $v \leq e + 1$. Consequently, if a pattern contributes nontrivially, then

$$r + s \leq k + 1.$$

Moreover, for a fixed identification pattern with parameters (r, s) , the number of labelings is of order $(1 + o(1))p^r n^s$, and the contribution to $\mathbb{E}M_{p,k}$ scales as

$$O\left(\frac{p^r n^s}{pn^k}\right) = O(p^{r-1} n^{s-k}) = O(y^{r-1} n^{r+s-k-1}).$$

In the limit $p, n \rightarrow \infty$ with $p/n \rightarrow y$, only patterns with $r + s = k + 1$ can contribute to $\lim \mathbb{E}M_{p,k}$. For such a pattern, necessarily $e = k$ and the associated bipartite graph is a tree. Moreover, every distinct edge is traversed exactly twice, and

$$\mathbb{E} \left[\prod_{t=1}^k x_{i_t j_t} x_{i_{t+1} j_t} \right] = 1.$$

Lemma 4. *The number of surviving patterns with exactly r distinct row vertices equals $N(k, r)$.*

Proof. See [BS10, Lemma 3.4]. $\square$

For a fixed tree pattern with r distinct row vertices and s distinct column vertices, its contribution is $\frac{p^r n^s}{pn^k} = (1 + o(1))y^{r-1}$ (since $r + s = k + 1$). Summing over patterns grouped by r yields

$$\lim_p \mathbb{E}M_{p,k} = \sum_{r=1}^k (\# \text{patterns with } r \text{ row vertices}) y^{r-1} = \sum_{r=1}^k N(k, r) y^{r-1} = m_k(y).$$

It's a similar argument as in the proof of Theorem 1 in the Wigner matrix case to upgrade from convergence in expectation to convergence in probability. We omit the details.

1.3 Spectral norm bound with the high moment method

We have seen that the moment method is a combinatorial way of proving convergence to the limiting spectral distribution. Another application of the moment method is to study the spectral norm of a random matrix. Let A be an $n \times n$ Hermitian matrix. The spectral norm of A is

$$\|A\| = \sup_{\|x\|=1} \|Ax\| = \max\{|\lambda_1(A)|, |\lambda_n(A)|\}.$$

In particular, the following inequality holds:

$$\frac{1}{n} \operatorname{tr}(A^{2k}) \leq \|A\|^{2k} \leq \operatorname{tr}(A^{2k}).$$

Taking the $1/2k$ -th power yields

$$n^{-1/(2k)} [\operatorname{tr}(A^{2k})]^{1/(2k)} \leq \|A\| \leq [\operatorname{tr}(A^{2k})]^{1/(2k)}.$$

If $k = \omega(\log n)$, then $n^{-1/(2k)} = 1 + o(1)$. This suggests a good estimate of $\operatorname{tr}(A^{2k})$ for a growing sequence of $k = \omega(\log n)$ gives a sharp two-sided bound on $\|A\|$.

Now assume A is a Wigner matrix. From the semicircle law, we know $\|A\| \geq (2 + o(1))\sqrt{n}$ with high probability. But the semicircle law itself does not imply $\|A\|/\sqrt{n} \rightarrow 2$ in probability.

Different from the ϵ -net arguments in [Ver10], the moment method can be used to obtain sharp constants for Wigner matrices. We will introduce a simplified version of a bound obtained by Vu [Vu07].

Theorem 3. Assume A is a symmetric random matrix, where $a_{ij}, 1 \leq i \leq j \leq n$ are independent such that $|a_{ij}| \leq 1, \mathbb{E}a_{ij} = 0, \text{Var}(a_{ij}) = 1$. Then, with high probability $1 - n^{-C_1}$

$$\|A\| \leq 2\sqrt{n} + C_2 n^{1/3} \log n$$

for some absolute constants $C_1, C_2 > 0$.

Proof. We have

$$\mathbb{E} \text{tr} A^k = \sum_{i_1, \dots, i_k \in [n]} \mathbb{E}(a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_{k-1} i_k} a_{i_k i_1}).$$

For $1 \leq p \leq k$, let $E(n, k, p)$ be the sum of $\mathbb{E}(a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_{k-1} i_k} a_{i_k i_1})$ where the number of distinct indices among $i_1, \dots, i_k$ is p . We have $p \leq k/2 + 1$ by independence. Let $W(n, k, p)$ be the number of k -step walks in K_n using p vertices, where each edge in the walk is visited at least twice. Then

$$W(n, k, p) \leq n(n-1) \cdots (n-p+1) W'(k, p),$$

where $W'(k, p)$ denotes the number of k -step walks with a fix labeling $v_1, \dots, v_p$.

Lemma 5 (Lemma 2.1 in [Vu07]). *We have*

$$W'(k, p) \leq \binom{k}{2p-2} p^{2(k-2p+2)} 2^{2p-2}.$$

From this lemma, for a given p , we have

$$E(n, k, p) \leq n(n-1) \cdots (n-p+1) \binom{k}{2p-2} p^{2(k-2p+2)} 2^{2p-2} := S(n, k, p).$$

One can check that $S(n, k, p-1) \leq \frac{k^6}{4n} S(n, k, p)$. So for $k \leq n^{1/6}$, $S(n, k, p-1) \leq \frac{1}{2} S(n, k, p)$. We have

$$\begin{aligned} \mathbb{E} \text{tr} A^k &= \sum_{p=1}^{k/2+1} E(n, k, p) \leq \sum_{p=1}^{k/2+1} S(n, k, p) \leq 2S(n, k, k/2+1) \\ &= 2^k n(n-1) \cdots (n-k/2) \leq 2n(2\sqrt{n})^k. \end{aligned}$$

Now take $k = 2\lfloor n^{1/6} \rfloor$. We have by Markov's inequality,

$$\begin{aligned} \mathbb{P}(\|A\| \geq 2\sqrt{n} + cn^{1/3} \log n) &\leq \frac{\mathbb{E}\|A^k\|}{(2\sqrt{n} + cn^{1/3} \log n)^k} \\ &\leq 2n \left(\frac{2\sqrt{n}}{2\sqrt{n} + cn^{1/3} \log n} \right)^k \\ &= 2n \left(1 + \frac{c}{2} \log(n) n^{-1/6} \right)^{-k} \\ &\leq 2n \exp(-kc/4 \cdot (\log n) n^{-1/6}) \\ &\leq 2n \exp(-c/3 \cdot \log n) \leq 2n^{1-c/3} \end{aligned}$$

where we use $(1+x)^{-k} \leq \exp(-kx/2)$ for $x \in (0, 1)$ in the third inequality. Taking $c > 3$ finishes the proof. $\square$

A more refined estimate of $W'(k, p)$ is available in [Vu07], which improves the error term $n^{1/3} \log n$ to $n^{1/4} \log n$.

Remark 2 (Comparison with the ϵ -net argument). Under the same setting, one can apply the concentration of spectral norm for sub-Gaussian random matrices using the ϵ -net [Ver10], which gives $\|A\| \leq C\sqrt{n}$ with probability $1 - \exp(-cn)$. The tail bound is better, but one can not reach the $2\sqrt{n}$ sharp leading order term using the net argument.

Recent development with high-moment method The high-moment method is still a powerful way to get precise estimates of the extreme singular values of random matrices. For example, sharp bounds for inhomogeneous Gaussian random matrices [LvHY18, BvH24a], random regular graphs [Bor20], and sparse random matrices [BGBK20].

1.4 Stieltjes transform

The Stieltjes transform is a powerful tool in functional analysis and probability theory, particularly in Random Matrix Theory. It maps a probability measure (or distribution) to a complex-valued function.

Definition 5. For a probability measure μ supported on the real line $\mathbb{R}$, its Stieltjes transform $m_\mu(z)$ is defined for all complex numbers z outside the support of μ (typically $z \in \mathbb{C} \setminus \mathbb{R}$) as:

$$m_\mu(z) = \int_{\mathbb{R}} \frac{1}{x - z} d\mu(x)$$

In the context of an $N \times N$ matrix H with eigenvalues $\lambda_1, \dots, \lambda_N$, the measure is the Empirical Spectral Distribution (ESD), $\mu_N = \frac{1}{N} \sum_{i=1}^N \delta_{\lambda_i}$. The Stieltjes transform becomes the normalized trace of the *resolvent* matrix $(H - zI)^{-1}$:

$$m_N(z) = \frac{1}{N} \sum_{i=1}^N \frac{1}{\lambda_i - z} = \frac{1}{N} \text{Tr}((H - zI)^{-1})$$

The crucial utility of the transform is that we can recover the probability density $\rho(x)$ of the measure μ from $m_\mu(z)$ using the *inversion formula*:

$$\rho(x) = \frac{1}{\pi} \lim_{\epsilon \rightarrow 0^+} \text{Im}[m_\mu(x + i\epsilon)]$$

Relation between moments and the Stieltjes transform Here is the derivation of how the Stieltjes transform encodes the moments. Let μ be a probability measure with compact support, and let M_k be the k -th moment of the distribution:

$$M_k = \int_{\mathbb{R}} x^k d\mu(x)$$

If we consider a complex number z such that $|z|$ is much larger than any x in the support of the distribution (i.e., $|z| \rightarrow \infty$), we can expand $\frac{1}{x-z}$ using a geometric series:

$$\frac{1}{x-z} = -\frac{1}{z} \left(\frac{1}{1-\frac{x}{z}} \right) = -\frac{1}{z} \sum_{k=0}^{\infty} \left(\frac{x}{z} \right)^k = -\sum_{k=0}^{\infty} \frac{x^k}{z^{k+1}}$$

Substituting this series back into the integral definition:

$$m(z) = \int_{\mathbb{R}} \left(-\sum_{k=0}^{\infty} \frac{x^k}{z^{k+1}} \right) d\mu(x)$$

Assuming we can swap the sum and the integral (valid for $|z| > \sup |x|$), we get:

$$m(z) = -\sum_{k=0}^{\infty} \frac{1}{z^{k+1}} \underbrace{\int_{\mathbb{R}} x^k d\mu(x)}_{M_k}.$$

This yields the fundamental relationship:

$$m(z) = -\frac{M_0}{z} - \frac{M_1}{z^2} - \frac{M_2}{z^3} - \dots = -\sum_{k=0}^{\infty} \frac{M_k}{z^{k+1}}$$

Since μ is a probability measure, the 0-th moment $M_0 = 1$. This shows that Stieltjes Transform $m(z)$ encodes moments M_k as coefficients of $\frac{1}{z^{k+1}}$ in a Laurent expansion near ∞ . Therefore, one can recover all moment information from $m(z)$.

Convergence of probability measure via Stieltjes Transform The fundamental theorem linking the convergence of Stieltjes transforms to the convergence of probability measures is the *Continuity Theorem*. This allows us to prove weak convergence of spectral distributions by studying the pointwise limit of their transforms.

Theorem 4 (Continuity Theorem for Stieltjes Transforms, Exercise 2.4.10 in [Tao23]). *Let $\{\mu_n\}_{n \geq 1}$ be a sequence of probability measures on the real line $\mathbb{R}$ and μ be a probability measure. Let $m_n(z)$ denote the Stieltjes transform of μ_n :*

$$m_n(z) = \int_{\mathbb{R}} \frac{1}{x-z} d\mu_n(x), \quad z \in \mathbb{C} \setminus \mathbb{R}$$

and $m(z)$ be the Stieltjes transform of μ . Suppose for every fixed $z \in \mathbb{C}^+$:

$$\lim_{n \rightarrow \infty} m_n(z) = m(z).$$

Then

$$\mu_n \xrightarrow{w} \mu.$$

1.5 Marchenko-Pastur law with relaxed dependence

We will present a short proof of the Marchenko-Pastur law using the Stieltjes transform due to Yaskov [Yas16].

Theorem 5. Let X be a $p \times n$ random matrix whose columns $\mathbf{x}_1, \dots, \mathbf{x}_n$ are i.i.d. with the same distribution as $\mathbf{x} \in \mathbb{R}^p$. Assume $\frac{p}{n} \rightarrow c$ and

$$\frac{1}{p} \left(\mathbf{x}^\top A \mathbf{x} - \text{tr}(A) \right) \rightarrow 0 \quad \text{in probability.} \quad (3)$$

Then the empirical spectral distribution of $\frac{1}{n} X X^\top$ converges weakly almost surely to the Marchenko-Pastur law μ_c .

Remark 3. The condition 3 is a weak form of concentration of $\mathbf{x}^\top A \mathbf{x}$ around $\text{tr}(A)$. This can be verified if $\mathbb{E} \mathbf{x}^\top A \mathbf{x} = \text{tr}(A)$ and $\text{Var}(\mathbf{x}^\top A \mathbf{x}) = o(p^2)$. A non-asymptotic version of (3) can be obtained for random vectors with independent sub-Gaussian entries due to Hanson-Wright inequality [RV13].

We will use the following lemma that gives the inverse of the rank-one perturbation of a matrix:

Lemma 6 (Sherman-Morrison Formula). Let $A \in \mathbb{R}^{n \times n}$ be an invertible matrix, and let $u, v \in \mathbb{R}^n$ be column vectors. If $1 + v^\top A^{-1} u \neq 0$, then the matrix sum $A + uv^\top$ is invertible, and its inverse is given by:

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1} u v^\top A^{-1}}{1 + v^\top A^{-1} u}.$$

The following lemmas are linear algebra facts:

Lemma 7. Let C be a $p \times p$ real symmetric positive semi-definite matrix and $x \in \mathbb{R}^p$. If $z \in \mathbb{C}$ is such that $v = \text{Im}(z) > 0$, then:

1. $\|(C - zI_p)^{-1}\| \leq \frac{1}{v}$
2. $|\text{tr}(C + xx^\top - zI_p)^{-1} - \text{tr}(C - zI_p)^{-1}| \leq \frac{1}{v}$
3. $|x^\top (C + xx^\top - zI_p)^{-1} x| \leq 1 + \frac{|z|}{v}$
4. $\text{Im}(z + z \text{tr}(C - zI_p)^{-1}) \geq v$ and $\text{Im}(\text{tr}(C - zI_p)^{-1}) > 0$
5. $\text{Im}(z + z x^\top (C - zI_p)^{-1} x) \geq v$.

Proof of Theorem 5. We define the Stieltjes transform of the empirical measure μ_n as $s_n(z)$ and the limit measure $\mu_{MP,c}$ as $s(z)$ for $z \in \mathbb{C}$ with $\text{Im}(z) > 0$.

$$s_n(z) = \frac{1}{p} \text{tr}(n^{-1} X X^\top - zI_p)^{-1}$$

By the Stieltjes continuity theorem, proving $\mu_{pn} \Rightarrow \mu_c$ almost surely reduces to proving $s_n(z) \rightarrow s(z)$ almost surely for $z \in \mathbb{C}$ with $\text{Im}(z) > 0$.

Step 1: Reduction to Expectation

Lemma 8 (Lemma 4.1 from [Ada11]). Let A be any $n \times N$ random matrix with independent rows and let $s_n : \mathbb{C}^+ \rightarrow \mathbb{C}$ be the Stieltjes transform of the spectral measure of $H = AA^*$. Then for any $z = x + iy \in \mathbb{C}_+$ and any $\varepsilon > 0$,

$$\mathbb{P}(|s_n(z) - \mathbb{E}s_n(z)| \geq \varepsilon) \leq C \exp(-c\varepsilon^2 ny^2).$$

Proof. The proof is based on martingale concentration. Lemma 4.1 from [Ada11]. $\square$

Lemma 8 above shows that $s_n(z)$ concentrates around its mean:

$$s_n(z) - \mathbb{E}s_n(z) \rightarrow 0 \quad \text{a.s.}$$

Thus, we focus on the convergence of $\mathbb{E}s_n(z)$ to $s(z)$.

Step 2: The Sherman-Morrison Formula Let

$$A_n = XX^\top = \sum_{k=1}^n \mathbf{x}_k \mathbf{x}_k^\top.$$

Consider the matrix with one additional column: $B_n = A_n + \mathbf{x}_{n+1} \mathbf{x}_{n+1}^\top = \sum_{k=1}^{n+1} \mathbf{x}_k \mathbf{x}_k^\top$. We start with the identity:

$$p = \text{Tr}((B_n - znI_p)(B_n - znI_p)^{-1}).$$

Expanding the trace term:

$$p = \sum_{k=1}^{n+1} \mathbf{x}_k^\top (B_n - znI_p)^{-1} \mathbf{x}_k - zn \text{Tr}(B_n - znI_p)^{-1}.$$

Taking expectations and using the independence of the columns $\{\mathbf{x}_k\}_{k=1}^{n+1}$, the sum simplifies to

$$p = (n+1)\mathbb{E}[\mathbf{x}_{n+1}^\top (B_n - znI_p)^{-1} \mathbf{x}_{n+1}] - zn\mathbb{E}[\text{Tr}(B_n - znI_p)^{-1}]. \quad (4)$$

We define the scaled transform

$$S_n(z) = \text{Tr}(A_n - znI_p)^{-1}.$$

Note that

$$S_n(z) = \frac{p}{n} s_n(z).$$

One can obtain from Lemma 7 (2) that

$$\mathbb{E}[\text{Tr}(B_n - znI_p)^{-1}] = \mathbb{E}S_n(z) + O(1/n). \quad (5)$$

To isolate the dependence on $\mathbf{x}_{n+1}$, we use the Sherman-Morrison formula in Lemma 6. This relates the resolvent of B_n to the resolvent of A_n :

$$\begin{aligned} (B_n - znI_p)^{-1} &= (A_n + \mathbf{x}_{n+1} \mathbf{x}_{n+1}^\top - znI_p)^{-1} \\ &= (A_n - znI_p)^{-1} - \frac{(A_n - znI_p)^{-1} \mathbf{x}_{n+1} \mathbf{x}_{n+1}^\top (A_n - znI_p)^{-1}}{1 + \mathbf{x}_{n+1}^\top (A_n - znI_p)^{-1} \mathbf{x}_{n+1}}. \end{aligned}$$

Therefore,

$$\mathbf{x}_{n+1}^\top (B_n - znI_p)^{-1} \mathbf{x}_{n+1} = \frac{\mathbf{x}_{n+1}^\top (A_n - znI_p)^{-1} \mathbf{x}_{n+1}}{1 + \mathbf{x}_{n+1}^\top (A_n - znI_p)^{-1} \mathbf{x}_{n+1}}. \quad (6)$$

Under (3), since A_n and $\mathbf{x}_{n+1}$ are independent, the quadratic form $\mathbf{x}_{n+1}^\top (A_n - znI_p)^{-1} \mathbf{x}_{n+1}$ satisfies

$$\mathbf{x}_{n+1}^\top (A_n - znI_p)^{-1} \mathbf{x}_{n+1} - \text{Tr}(A_n - znI_p)^{-1} = \mathbf{x}_{n+1}^\top (A_n - znI_p)^{-1} \mathbf{x}_{n+1} - S_n(z) \rightarrow 0$$

in probability.

Step 3: The Quadratic Equation Since $S_n(z) - \mathbb{E}S_n(z) \rightarrow 0$ in probability. Substituting this into (6) (with additional justifications) we get:

$$\mathbb{E}[\mathbf{x}_{n+1}^\top (B_n - znI_p)^{-1} \mathbf{x}_{n+1}] = \frac{\mathbb{E}S_n(z)}{1 + \mathbb{E}S_n(z)} + o(1).$$

Substituting back into the dimension identity (4) together with (5),

$$\frac{\mathbb{E}S_n(z)}{1 + \mathbb{E}S_n(z)} - z\mathbb{E}S_n(z) = c + o(1).$$

Therefore $\mathbb{E}S_n(z)$ approximately satisfies the following equation:

$$c = \frac{S}{1 + S} - zS$$

Rearranging gives the equation:

$$zS^2 + (z - 1 + c)S + c = 0.$$

This quadratic equation has a unique solution $S(z) = cs(z)$ with $\text{Im}(S(z)) \geq 0$, where

$$s(z) = \frac{-(z - 1 + c) + \sqrt{(z - 1 + c)^2 - 4cz}}{2cz}$$

corresponds to the Stieltjes transform of the Marchenko-Pastur distribution. A stability argument implies $\mathbb{E}S_n(z) \rightarrow S(z)$ hence $\mathbb{E}s_n(z) \rightarrow s(z)$. This finishes the proof. $\square$

Extension Sample covariance matrices with independent column vectors are common settings in high-dimensional statistics, in particular in covariance estimation problems [KL17]. Beyond the isotropic case when $\mathbb{E}\mathbf{x}\mathbf{x}^\top = \Sigma$, a deformed Marchenko-Pastur law can be derived [BZ08, KY17].

1.6 Semicircle law with Stieltjes transform

A similar strategy can be used to derive the semicircle law with the Stieltjes transform. We sketch the major steps below. See [BZ08] for more details.

Let W_n be an $n \times n$ standard real symmetric Wigner matrix. Assume the entries w_{ij} (for $i \leq j$) are independent random variables with mean 0 and variance 1, with all moments bounded.

$$X_n = \frac{1}{\sqrt{n}} W_n$$

We define the Stieltjes transform of the ESD of X_n as:

$$m_n(z) = \frac{1}{n} \text{Tr}(X_n - zI_n)^{-1}, \quad \text{for } \text{Im}(z) > 0.$$

Let $G(z) = (X_n - zI_n)^{-1}$ be the resolvent matrix. Unlike the rank-1 update method used for covariance matrices, we analyze the diagonal entries $G_{kk}(z)$ using the block inversion formula (Schur complement).

Lemma 9 (Block Inversion Formula). *Consider a block matrix M partitioned as:*

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

where A is $p \times p$ and D is $q \times q$. If M is invertible and A is invertible, the inverse M^{-1} can be expressed using the Schur complement $S_A = D - CA^{-1}B$:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BS_A^{-1}CA^{-1} & -A^{-1}BS_A^{-1} \\ -S_A^{-1}CA^{-1} & S_A^{-1} \end{pmatrix}$$

With Lemma 9, the k -th diagonal entry is given by:

$$G_{kk}(z) = \frac{1}{\frac{w_{kk}}{\sqrt{n}} - z - \mathbf{h}_k^\top G^{(k)}(z) \mathbf{h}_k},$$

where:

- $w_{kk}/\sqrt{n}$ is the k -th diagonal element of X_n .
- $\mathbf{h}_k$ is the k -th column of X_n with the k -th element removed (scaled by $1/\sqrt{n}$).
- $G^{(k)}(z)$ is the resolvent of the $(n-1) \times (n-1)$ minor matrix $X_n^{(k)}$ (where the k -th row and column are removed).

We apply the concentration of quadratic forms. The term $\mathbf{h}_k^\top G^{(k)}(z) \mathbf{h}_k$ involves a random vector $\mathbf{h}_k$ independent of the matrix $G^{(k)}$. For large n :

$$\mathbf{h}_k^\top G^{(k)}(z) \mathbf{h}_k \approx \mathbb{E}[\mathbf{h}_k^\top G^{(k)}(z) \mathbf{h}_k] = \frac{1}{n} \text{Tr}(G^{(k)}(z)).$$

As $n \rightarrow \infty$, the diagonal term $w_{kk}/\sqrt{n} \rightarrow 0$, and the trace of the minor is asymptotically equivalent to the full trace: $\frac{1}{n} \text{Tr}(G^{(k)}(z)) \approx m_n(z)$. Substituting these into the Schur complement formula:

$$G_{kk}(z) \approx \frac{1}{-z - m_n(z)}.$$

Since $m_n(z) = \frac{1}{n} \sum_{k=1}^n G_{kk}(z)$, averaging over $k \in [n]$,

$$m_n(z) \approx \frac{1}{-z - m_n(z)}.$$

The Stieltjes transform $s(z)$ of μ_{sc} satisfies the quadratic equation:

$$m(z)^2 + zm(z) + 1 = 0. \tag{7}$$

Therefore, a stability argument gives $m_n(z) = m(z) + o(1)$ with high probability.

Remark 4. The strategy above was carried out rigorously in [TV11] to prove the local law of Wigner matrices.

1.7 Local law and eigenvector delocalization

Local law and eigenvector delocalization of random matrices are important topics in mathematical physics [BGK16]. We study spectral statistics of large random matrices at small scale via *local laws*. These are high-probability estimates for the resolvent (Green function)

$$G(z) := (H - z)^{-1}, \quad z = E + i\eta \in \mathbb{C}_+,$$

down to scales η as small as $N^{-1+\delta}$ (almost the typical eigenvalue spacing N^{-1} in the bulk). We only state some important results, and more details can be found in [BGK16].

Definition 6 (Real symmetric Wigner). Let $H = (h_{ij})_{1 \leq i, j \leq N}$ be real symmetric with

$$h_{ij} = h_{ji}, \quad \mathbb{E}h_{ij} = 0, \quad \mathbb{E}h_{ij}^2 = \frac{1}{N} \quad (i \neq j), \quad \mathbb{E}h_{ii}^2 = \frac{1}{N}.$$

Assume $(\sqrt{N}h_{ij})$ has a uniformly sub-Gaussian tail.

Lemma 10. *Let eigenvalues and orthonormal eigenvectors be*

$$Hu_\alpha = \lambda_\alpha u_\alpha, \quad \alpha = 1, \dots, N.$$

Then for $z \in \mathbb{C} \setminus \{\lambda_\alpha\}$,

$$G(z) = \sum_{\alpha=1}^N \frac{u_\alpha u_\alpha^\top}{\lambda_\alpha - z}, \quad G_{ij}(z) = \sum_{\alpha=1}^N \frac{u_\alpha(i) u_\alpha(j)}{\lambda_\alpha - z}.$$

In particular,

$$\operatorname{Im} G_{ii}(E + i\eta) = \sum_{\alpha=1}^N \frac{\eta |u_\alpha(i)|^2}{(\lambda_\alpha - E)^2 + \eta^2}. \quad (8)$$

Proof. Since H is Hermitian, it admits the spectral decomposition

$$H = \sum_{\alpha=1}^N \lambda_\alpha u_\alpha u_\alpha^*,$$

where $\{u_\alpha\}$ form an orthonormal basis of $\mathbb{C}^N$ (or $\mathbb{R}^N$). For $z \in \mathbb{C} \setminus \mathbb{R}$, the resolvent can therefore be written as

$$G(z) = (H - z)^{-1} = \sum_{\alpha=1}^N \frac{u_\alpha u_\alpha^*}{\lambda_\alpha - z}.$$

Taking the (i, i) -entry yields

$$G_{ii}(z) = \sum_{\alpha=1}^N \frac{|u_\alpha(i)|^2}{\lambda_\alpha - z}.$$

Now set $z = E + i\eta$ with $\eta > 0$. For each α ,

$$\frac{1}{\lambda_\alpha - (E + i\eta)} = \frac{\lambda_\alpha - E + i\eta}{(\lambda_\alpha - E)^2 + \eta^2},$$

and hence

$$\operatorname{Im} \frac{1}{\lambda_\alpha - (E + i\eta)} = \frac{\eta}{(\lambda_\alpha - E)^2 + \eta^2}.$$

Taking imaginary parts term by term gives

$$\operatorname{Im} G_{ii}(E + i\eta) = \sum_{\alpha=1}^N |u_\alpha(i)|^2 \operatorname{Im} \frac{1}{\lambda_\alpha - (E + i\eta)} = \sum_{\alpha=1}^N \frac{\eta |u_\alpha(i)|^2}{(\lambda_\alpha - E)^2 + \eta^2},$$

as claimed. $\square$

Definition 7 (Stochastic domination). For nonnegative random variables $X = X^{(N)}$ and deterministic $Y = Y^{(N)}$, we write

$$X \prec Y$$

if for all $\epsilon > 0$ and $D > 0$ there exists $N_0(\epsilon, D)$ such that for $N \geq N_0$,

$$\mathbb{P}(X > N^\epsilon Y) \leq N^{-D}.$$

Fix a small constant $\tau \in (0, 1)$ (bulk cutoff) and consider energies $E \in [-2 + \tau, 2 - \tau]$. Define the *spectral domain*

$$\mathcal{D}(\tau) := \left\{ z = E + i\eta : E \in [-2 + \tau, 2 - \tau], \eta \in [N^{-1+\tau}, 1] \right\}.$$

One can also include edge regimes with a different η lower bound; we keep the bulk for simplicity.

Theorem 6 (Averaged local semicircle law). *For Wigner H , uniformly for $z \in \mathcal{D}(\tau)$,*

$$|m_N(z) - m_{\text{sc}}(z)| \prec \frac{1}{N\eta}.$$

The averaged local law implies local eigenvalue counting estimates in windows of size η .

Theorem 7 (Local eigenvalue counting from the averaged local law). *Assume the averaged local semicircle law holds on $\mathcal{D}(\tau)$, i.e.*

$$m_N(z) := \frac{1}{N} \operatorname{Tr}(H - z)^{-1}, \quad |m_N(z) - m_{\text{sc}}(z)| \prec \frac{1}{N\eta}, \quad z \in \mathcal{D}(\tau).$$

Then uniformly for all $E \in [-2 + \tau, 2 - \tau]$ and all $\eta \in [N^{-1+\tau}, 1]$,

$$|\#\{\alpha : \lambda_\alpha \in [E - \eta, E + \eta]\} - 2N\eta \rho_{\text{sc}}(E)| \prec N\eta \left(\eta + \frac{1}{N\eta} \right),$$

where $\rho_{\text{sc}}(x) = \frac{1}{2\pi} \sqrt{(4 - x^2)_+}$ is the semicircle density. Equivalently,

$$\frac{1}{N\eta} \#\{\alpha : \lambda_\alpha \in [E - \eta, E + \eta]\} = 2\rho_{\text{sc}}(E) + \mathcal{O}_\prec \left(\eta + \frac{1}{N\eta} \right).$$

Theorem 8 (Entrywise local semicircle law). *For Wigner H , uniformly for $z \in \mathcal{D}(\tau)$,*

$$\max_{i,j} |G_{ij}(z) - \delta_{ij} m_{\text{sc}}(z)| \prec \Psi(z),$$

where the control parameter is

$$\Psi(z) := \sqrt{\frac{\operatorname{Im} m_{\text{sc}}(z)}{N\eta}} + \frac{1}{N\eta}.$$

In the bulk $\operatorname{Im} m_{\text{sc}}(z) \asymp 1$, so $\Psi(z) \asymp (N\eta)^{-1/2}$.

Eigenvector delocalization We now show how the entrywise local law implies that eigenvectors are not concentrated on few coordinates.

Theorem 9 (Eigenvector ℓ^∞ -delocalization in the bulk). *Fix $\tau \in (0, 1)$. Let u_α be an eigenvector with eigenvalue $\lambda_\alpha \in [-2 + \tau, 2 - \tau]$. Then for any $\epsilon > 0$,*

$$\|u_\alpha\|_\infty := \max_i |u_\alpha(i)| \prec N^{-1/2}.$$

More precisely, $\max_i |u_\alpha(i)| \leq N^{-1/2+\epsilon}$ with overwhelming probability.

Proof sketch. Fix α with λ_α in the bulk. Choose $\eta := N^{-1+\tau}$ and set $E := \lambda_\alpha$. From (8) we have, for every i ,

$$\operatorname{Im} G_{ii}(\lambda_\alpha + i\eta) = \sum_{\beta=1}^N \frac{\eta |u_\beta(i)|^2}{(\lambda_\beta - \lambda_\alpha)^2 + \eta^2} \geq \frac{\eta |u_\alpha(i)|^2}{0 + \eta^2} = \frac{|u_\alpha(i)|^2}{\eta}.$$

Hence

$$|u_\alpha(i)|^2 \leq \eta \operatorname{Im} G_{ii}(\lambda_\alpha + i\eta). \quad (9)$$

By the entrywise local law, for $z = \lambda_\alpha + i\eta \in \mathcal{D}(\tau)$,

$$G_{ii}(z) = m_{\text{sc}}(z) + O_{\prec}(\Psi(z)).$$

In the bulk, $\operatorname{Im} m_{\text{sc}}(z) \asymp 1$ uniformly for $\eta \in [N^{-1+\tau}, 1]$ from (7), and $\Psi(z) \ll 1$. Therefore $\operatorname{Im} G_{ii}(z) = O_{\prec}(1)$. Plug into (9):

$$|u_\alpha(i)|^2 = O_{\prec}(\eta) = O_{\prec}(N^{-1+\tau}).$$

Taking square roots and then a union bound over i and α ,

$$\max_i |u_\alpha(i)| \prec N^{-1/2+\tau/2}.$$

Since $\tau > 0$ can be taken arbitrarily small, this gives optimal eigenvector delocalization. $\square$

Remark 5. This argument gives delocalization for eigenvectors whose eigenvalues lie in the regime where the local law holds. At the spectral edges, one typically still has delocalization, but the proof uses edge local laws [BGK16]. A more refined argument can obtain $O(\sqrt{\log n}/n)$ bound for eigenvector delocalization in the bulk, see [VW15]. A sharp eigenvector delocalization with explicit constant can be found in [BL22].

2 Community detection in the stochastic block model

In this section, we will introduce the problem of community detection in the stochastic block model, with a focus on spectral clustering. A good survey on this topic is [Abb18].

The *two-block stochastic block model* (SBM) is a random graph model with two latent communities. Let $[n] := \{1, \dots, n\}$. Each vertex $i \in [n]$ is assigned an unobserved label

$$\sigma_i \in \{+1, -1\}.$$

A common choice is the *balanced* prior, where $\sigma_1, \dots, \sigma_n$ are i.i.d. with

$$\mathbb{P}(\sigma_i = +1) = \mathbb{P}(\sigma_i = -1) = \frac{1}{2}.$$

Conditional on $\sigma = (\sigma_1, \dots, \sigma_n)$, the adjacency matrix $A = (A_{ij})_{i,j \in [n]}$ of an undirected simple graph is generated as follows: for each $1 \leq i < j \leq n$, independently,

$$\mathbb{P}(A_{ij} = 1 \mid \sigma) = \begin{cases} p, & \text{if } \sigma_i = \sigma_j, \\ q, & \text{if } \sigma_i \neq \sigma_j, \end{cases} \quad A_{ji} = A_{ij}, \quad A_{ii} = 0.$$

The model is *assortative* if $p > q$ (more edges within communities) and *disassortative* if $p < q$.

Let $\mathbf{1} \in \mathbb{R}^n$ be the all-ones vector and write $\sigma \in \{+1, -1\}^n$ for the label vector. Ignoring diagonal effects, the conditional expectation of the adjacency matrix has the rank-two form

$$\mathbb{E}[A \mid \sigma] \approx \frac{p+q}{2} \mathbf{1}\mathbf{1}^\top + \frac{p-q}{2} \sigma\sigma^\top.$$

The second term represents the community signal.

Community detection Given the observed graph A , the goal is to infer the hidden labels $\sigma \in \{+1, -1\}^n$ up to a global sign flip (since σ and $-\sigma$ generate the same distribution).

A standard metric for an estimator $\hat{\sigma} \in \{+1, -1\}^n$ is the *overlap*

$$\text{ov}(\hat{\sigma}, \sigma) := \frac{1}{n} \max_{s \in \{\pm 1\}} \sum_{i=1}^n \mathbf{1}\{\hat{\sigma}_i = s \sigma_i\} = \frac{1}{2} \left(1 + \frac{1}{n} |\langle \hat{\sigma}, \sigma \rangle| \right).$$

Random guessing yields $\text{ov}(\hat{\sigma}, \sigma) = \frac{1}{2} + o(1)$. Any $\text{ov}(\hat{\sigma}, \sigma) > \frac{1}{2}$ bounded away from $\frac{1}{2}$ is called *weak recovery* (or *detection*). Two commonly studied scalings are:

- *Constant-degree regime (detection/partial recovery)*: $p = \frac{a}{n}$ with constants a, b . The detection threshold is

$$(a - b)^2 > 2(a + b),$$

established in [BLM18, MNS15a].

- *Logarithmic-degree regime (exact recovery)*: $p = \frac{a \log n}{n}$ and $q = \frac{b \log n}{n}$. The exact recovery threshold is

$$(\sqrt{a} - \sqrt{b}) \geq 2,$$

established in [ABH16]. A simple spectral method to achieve this threshold is proved in [AFWZ20].

2.1 Matrix concentration inequality

A first step we need to show is that the adjacency matrix A of an SBM is close to its expectation. This guarantees the signal information is not lost.

Theorem 10 (Matrix Bernstein inequality). *Let $\{S_k\}_{k=1}^N$ be independent random matrices in $\mathbb{R}^{d_1 \times d_2}$ such that $\mathbb{E}S_k = 0$ and $\|S_k\| \leq L$ almost surely for all k . Define*

$$Z := \sum_{k=1}^N S_k, \quad v(Z) := \max\left\{\left\|\mathbb{E}(ZZ^\top)\right\|, \left\|\mathbb{E}(Z^\top Z)\right\|\right\}.$$

Then for every $t \geq 0$,

$$\mathbb{P}(\|Z\| \geq t) \leq (d_1 + d_2) \exp\left(-\frac{t^2/2}{v(Z) + Lt/3}\right),$$

and also

$$\mathbb{E}\|Z\| \leq \sqrt{2v(Z) \log(d_1 + d_2)} + \frac{L}{3} \log(d_1 + d_2).$$

The matrix Bernstein inequality [T⁺15] is a widely used inequality that obtains spectral norm concentration bounds up to a $\log(d_1 + d_2)$ factor. For example, applying it to a Wigner W matrix with bounded entries yields $\|W\| = O(\sqrt{n \log n})$ with high probability.

Matrix Bernstein for unbounded entries When the random matrix has unbounded entries, the following generalization is useful if one can control the moments of each component.

Lemma 11 ([Tro12], Theorem 6.2). *Let Z_k be independent Hermitian matrices of size $n \times n$. Assume*

$$\mathbb{E}Z_i = 0, \quad \|\mathbb{E}[Z_i^p]\| \leq \frac{1}{2}p!R^{p-2}a^2,$$

for any integer $p \geq 2$. Then for all $t \geq 0$,

$$\mathbb{P}\left(\left\|\sum_{i=1}^{d_1} Z_i\right\| \geq t\right) \leq n \exp\left(-\frac{t^2}{2d_1a^2 + 2Rt}\right).$$

Another strategy is to truncate the unbounded entries, see for example [HSS16, Proposition A.7].

Remark 6 (Removing the $\log n$ factor). Recent breakthroughs in [BBVH23, BvH24b] removed the extra $\log(n)$ factor in the matrix Bernstein's inequality in many cases.

Concentration of random graphs For adjacency matrices, a direct Bernstein application yields $\|A - \mathbb{E}A\| \lesssim \sqrt{d \log n} + \log n$ where $d = np$ is the expected degree scale. In many random graph models, the true scale is $\Theta(\sqrt{d})$; obtaining this often requires more refined arguments. Below is a commonly used bound for the concentration of the random graph $G(n, p)$ due to [FO05]. When applying this to the inhomogeneous random graph $G(n, p_{ij})$ where each edge ij is connected with probability p_{ij} . The following generalization holds. See [LR15, LLV17].

Theorem 11 (Concentration of random graphs). *Let $G \sim G(n, p_{ij})$. For any $c > 0$. Let $np_{\max} \geq c \log n$. Then for any $K > 0$, there exists constant C depending on c, K such that with probability $1 - n^{-K}$,*

$$\|A - \mathbb{E}A\| \leq C\sqrt{np_{\max}}.$$

Lemma 12 (Weyl inequality). *The k -th largest eigenvalues of symmetric matrices A and B (of the same dimensions) satisfy*

$$|\lambda_k(A) - \lambda_k(B)| \leq \|A - B\|.$$

Similarly, the k -th largest singular values of general rectangular matrices satisfy

$$|s_k(A) - s_k(B)| \leq \|A - B\|.$$

In our SBM example, $\|A - \mathbb{E}A\| \leq C\sqrt{np}$ with high probability as long as $np \geq c \log n$. From Weyl's inequality,

$$|\lambda_k(A) - \lambda_k(\mathbb{E}A)| \leq \|A - \mathbb{E}A\| = O(\sqrt{np}).$$

Therefore $\lambda_1(A), \lambda_2(A)$ are concentrated around $n(p+q)/2, n(p-q)/2$, respectively. And all other eigenvalues are concentrated around 0. As long as $n(p-q)/2 \gg \sqrt{np}$, we should see a separation of eigenvalues for A . See Figure 2.

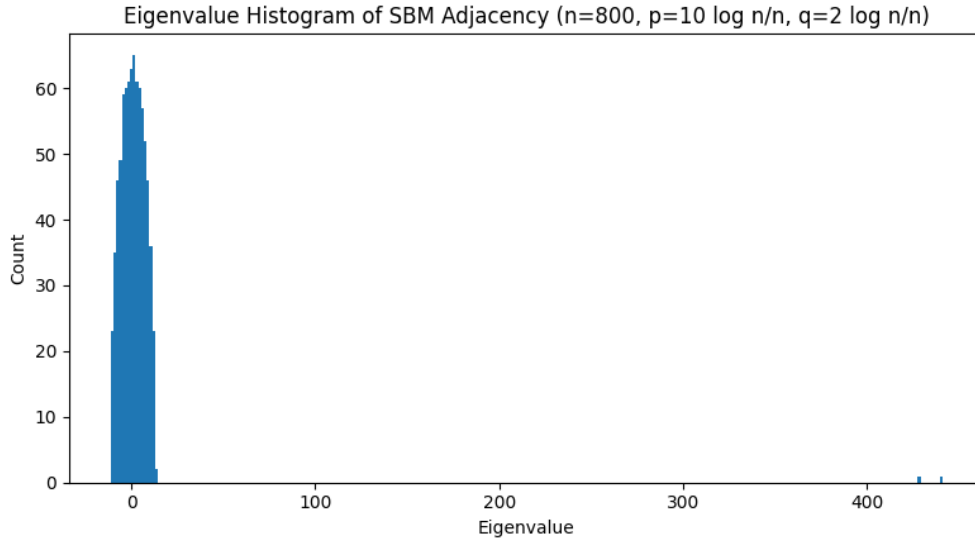


Figure 2: Eigenvalue histogram of an SBM. Two outliers outside the bulk around $\lambda_1(\mathbb{E}A), \lambda_2(\mathbb{E}A)$.

2.2 Davis-Kahan inequality

An introduction to the spectral method based on Davis-Kahan can be found in [CCFM21]. We state the version from [Ver18].

Theorem 12 (Davis-Kahan inequality). *Consider two symmetric matrices $A, B \in \mathbb{R}^{n \times n}$ with spectral decompositions*

$$A = \sum_{i=1}^n \lambda_i u_i u_i^\top, \quad B = \sum_{i=1}^n \mu_i v_i v_i^\top,$$

where the eigenvalues are ordered as $\lambda_1 \geq \dots \geq \lambda_n$ and $\mu_1 \geq \dots \geq \mu_n$. Assume that the k -th eigenvalue of A is δ -separated from the rest:

$$\min_{i: i \neq k} |\lambda_k - \lambda_i| = \delta > 0.$$

Then the corresponding unit eigenvectors u_k and v_k satisfy

$$\sin \angle(u_k, v_k) \leq \frac{2\|A - B\|}{\delta}.$$

Moreover, the ℓ_2 distance between eigenvectors, up to a sign flip, obeys

$$\min\{\|u_k - v_k\|_2, \|u_k + v_k\|_2\} \leq \frac{2\sqrt{2}\|A - B\|}{\delta}.$$

We prove the following lemma about projection matrices first.

Lemma 13 (Davis–Kahan inequality for spectral projections). *Consider two symmetric matrices A and B with spectral decompositions*

$$A = \sum_{i=1}^n \lambda_i u_i u_i^\top \quad \text{and} \quad B = \sum_{j=1}^n \mu_j v_j v_j^\top.$$

Let I, J be two δ -separated subsets of $\mathbb{R}$, with I being an interval. Then the spectral projections

$$P = \sum_{i:\lambda_i \in I} u_i u_i^\top \quad \text{and} \quad Q = \sum_{j:\mu_j \in J} v_j v_j^\top$$

satisfy

$$\|QP\| \leq \frac{\|A - B\|}{\delta}.$$

Proof. Without loss of generality, assume that the interval I is finite and closed. Adding the same multiple of the identity to A and B , we may center I as

$$I = [-r, r],$$

so that $|\lambda_i| \leq r$ for $\lambda_i \in I$ and $|\mu_j| \geq r + \delta$ for $\mu_j \in J$. The idea is to study how P and Q interact through

$$H := B - A.$$

We have

$$\|H\| \geq \|QHP\| = \|QBP - QAP\| \geq \|QBP\| - \|QAP\|. \quad (10)$$

The spectral projection Q commutes with B , so

$$\|QBP\| = \|BQP\| \geq (r + \delta)\|QP\|. \quad (11)$$

To see the last inequality, note that the image of Q is spanned by orthonormal vectors v_j with $|\mu_j| \geq r + \delta$. The matrix B maps each such vector v_j to $\mu_j v_j$, thus scaling it by at least $r + \delta$. Hence, for any x ,

$$\|BQP x\|_2 \geq (r + \delta)\|QP x\|_2.$$

Taking the supremum over unit vectors x yields (11). Also, since

$$AP = PAP = \sum_{i:\lambda_i \in I} \lambda_i u_i u_i^\top,$$

we obtain

$$\|QAP\| = \|QPAP\| \leq \|QP\| \|AP\| \leq r\|QP\|, \quad (12)$$

because

$$\|AP\| = \max_{i:\lambda_i \in I} |\lambda_i| \leq r.$$

Putting the bounds (11) and (12) into (10), we get

$$\|H\| \geq \delta\|QP\|,$$

which completes the proof. $\square$

Proof of Theorem 12. We can assume that

$$\varepsilon := \|A - B\| \leq \delta/2,$$

otherwise the conclusion is trivial. By Weyl's inequality (Lemma 12), we have

$$|\lambda_j - \mu_j| \leq \varepsilon \quad \text{for each } j,$$

so

$$\min_{j:j \neq k} |\lambda_k - \mu_j| \geq \min_{j:j \neq k} |\lambda_k - \lambda_j| - \varepsilon = \delta - \varepsilon \geq \delta/2.$$

Apply Lemma 13 for the $\delta/2$ -separated subsets

$$I = \{\lambda_k\} \quad \text{and} \quad J = \{\mu_j : j \neq k\}$$

to get

$$\|QP\| \leq \frac{2\varepsilon}{\delta}.$$

To finish the proof, recall that $P = u_k u_k^\top$, $Q = I - v_k v_k^\top$. A short computation shows that

$$\|QP\| = \|Qu_k\|_2 = \sqrt{1 - \langle u_k, v_k \rangle^2} = \sin \angle(u_k, v_k).$$

Without loss of generality, assume $\langle u_k, v_k \rangle = \cos(\alpha)$, $\alpha \in [0, \pi/2]$ (otherwise take $-u_k$). Then

$$\|u_k - v_k\|^2 = 2(1 - \cos \alpha) = 4 \sin^2(\alpha/2).$$

And $\|u - v\| = 2 \sin(\alpha/2)$. Since $\sin \alpha = 2 \sin(\alpha/2) \cos(\alpha/2) \geq \sqrt{2} \sin(\alpha/2)$. Thus $\|u_k - v_k\| = 2 \sin(\alpha/2) \leq \sqrt{2} \sin(\alpha)$. This implies

$$\min\{\|u_k - v_k\|_2, \|u_k + v_k\|_2\} \leq \frac{2\sqrt{2}\|A - B\|}{\delta}.$$

$\square$

With the concentration of random graphs and Davis-Kahan inequality, we can prove a recovery result of a spectral method.

2.3 Weak recovery with the spectral method

Denote $p = \frac{a}{n}$, $q = \frac{b}{n}$ and $d = n(\frac{p+q}{2})$. For simplicity², we assume

$$\mathbb{E}A = d\mathbf{1}\mathbf{1}^\top + \frac{n(p-q)}{2}\sigma\sigma^\top.$$

Signal-plus-noise decomposition. Let

$$\tilde{A} := A - \frac{\hat{d}}{n}\mathbf{1}\mathbf{1}^\top,$$

where $\hat{d} = \sum_{ij} A_{ij}/n$ is the average degree of all vertices. Define the unit vector

$$u := \frac{\sigma}{\sqrt{n}}, \quad u \perp \mathbf{1}.$$

A direct computation gives

$$\mathbb{E}[\tilde{A} \mid \sigma] = \frac{a-b}{2n}\sigma\sigma^\top = \frac{a-b}{2}uu^\top =: M.$$

Thus M has eigenvalue $\lambda_1(M) = (a-b)/2$ with eigenvector u , and all other eigenvalues equal to 0. The eigengap is therefore

$$\delta = \frac{a-b}{2}.$$

Write

$$\tilde{A} = M + E, \quad E := \tilde{A} - \mathbb{E}[\tilde{A}] = A - \mathbb{E}A + \frac{d-\hat{d}}{n}\mathbf{1}\mathbf{1}^\top.$$

Matrix concentration. By Theorem 11 for inhomogeneous Erdős–Rényi graphs, if $d \gtrsim \log n$, then with high probability,

$$\|E\| \lesssim \sqrt{d},$$

where we use $d - \hat{d} = O(\sqrt{d})$ when $d \gtrsim \log n$ (one can check by Chernoff bound).

Eigenvector perturbation Let $\hat{v}$ be the leading eigenvector of $\tilde{A}$. By the Davis–Kahan $\sin \Theta$ theorem,

$$\sin \angle(\hat{v}, u) \leq \frac{\|E\|}{\delta} = \frac{2\|E\|}{a-b}.$$

Consequently, with high probability,

$$\sin \angle(\hat{v}, u) \lesssim \frac{\sqrt{d}}{a-b}.$$

²The diagonal term only changes the spectrum by pI , which is much smaller than the fluctuation $O(\sqrt{np})$.

From angle to misclassification. Let $\theta = \text{sign}(\langle \hat{v}, u \rangle)$. Using the angle-distance inequality for unit vectors,

$$\|\hat{v} - \theta u\|_2 \leq \sqrt{2} \sin \angle(\hat{v}, u).$$

Let m be the number of misclassified vertices under the estimator

$$\hat{\sigma}_i := \text{sign}(\hat{v}_i).$$

Then $\|\hat{v} - \theta u\|_2^2 \geq \frac{m}{n}$, which implies $\frac{m}{n} \leq 2 \sin^2 \angle(\hat{v}, u)$. Combining with the Davis–Kahan bound yields

$$\frac{m}{n} \lesssim \frac{d}{(a-b)^2}.$$

Theorem 13 (Weak recovery for SBM via spectral method). *Assume $a + b \gtrsim \log n$ and $a > b$. Then the spectral estimator*

$$\hat{\sigma}_i = \text{sign}(\hat{v}_i)$$

satisfies, with high probability,

$$\min_{\theta \in \{\pm 1\}} \frac{1}{n} |\{i : \hat{\sigma}_i \neq \theta \sigma_i\}| \lesssim \frac{(a+b)}{(a-b)^2}.$$

In particular, if $a - b \gg \sqrt{a + b}$, then the misclassification rate converges to zero.

Remark 7. The ratio $\frac{(a-b)^2}{2(a+b)}$ is called the signal-to-noise ratio. And the detection threshold is $\text{SNR} = \frac{(a-b)^2}{2(a+b)} > 1$. When $a, b = O(1)$, the misclassification rate is a function of the SNR [CRV15] and becomes $o(1)$ when SNR goes to infinity.

Partial recovery with the spectral method A similar argument (combining matrix concentration and Davis-Kahan inequality) applies, with an additional regularization step (removing high degree vertices above $2d$) when $a, b = O(1)$. See [CRV15, LLV17].

Theorem 14 (Concentration of regularized adjacency matrices). *Consider a random graph from the inhomogeneous Erdős–Rényi model $G(n, (p_{ij}))$, and let $d = \max_{i,j} np_{ij}$. For all high degree vertices of the graph (say, those with degrees larger than $2d$), reduce the weights of the edges incident to them in an arbitrary way, but so that all degrees of the new (weighted) graph become bounded by $2d$. Then, with high probability, the adjacency matrix A' of the new graph satisfies*

$$\|A' - \mathbb{E}A\| = O(\sqrt{d}).$$

By applying the spectral method on A' instead of A . Theorem 13 holds as long as

$$(a-b)^2 \gg (a+b).$$

And partial recovery can be achieved when $\frac{(a-b)^2}{2(a+b)}$ is sufficiently large.

2.4 Exact recovery with the spectral method

From the discussion above, the ℓ_2 perturbation analysis through Davis-Kahan inequality can only give an estimate on $\min_{\theta \in \pm 1} \|v_2(A) - \theta \sigma\|$. If we want to find an algorithm that recovers σ exactly up to a sign flip, we would like a bound on $\min_{\theta \in \pm 1} \|v_2(A) - \theta \sigma\|_\infty$. This is achieved by a more refined eigenvector perturbation analysis in [AFWZ20].

A vanilla spectral method Since v_2 reveals σ by signs, a natural estimator is:

1. Compute v_2 , an unit eigenvector of A associated with the second-largest eigenvalue λ_2 .
2. Output $\hat{\sigma} = \text{sign}(v_2)$ entrywise.

Exact recovery is equivalent to: for some global sign $s \in \{\pm 1\}$,

$$\min_{i \in [n]} s \sigma_i (v_2)_i > 0.$$

A stronger (and convenient) target is a *margin*:

$$\sqrt{n} \min_i s \sigma_i (v_2)_i \geq \eta \quad \text{for some constant } \eta > 0.$$

The Davis–Kahan theorem controls an *angle* (or an ℓ_2 -type error):

$$\sin \angle(v_2, \sigma) \lesssim \frac{\|A - \mathbb{E}A\|}{\delta}.$$

Here $\delta \asymp \lambda_2 \asymp \log n$, while $\|A - \mathbb{E}A\|$ is typically $\mathcal{O}(\sqrt{\log n})$, so Davis–Kahan gives only

$$\sin \angle(v_2, \sigma) = \mathcal{O}\left(\frac{1}{\sqrt{\log n}}\right).$$

That does *not* imply all coordinates have the correct sign: a small ℓ_2 error can still allow a few coordinates to flip sign, and exact recovery is a *max/entrywise* event.

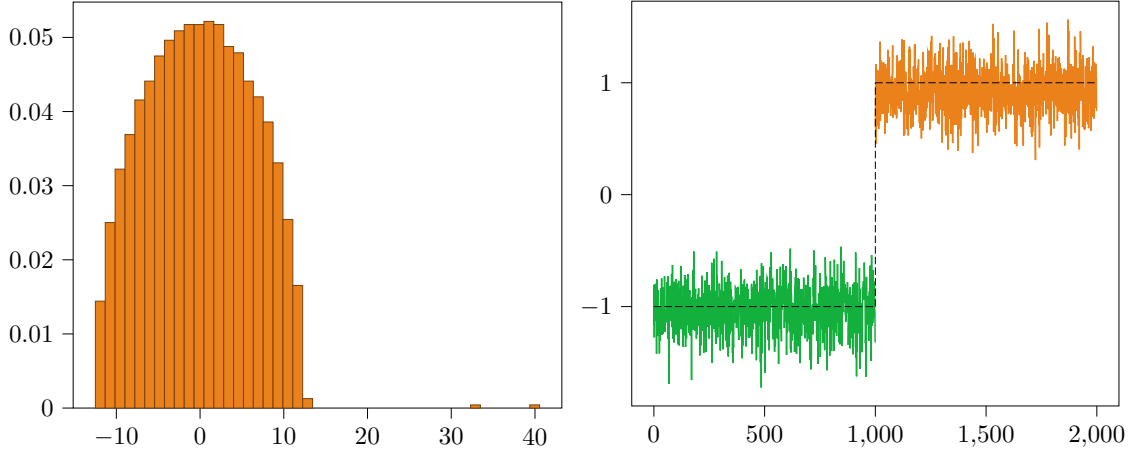


Figure 3: Eigenvalues and the second eigenvector of A above the exact recovery threshold

The seminal work [AFWZ20] proved the following.

Theorem 15 (Exact recovery by the vanilla spectral method). *Let $A \sim \text{SBM}\left(n, \frac{a \log n}{n}, \frac{b \log n}{n}\right)$ with $a > b > 0$ and two balanced communities with labels $\sigma \in \{\pm 1\}^n$. If $\sqrt{a} - \sqrt{b} > \sqrt{2}$, then the estimator*

$$\hat{\sigma}_i := \text{sign}((v_2)_i), \quad i \in [n],$$

where v_2 is a unit eigenvector of A associated with its second-largest eigenvalue, achieves exact recovery:

$$\mathbb{P}(\hat{\sigma} = \sigma \text{ or } \hat{\sigma} = -\sigma) = 1 - o(1).$$

For the rest of this section, we sketch their proof.

Key decomposition We need *entrywise* control (an ℓ_∞ statement), not just an ℓ_2 statement. Let

$$A^* := \mathbb{E}[A \mid \sigma], \quad v_2^* := \frac{\sigma}{\sqrt{n}}, \quad \lambda_2^* := \frac{a-b}{2} \log n,$$

so that $A^* v_2^* = \lambda_2^* v_2^*$. We analyze the *linearization*

$$\frac{Av_2^*}{\lambda_2^*} = v_2^* + \frac{(A - A^*)v_2^*}{\lambda_2^*},$$

which is linear in A . The key insight is that $\frac{Av_2^*}{\lambda_2^*}$ is a better approximation of v_2 .

Fix $s \in \{\pm 1\}$. Write

$$v_2 - sv_2^* = s \underbrace{\left(\frac{Av_2^*}{\lambda_2^*} - v_2^* \right)}_{\text{first-order / linear term}} + \underbrace{\left(v_2 - \frac{sAv_2^*}{\lambda_2^*} \right)}_{\text{higher-order remainder}}.$$

Higher-order term With leave-one-out analysis [CCFM21], we can prove the following:

Lemma 14 (Remainder is small in ℓ_∞). *If $A \sim \text{SBM}\left(n, \frac{a \log n}{n}, \frac{b \log n}{n}\right)$, then with probability $1 - \mathcal{O}(n^{-3})$,*

$$\min_{s \in \{\pm 1\}} \left\| v_2 - s \frac{Av_2^*}{\lambda_2^*} \right\|_\infty \leq \frac{C}{\sqrt{n} \log \log n},$$

where $C = C(a, b)$ depends only on a, b .

The signal level in v_2^* is $\min_i |(v_2^*)_i| = 1/\sqrt{n}$. The remainder above is smaller by a factor $\log \log n$. So *sign recovery* reduces to analyzing the signs of $\frac{Av_2^*}{\lambda_2^*}$.

Linear term The following lemma shows $\frac{Av_2^*}{\lambda_2^*}$ always has the correct sign, and each entry is of order $1/\sqrt{n}$.

Lemma 15. *With high probability, the following holds:*

$$\sqrt{n} \sigma_i \frac{(Av_2^*)_i}{\lambda_2^*} \geq \frac{2\varepsilon}{a-b} \text{ for all } i.$$

Recall $(v_2^*)_j = \sigma_j/\sqrt{n}$. Then for any vertex i ,

$$\sigma_i (Av_2^*)_i = \sum_{j=1}^n A_{ij} \frac{\sigma_j}{\sqrt{n}} = \frac{1}{\sqrt{n}} \left(\sum_{j: \sigma_j = \sigma_i} A_{ij} - \sum_{j: \sigma_j \neq \sigma_i} A_{ij} \right).$$

Define within- and cross-degrees

$$D_i^{\text{in}} := \sum_{j:\sigma_j=\sigma_i} A_{ij}, \quad D_i^{\text{out}} := \sum_{j:\sigma_j \neq \sigma_i} A_{ij}.$$

Then

$$\sigma_i(Av_2^*)_i = \frac{1}{\sqrt{n}}(D_i^{\text{in}} - D_i^{\text{out}}).$$

Dividing by $\lambda_2^* = \frac{a-b}{2} \log n$ and multiplying by $\sqrt{n}$ gives

$$\sqrt{n} \sigma_i \frac{(Av_2^*)_i}{\lambda_2^*} = \frac{2}{(a-b) \log n} (D_i^{\text{in}} - D_i^{\text{out}}). \quad (13)$$

A sharp tail bound (Chernoff exponent). The paper proves the following useful tail inequality for the difference of two binomials in this regime.

Lemma 16. *Let $W_1, \dots, W_{n/2}$ be i.i.d. Bernoulli($\frac{a \log n}{n}$) and $Z_1, \dots, Z_{n/2}$ be i.i.d. Bernoulli($\frac{b \log n}{n}$) independent of the W_i 's. Then for any $\varepsilon \in \mathbb{R}$,*

$$\mathbb{P} \left(\sum_{i=1}^{n/2} W_i - \sum_{i=1}^{n/2} Z_i \leq \varepsilon \log n \right) \leq n^{-(\sqrt{a}-\sqrt{b})^2/2 + \varepsilon \log(a/b)/2}.$$

This concentration inequality was also established in [ABH16, MNS16]. The exponent $(\sqrt{a} - \sqrt{b})^2/2$ is exactly what drives the exact recovery threshold: to union bound over n vertices we need the per-vertex failure probability to be $o(1/n)$, i.e. exponent > 1 .

Proof. Let $\lambda = -\log(a/b)/2 < 0$, and we apply Markov's inequality to the moment generating function,

$$\mathbb{P} \left(\sum_{i=1}^{n/2} W_i - \sum_{i=1}^{n/2} Z_i \leq \varepsilon \log n \right) = \mathbb{P} \left(e^{\lambda \sum_{i=1}^{n/2} (W_i - Z_i)} \geq e^{\lambda \varepsilon \log n} \right) \leq n^{-\lambda \varepsilon} \mathbb{E} e^{\lambda \sum_{i=1}^{n/2} (W_i - Z_i)}.$$

Observe that $e^\lambda = \sqrt{b/a}$, so

$$\log \mathbb{E} e^{\lambda W_i} = \log \left(e^\lambda \cdot \frac{a \log n}{n} + 1 - \frac{a \log n}{n} \right) \leq (\sqrt{ab} - a) \frac{\log n}{n},$$

where we used the fact that $\log(1+x) \leq x$ for any $x > -1$. Similarly we have

$$\log \mathbb{E} e^{-\lambda Z_i} \leq (\sqrt{ab} - b) \frac{\log n}{n}.$$

By independence, this leads to

$$\begin{aligned} \log \mathbb{P} \left(\sum_{i=1}^{n/2} W_i - \sum_{i=1}^{n/2} Z_i \leq \varepsilon \log n \right) &\leq \varepsilon \frac{\log(a/b)}{2} \log n + \frac{n}{2} \left(\log \mathbb{E} e^{\lambda W_i} + \log \mathbb{E} e^{-\lambda Z_i} \right) \\ &\leq \left[\varepsilon \frac{\log(a/b)}{2} - \frac{(\sqrt{a} - \sqrt{b})^2}{2} \right] \log n, \end{aligned}$$

which is exactly the desired inequality. $\square$

From tail bound to uniform positivity. Assume $\sqrt{a} - \sqrt{b} > \sqrt{2}$, i.e. $(\sqrt{a} - \sqrt{b})^2/2 > 1$. Choose a small constant $\varepsilon > 0$ such that

$$(\sqrt{a} - \sqrt{b})^2/2 - \varepsilon \log(a/b)/2 > 1.$$

Then for each fixed i , the lemma implies

$$\mathbb{P}\left(D_i^{\text{in}} - D_i^{\text{out}} \leq \varepsilon \log n\right) \leq o(1/n).$$

A union bound yields

$$\mathbb{P}\left(\exists i : D_i^{\text{in}} - D_i^{\text{out}} \leq \varepsilon \log n\right) = o(1).$$

Hence, with high probability, for all i ,

$$D_i^{\text{in}} - D_i^{\text{out}} \geq \varepsilon \log n.$$

Plugging into (13) gives a uniform lower bound:

$$\sqrt{n} \sigma_i \frac{(Av_2^*)_i}{\lambda_2^*} \geq \frac{2\varepsilon}{a-b} \quad \text{for all } i.$$

Transfer from the linearized vector to v_2 . By the entrywise approximation Lemma 14, with high probability we can pick $s \in \{\pm 1\}$ such that

$$\left\| v_2 - s \frac{Av_2^*}{\lambda_2^*} \right\|_\infty \leq \frac{C}{\sqrt{n} \log \log n}.$$

Multiply by $\sqrt{n}$ and combine the lower bounds:

$$\sqrt{n} \min_i s \sigma_i (v_2)_i \geq \sqrt{n} \min_i \sigma_i \frac{(Av_2^*)_i}{\lambda_2^*} - \frac{C}{\log \log n} \geq \frac{2\varepsilon}{a-b} - o(1).$$

For large n the RHS is positive, so all coordinates satisfy $s \sigma_i (v_2)_i > 0$. Therefore $\text{sign}(v_2) = s \sigma$ and exact recovery holds.

2.5 Leave-one-out analysis for ℓ_∞ -norm perturbation bound

Proof of Lemma 14 We need the following lemmas before the proof:

Lemma 17 (Row-wise weighted Bernstein with a logarithmic gain). *Let $X_1, \dots, X_n$ be independent Bernoulli random variables with $\mathbb{P}(X_j = 1) = p_j$ and $p_{\max} := \max_j p_j$. Let $w \in \mathbb{R}^n$ be deterministic and set $S := \sum_{j=1}^n w_j (X_j - \mathbb{E}X_j)$. Then for any $\alpha \geq 0$,*

$$\mathbb{P}\left(|S| \geq \frac{(2+\alpha)p_{\max}n}{1 \vee \log\left(\frac{\sqrt{n}\|w\|_\infty}{\|w\|_2}\right)} \|w\|_\infty\right) \leq 2e^{-\alpha n p_{\max}}.$$

Proof. By homogeneity, assume $\|w\|_\infty = 1$ (otherwise rescale). Fix $\lambda > 0$ and use Markov:

$$\mathbb{P}(S \geq t) \leq e^{-\lambda t} \mathbb{E} e^{\lambda S} = e^{-\lambda t} \prod_{j=1}^n \mathbb{E} \exp(\lambda w_j (X_j - \mathbb{E} X_j)).$$

For each j ,

$$\mathbb{E} e^{\lambda w_j (X_j - \mathbb{E} X_j)} = e^{-\lambda w_j p_j} ((1 - p_j) + p_j e^{\lambda w_j}),$$

so using $\log(1 + x) \leq x$ and $p_j \leq p_{\max}$,

$$\log \mathbb{E} e^{\lambda w_j (X_j - \mathbb{E} X_j)} \leq p_{\max} (e^{\lambda w_j} - 1 - \lambda w_j).$$

Since $|w_j| \leq 1$, the elementary inequality $e^x \leq 1 + x + \frac{e^\lambda}{2} x^2$ for $|x| \leq \lambda$ yields $e^{\lambda w_j} - 1 - \lambda w_j \leq \frac{e^\lambda}{2} \lambda^2 w_j^2$. Hence

$$\log \mathbb{E} e^{\lambda S} \leq \frac{p_{\max} e^\lambda}{2} \lambda^2 \|w\|_2^2.$$

Therefore

$$\log \mathbb{P}(S \geq t) \leq -\lambda t + \frac{p_{\max} e^\lambda}{2} \lambda^2 \|w\|_2^2.$$

Let $x := \frac{\sqrt{n}}{\|w\|_2} \geq 1$ and choose

$$\lambda := 1 \vee \log x.$$

Then $e^\lambda \leq ex$ and $(1 \vee \log x)^2 \leq x$ for $x \geq 1$, so

$$\frac{p_{\max} e^\lambda}{2} \lambda^2 \|w\|_2^2 \leq \frac{ep_{\max}}{2} x \lambda^2 \|w\|_2^2 = \frac{ep_{\max}}{2} \lambda^2 \sqrt{n} \|w\|_2 \leq \frac{ep_{\max}}{2} n.$$

Now set $t := \frac{(2+\alpha)p_{\max}n}{\lambda}$. Then

$$\log \mathbb{P}(S \geq t) \leq -(2 + \alpha)p_{\max}n + \frac{e}{2}p_{\max}n \leq -\alpha p_{\max}n.$$

Thus $\mathbb{P}(S \geq t) \leq e^{-\alpha p_{\max}n}$. The same bound holds for $\mathbb{P}(-S \geq t)$, hence $\mathbb{P}(|S| \geq t) \leq 2e^{-\alpha p_{\max}n}$. Undoing $\|w\|_\infty = 1$ gives the claim. $\square$

Lemma 18 (Residual-to-angle bound). *Let M be symmetric with eigenvalues $\nu_1 \geq \dots \geq \nu_n$ and orthonormal eigenvectors $u_1, \dots, u_n$. Fix $\theta \in \mathbb{R}$ and let v be a unit vector. Let $k \in [n]$ minimize $|\nu_k - \theta|$ (break ties arbitrarily) and define*

$$\text{gap}(\theta; M) := \min_{\ell \neq k} |\nu_\ell - \theta|.$$

If $\text{gap}(\theta; M) > 0$, then

$$\sin \angle(u_k, v) \leq \frac{\|(M - \theta I)v\|_2}{\text{gap}(\theta; M)}.$$

Since $\|(M - \theta I)v\|_2 \leq \|M - \theta I\|$, this is better than the Davis-Kahan inequality in Theorem 12.

Proof. Expand $v = \sum_{\ell=1}^n c_\ell u_\ell$ with $\sum_{\ell} c_\ell^2 = 1$. Then

$$(M - \theta I)v = \sum_{\ell=1}^n c_\ell (\nu_\ell - \theta) u_\ell, \quad \|(M - \theta I)v\|_2^2 = \sum_{\ell=1}^n c_\ell^2 (\nu_\ell - \theta)^2.$$

Let k be the index minimizing $|\nu_k - \theta|$. Then

$$\|(M - \theta I)v\|_2^2 \geq \sum_{\ell \neq k} c_\ell^2 \text{gap}(\theta; M)^2 = \text{gap}(\theta; M)^2 (1 - c_k^2).$$

Since $\sin \angle(u_k, v) = \sqrt{1 - c_k^2}$, the result follows. $\square$

We will condition on the following three good properties:

Lemma 19 (Spectral norm concentration). *Let $A \sim \text{SBM}(n, p, q, J)$ with $p = a \frac{\log n}{n}$ and $q = b \frac{\log n}{n}$, $a > b > 0$. Let $A^* = \mathbb{E}[A \mid \sigma]$. Then there exist constants $c_1, c_2 > 0$ depending only on (a, b) such that*

$$\mathbb{P}\left(\|A - A^*\|_2 \geq c_1 \sqrt{\log n}\right) \leq c_2 n^{-4}.$$

Lemma 20 (Maximum degree bound). *Under the same model, there exists $C = C_d(a, b)$ such that*

$$\mathbb{P}\left(d_{\max} := \max_{i \in [n]} \sum_j A_{ij} \geq C \log n\right) \leq n^{-4}.$$

Lemma 21 (Leave-one-out delocalization). *Let $A^{(m)}$ be obtained from A by setting the m -th row and column to zero, and let $v_2^{(m)}$ be a unit eigenvector of $A^{(m)}$ corresponding to its second-largest eigenvalue. Then there exists $C_\infty = C_\infty(a, b) > 0$ such that*

$$\mathbb{P}\left(\max_{m \in [n]} \|v_2^{(m)}\|_\infty \leq \frac{C_\infty}{\sqrt{n}}\right) \geq 1 - \mathcal{O}(n^{-3}).$$

The proof of Lemma 21 is deferred to the next subsection.

Now we are ready to begin the proof of Lemma 14.

Proof of Lemma 14. Let $s := \text{sign}(\langle v_2, v_2^* \rangle) \in \{\pm 1\}$ and set $u := s v_2$ so that $\langle u, v_2^* \rangle \geq 0$. It suffices to prove

$$\left\|u - \frac{A v_2^*}{\lambda_2^*}\right\|_\infty \leq \frac{C}{\sqrt{n} \log \log n}.$$

Step 1: define the high-probability event. Let $\mathcal{E}$ be the intersection of the following events:

$$(E1) \quad \|A - A^*\|_2 \leq c_1 \sqrt{\log n} \text{ (Lemma 19),}$$

$$(E2) \quad d_{\max} \leq C \log n \text{ (Lemma 20),}$$

$$(E3) \quad \max_{m \in [n]} \|v_2^{(m)}\|_\infty \leq C_\infty / \sqrt{n} \text{ (Lemma 21).}$$

By Lemmas 19–21, we have $\mathbb{P}(\mathcal{E}) \geq 1 - \mathcal{O}(n^{-3})$. On $\mathcal{E}$, Weyl's inequality implies $|\lambda_2 - \lambda_2^*| \leq \|A - A^*\|_2 \leq c_1 \sqrt{\log n}$. Since $\lambda_2^* = \Theta(\log n)$ for fixed (a, b) , for all large n there exists $c_0 > 0$ such that

$$|\lambda_2| \geq c_0 \log n, \quad |\lambda_2^*| \geq c_0 \log n. \quad (14)$$

Step 2: leave-one-out eigenvectors. For each $m \in [n]$, let $A^{(m)}$ be obtained from A by setting the m -th row and column to zero. Let $v_2^{(m)}$ be a unit eigenvector of $A^{(m)}$ for its second-largest eigenvalue $\lambda_2^{(m)}$, with sign fixed so that $\langle v_2^{(m)}, v_2^* \rangle \geq 0$. Note that $v_2^{(m)}$ is independent of the m -th row/column of A .

Step 3: a coordinate decomposition. Fix $m \in [n]$. From the eigenvector equation $Au = \lambda_2 u$,

$$\lambda_2 u_m = (Au)_m = \sum_j A_{mj} u_j.$$

Insert $v_2^{(m)}$:

$$\sum_j A_{mj} u_j = \sum_j A_{mj} (v_2^{(m)})_j + \sum_j A_{mj} (u_j - (v_2^{(m)})_j).$$

Therefore

$$u_m - \left(\frac{A v_2^*}{\lambda_2^*} \right)_m = T_{m,1} + T_{m,2} + T_{m,3}, \quad (15)$$

where

$$\begin{aligned} T_{m,1} &:= \frac{1}{\lambda_2} \sum_j A_{mj} ((v_2^{(m)})_j - (v_2^*)_j), \\ T_{m,2} &:= \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_2^*} \right) \sum_j A_{mj} (v_2^*)_j, \\ T_{m,3} &:= \frac{1}{\lambda_2} \sum_j A_{mj} (u_j - (v_2^{(m)})_j). \end{aligned}$$

We bound $\max_m |T_{m,k}|$ for $k = 1, 2, 3$ on $\mathcal{E}$.

Step 4: bound $T_{m,2}$. On $\mathcal{E}$, $|\lambda_2 - \lambda_2^*| \leq c_1 \sqrt{\log n}$ and (14) holds, so

$$\left| \frac{1}{\lambda_2} - \frac{1}{\lambda_2^*} \right| = \frac{|\lambda_2 - \lambda_2^*|}{|\lambda_2 \lambda_2^*|} \leq \frac{C}{(\log n)^{3/2}}.$$

Also $\left| \sum_j A_{mj} (v_2^*)_j \right| \leq d_m \|v_2^*\|_\infty \leq \frac{C \log n}{\sqrt{n}}$ on $\mathcal{E}$. Hence

$$|T_{m,2}| \leq \frac{C}{(\log n)^{3/2}} \cdot \frac{\log n}{\sqrt{n}} = \frac{C}{\sqrt{n} \sqrt{\log n}}.$$

Step 5: bound $T_{m,3}$. We first show $\|u - v_2^{(m)}\|_2 \leq C/\sqrt{n}$ on $\mathcal{E}$. Consider $v_2^{(m)}$ as an approximate eigenvector of A :

$$(A - \lambda_2^{(m)} I) v_2^{(m)} = (A - A^{(m)}) v_2^{(m)}.$$

Since $A^{(m)}$ has the m -th row/column zero, for any nonzero eigenvalue we have $(v_2^{(m)})_m = 0$, and therefore

$$(A - A^{(m)})v_2^{(m)} = e_m \sum_j A_{mj}(v_2^{(m)})_j.$$

Thus, on $\mathcal{E}$,

$$\|(A - \lambda_2^{(m)}I)v_2^{(m)}\|_2 = \left| \sum_j A_{mj}(v_2^{(m)})_j \right| \leq d_m \|v_2^{(m)}\|_\infty \leq \frac{C \log n}{\sqrt{n}}.$$

Moreover, $\lambda_2^{(m)}$ is within $O(\sqrt{\log n})$ of $\lambda_2^* = \Theta(\log n)$ (since $\|A^{(m)} - A^*\|_2 \leq \|A - A^*\|_2 + \|A - A^{(m)}\|_2 \leq C\sqrt{\log n}$ on $\mathcal{E}$), and λ_2^* is $\Theta(\log n)$, so for all large n the gap $\text{gap}(\lambda_2^{(m)}; A)$ is at least $c_0 \log n$. Applying Lemma 18 with $M = A$, $\theta = \lambda_2^{(m)}$, and $v = v_2^{(m)}$ gives

$$\sin \angle(u, v_2^{(m)}) \leq \frac{C \log n / \sqrt{n}}{c_0 \log n} \leq \frac{C}{\sqrt{n}},$$

hence $\|u - v_2^{(m)}\|_2 \leq \sqrt{2} \sin \angle(u, v_2^{(m)}) \leq C/\sqrt{n}$. Now bound $T_{m,3}$ by Cauchy–Schwarz:

$$|T_{m,3}| \leq \frac{1}{|\lambda_2|} \|A_{m,\cdot}\|_2 \|u - v_2^{(m)}\|_2 \leq \frac{1}{c_0 \log n} \cdot \sqrt{d_m} \cdot \frac{C}{\sqrt{n}} \leq \frac{C}{\sqrt{n} \sqrt{\log n}}.$$

Step 6: bound $T_{m,1}$. Fix m and define $w^{(m)} := v_2^{(m)} - v_2^*$. Then

$$\lambda_2 T_{m,3} = \sum_j A_{mj} w_j^{(m)} = \sum_j (A_{mj} - A_{mj}^*) w_j^{(m)} + \sum_j A_{mj}^* w_j^{(m)}.$$

We bound the two pieces.

(i) *Mean part.* Since A^* has rank 2 with eigenvalues $\Theta(\log n)$ and eigenvectors with entries $\Theta(1/\sqrt{n})$, for any vector x one has the deterministic bound

$$|(A^* x)_m| \leq \frac{C \log n}{\sqrt{n}} \|x\|_2.$$

Apply this with $x = w^{(m)}$:

$$\left| \sum_j A_{mj}^* w_j^{(m)} \right| = |(A^* w^{(m)})_m| \leq \frac{C \log n}{\sqrt{n}} \|w^{(m)}\|_2.$$

By Davis–Kahan between $A^{(m)}$ and A^* (using $\|A^{(m)} - A^*\|_2 \leq C\sqrt{\log n}$ on $\mathcal{E}$ and eigengap $\Theta(\log n)$), we have

$$\|w^{(m)}\|_2 = \|v_2^{(m)} - v_2^*\|_2 \leq C/\sqrt{\log n}.$$

Hence

$$\left| \sum_j A_{mj}^* w_j^{(m)} \right| \leq \frac{C \sqrt{\log n}}{\sqrt{n}}.$$

Dividing by $|\lambda_2| \geq c_0 \log n$ yields a contribution $\leq C/(\sqrt{n} \sqrt{\log n})$.

(ii) *Fluctuation part.* Condition on $A^{(m)}$. Then $w^{(m)}$ is deterministic and independent of the random row $\{A_{mj}\}_j$. Also all these Bernoulli variables have parameters $\leq p = a \frac{\log n}{n}$, so we may take $p_{\max} = p$ in Lemma 17. On $\mathcal{E}$, by Lemma 21 we have

$$\|w^{(m)}\|_{\infty} \leq \|v_2^{(m)}\|_{\infty} + \|v_2^{\star}\|_{\infty} \leq \frac{C}{\sqrt{n}}.$$

Also $\|w^{(m)}\|_2 \leq C/\sqrt{\log n}$ as above, so

$$\frac{\sqrt{n}\|w^{(m)}\|_{\infty}}{\|w^{(m)}\|_2} \geq c\sqrt{\log n} \implies 1 \vee \log\left(\frac{\sqrt{n}\|w^{(m)}\|_{\infty}}{\|w^{(m)}\|_2}\right) \geq c \log \log n.$$

Apply Lemma 17 with $\alpha = 5/a$, so that $2e^{-\alpha np} = 2e^{-5 \log n} = 2n^{-5}$. We obtain, conditional on $A^{(m)}$,

$$\left| \sum_j (A_{mj} - A_{mj}^{\star}) w_j^{(m)} \right| \leq \frac{C(pn)}{\log \log n} \|w^{(m)}\|_{\infty} \leq \frac{C \log n}{\log \log n} \cdot \frac{1}{\sqrt{n}}$$

with probability at least $1 - 2n^{-5}$. A union bound over $m \in [n]$ gives that the above holds simultaneously for all m with probability at least $1 - \mathcal{O}(n^{-4})$. Dividing by $|\lambda_2| \geq c_0 \log n$ yields

$$\max_{m \in [n]} \frac{1}{|\lambda_2|} \left| \sum_j (A_{mj} - A_{mj}^{\star}) w_j^{(m)} \right| \leq \frac{C}{\sqrt{n} \log \log n}.$$

Combining (i) and (ii), we conclude that on an event of probability at least $1 - \mathcal{O}(n^{-3})$,

$$\max_{m \in [n]} |T_{m,1}| \leq \frac{C}{\sqrt{n} \log \log n}.$$

Step 7: finish. Combining (15) with the bounds for $T_{m,1}$, $T_{m,2}$, and $T_{m,3}$ yields

$$\max_{m \in [n]} \left| u_m - \left(\frac{Av_2^{\star}}{\lambda_2^{\star}} \right)_m \right| \leq \frac{C}{\sqrt{n} \log \log n}$$

with probability at least $1 - \mathcal{O}(n^{-3})$. This is exactly

$$\min_{s \in \{\pm 1\}} \left\| v_2 - s \frac{Av_2^{\star}}{\lambda_2^{\star}} \right\|_{\infty} \leq \frac{C}{\sqrt{n} \log \log n}.$$

□

Proof of Lemma 21 Finally, it remains to prove the leave-one-out delocalization statement.

Proof of Lemma 21. Define the event

$$\mathcal{E} := \left\{ \|A - A^{\star}\|_2 \leq c_1 \sqrt{\log n} \right\} \cap \left\{ d_{\max} \leq C \log n \right\},$$

where $d_{\max} = \max_i \sum_j A_{ij}$. Then $\mathbb{P}(\mathcal{E}) \geq 1 - \mathcal{O}(n^{-3})$.

On $\mathcal{E}$, for every m we have

$$\|A^{(m)} - A^*\|_2 \leq \|A - A^*\|_2 + \|A - A^{(m)}\|_2 \leq c_1 \sqrt{\log n} + 2\sqrt{d_{\max}} \leq C\sqrt{\log n}.$$

Since A^* has eigenvalues $\lambda_1^* = \frac{a+b}{2} \log n$, $\lambda_2^* = \frac{a-b}{2} \log n$, and $\lambda_3^* = \dots = \lambda_n^* = 0$, Weyl's inequality yields for all m :

$$\begin{aligned} |\lambda_2^{(m)} - \lambda_2^*| &\leq \|A^{(m)} - A^*\|_2 \leq C\sqrt{\log n}, \\ \max\{|\lambda_3^{(m)}|, |\lambda_n^{(m)}|\} &\leq \|A^{(m)} - A^*\|_2 \leq C\sqrt{\log n}. \end{aligned}$$

There is a constant $c_0 = c_0(a, b) > 0$ such that

$$|\lambda_2^{(m)}| \geq c_0 \log n, \quad \text{gap}_m := \min\{\lambda_1^{(m)} - \lambda_2^{(m)}, \lambda_2^{(m)} - \lambda_3^{(m)}\} \geq c_0 \log n, \quad (16)$$

uniformly over m .

Step 2: leave-one-out (remove one more index). Fix $m \in [n]$ and an index $i \neq m$. Let $A^{(m,i)}$ be the matrix obtained by zeroing out both rows/columns m and i . Let $u_2^{(m,i)}$ be a unit eigenvector of $A^{(m,i)}$ for its second-largest eigenvalue $\lambda_2^{(m,i)}$. (Notice $(u_2^{(m,i)})_i = (u_2^{(m,i)})_m = 0$ automatically if $\lambda_2^{(m,i)} \neq 0$.)

Consider the vector $u_2^{(m)}$ but viewed as an *approximate* eigenvector of $A^{(m,i)}$. We compute the residual:

$$(A^{(m,i)} - \lambda_2^{(m)} I)u_2^{(m)} = (A^{(m,i)} - A^{(m)})u_2^{(m)}.$$

The difference $A^{(m)} - A^{(m,i)}$ only contains the i -th row/column (with the m -th already zero), so one checks

$$(A^{(m)} - A^{(m,i)})u_2^{(m)} = e_i (A^{(m)} u_2^{(m)})_i + (A^{(m)} e_i) (u_2^{(m)})_i = e_i \lambda_2^{(m)} (u_2^{(m)})_i + (A^{(m)} e_i) (u_2^{(m)})_i.$$

Hence

$$\|(A^{(m,i)} - \lambda_2^{(m)} I)u_2^{(m)}\|_2 \leq |\lambda_2^{(m)}| \cdot |(u_2^{(m)})_i| + \|A^{(m)} e_i\|_2 |(u_2^{(m)})_i|.$$

On $\mathcal{E}$, $\|A^{(m)} e_i\|_2 = \|A_{i,\cdot}^{(m)}\|_2 \leq \sqrt{d_i} \leq \sqrt{d_{\max}} \leq C\sqrt{\log n}$, and $|\lambda_2^{(m)}| \asymp \log n$, so

$$\|(A^{(m,i)} - \lambda_2^{(m)} I)u_2^{(m)}\|_2 \leq C \log n |(u_2^{(m)})_i|.$$

By (16), the spectral gap around $\lambda_2^{(m)}$ for $A^{(m,i)}$ is at least $c_0 \log n$ for all large n (Weyl again gives $|\lambda_2^{(m,i)} - \lambda_2^*| \leq C\sqrt{\log n}$ and $|\lambda_3^{(m,i)}| \leq C\sqrt{\log n}$, so the gap remains $\Theta(\log n)$). Therefore the residual-to-angle Lemma 18 implies there exists a sign $s_{m,i} \in \{\pm 1\}$ such that

$$\|u_2^{(m)} - s_{m,i} u_2^{(m,i)}\|_2 \leq C |(u_2^{(m)})_i|. \quad (17)$$

Step 3: bootstrap inequality for the ℓ_∞ norm. Define

$$B := \sqrt{n} \max_{m \in [n]} \|u_2^{(m)}\|_\infty.$$

Fix m and let $i \neq m$ be such that $|(u_2^{(m)})_i| = \|u_2^{(m)}\|_\infty = B/\sqrt{n}$. Since $u_2^{(m)}$ is an eigenvector of $A^{(m)}$, for $i \neq m$ we have

$$\lambda_2^{(m)}(u_2^{(m)})_i = \sum_j A_{ij}(u_2^{(m)})_j.$$

Insert the leave-one-out vector $u_2^{(m,i)}$:

$$\sum_j A_{ij}(u_2^{(m)})_j = \sum_j A_{ij} s_{m,i}(u_2^{(m,i)})_j + \sum_j A_{ij} \left((u_2^{(m)})_j - s_{m,i}(u_2^{(m,i)})_j \right).$$

Therefore combining the two equations above,

$$|(u_2^{(m)})_i| \leq \frac{1}{|\lambda_2^{(m)}|} \left| \sum_j A_{ij} s_{m,i}(u_2^{(m,i)})_j \right| + \frac{1}{|\lambda_2^{(m)}|} \left| \sum_j A_{ij} \left((u_2^{(m)})_j - s_{m,i}(u_2^{(m,i)})_j \right) \right|. \quad (18)$$

Step 3a: bound the second error term in (18). By Cauchy–Schwarz, $\|A_{i,\cdot}\|_2 \leq \sqrt{d_i} \leq C\sqrt{\log n}$ on $\mathcal{E}$, and by (17),

$$\left| \sum_j A_{ij} \left((u_2^{(m)})_j - s_{m,i}(u_2^{(m,i)})_j \right) \right| \leq \|A_{i,\cdot}\|_2 \|u_2^{(m)} - s_{m,i}u_2^{(m,i)}\|_2 \leq C\sqrt{\log n} \cdot |(u_2^{(m)})_i|.$$

Dividing by $|\lambda_2^{(m)}| \geq c_0 \log n$ (by (16)) yields

$$\frac{1}{|\lambda_2^{(m)}|} \left| \sum_j A_{ij} \left((u_2^{(m)})_j - s_{m,i}(u_2^{(m,i)})_j \right) \right| \leq \frac{C}{\sqrt{\log n}} |(u_2^{(m)})_i|. \quad (19)$$

Step 3b: bound the first term. Write

$$\sum_j A_{ij} s_{m,i}(u_2^{(m,i)})_j = (A^* s_{m,i} u_2^{(m,i)})_i + \sum_j (A_{ij} - A_{ij}^*) s_{m,i}(u_2^{(m,i)})_j.$$

Since A^* has rank 2 with eigenvalues $\lambda_1^*, \lambda_2^* = \Theta(\log n)$ and eigenvectors having entries of magnitude $1/\sqrt{n}$, we have the deterministic bound

$$|(A^* s_{m,i} u_2^{(m,i)})_i| \leq \frac{|\lambda_1^*| + |\lambda_2^*|}{\sqrt{n}} \leq C \frac{\log n}{\sqrt{n}}. \quad (20)$$

For the fluctuation term, note that $u_2^{(m,i)}$ is independent of the i -th row/column of A (those entries are removed in $A^{(m,i)}$). Also, by (17),

$$\|u_2^{(m,i)}\|_\infty \leq \|u_2^{(m)}\|_\infty + \|u_2^{(m)} - s_{m,i}u_2^{(m,i)}\|_2 \leq \frac{B}{\sqrt{n}} + |(u_2^{(m)})_i| \leq C \frac{B}{\sqrt{n}}.$$

Applying Lemma 17 conditionally on $A^{(m,i)}$ (because $A^{(m,i)}$ is independent of $\{A_{ij}, j \in [n]\}$), we obtain

$$\mathbb{P} \left(\left| \sum_j (A_{ij} - A_{ij}^*) s_{m,i}(u_2^{(m,i)})_j \right| \leq \frac{C \log n}{1 \vee \log B} \cdot \frac{B}{\sqrt{n}} \right) \geq 1 - 2n^{-5}.$$

A union bound over all (m, i) pairs (at most n^2) shows that with probability at least $1 - \mathcal{O}(n^{-3})$, simultaneously for all m and all $i \neq m$,

$$\left| \sum_j (A_{ij} - A_{ij}^*) s_{m,i}(u_2^{(m,i)})_j \right| \leq \frac{C \log n}{1 \vee \log B} \cdot \frac{B}{\sqrt{n}}. \quad (21)$$

Combining (20) and (21), we have

$$\left| \sum_j A_{ij} s_{m,i}(u_2^{(m,i)})_j \right| \leq C \frac{\log n}{\sqrt{n}} + \frac{C \log n}{1 \vee \log B} \cdot \frac{B}{\sqrt{n}}. \quad (22)$$

Step 4: close the bootstrap and solve for B . Plug (19) and (22) into (18), and use $|\lambda_2^{(m)}| \geq c_0 \log n$:

$$|(u_2^{(m)})_i| \leq \frac{C}{\sqrt{n}} + \frac{C}{1 \vee \log B} \cdot \frac{B}{\sqrt{n}} + \frac{C}{\sqrt{\log n}} |(u_2^{(m)})_i|.$$

For n large enough, $C/\sqrt{\log n} \leq 1/4$, so we can move the last term to the left:

$$\frac{3}{4} |(u_2^{(m)})_i| \leq \frac{C}{\sqrt{n}} + \frac{C}{1 \vee \log B} \cdot \frac{B}{\sqrt{n}}.$$

Multiply by $\sqrt{n}$ and recall $|(u_2^{(m)})_i| = B/\sqrt{n}$:

$$\frac{3}{4} B \leq C + \frac{C}{1 \vee \log B} B.$$

If $B \leq e$, then trivially $B \leq e$. If $B > e$, then $\log B \geq 1$ and the inequality becomes

$$\frac{3}{4} B \leq C + \frac{C}{\log B} B.$$

Choose B_0 so large that $C/\log B_0 \leq 1/4$. Then whenever $B \geq B_0$,

$$\frac{1}{2} B \leq C \quad \Rightarrow \quad B \leq 2C,$$

a contradiction if $B_0 > 2C$. Hence on the event where all the bounds above hold, we must have $B \leq \max\{e, B_0, 2C\} =: C_\infty(a, b)$. Therefore,

$$\max_{m \in [n]} \|u_2^{(m)}\|_\infty = \frac{B}{\sqrt{n}} \leq \frac{C_\infty}{\sqrt{n}}.$$

Combining with the probability bounds gives the claim. $\square$

Remark 8 (Beyond the 2-block case). The exact recovery threshold of k -block models is given in [AS15]. Exact recovery for more than 2 blocks with the vanilla spectral method is considered in [Zha24]. Spectral methods for partial recovery for k -blocks were considered in [CRV15].

Remark 9 ($\ell_{2,\infty}$ -perturbation bound). Eigenvector perturbation analysis under random noise can be much better than the deterministic bound from Davis-Kahan inequality. This phenomenon has been explored in many contexts using the leave-one-out analysis, including spectral methods for low-rank matrix recovery [DLS21, CLC⁺21], convex optimization [DC20], gradient descent analysis [MWCC18, CLPC22], and high-dimensional statistics [HL22b].

2.6 Detection threshold

Definition 8 (Symmetric sparse SBM with k groups). Fix integers n (vertices) and $k \geq 2$ (communities). Let labels $\sigma_i \in [k] := \{1, \dots, k\}$ be i.i.d. uniform. Conditional on $\sigma = (\sigma_1, \dots, \sigma_n)$, edges are independent and for $i \neq j$,

$$\mathbb{P}\{(i, j) \in E \mid \sigma_i = \sigma_j\} = \frac{a}{n}, \quad \mathbb{P}\{(i, j) \in E \mid \sigma_i \neq \sigma_j\} = \frac{b}{n},$$

where $a, b = \Theta(1)$ as $n \rightarrow \infty$ (constant expected degree).

A uniformly random vertex has expected degree

$$d = \frac{a + (k-1)b}{k}.$$

The *sparse* regime means $d = O(1)$ as $n \rightarrow \infty$. If $a > b$ the model is *assortative* (within-community edges are more likely); if $a < b$ it is *disassortative*.

Conditioned on a vertex label, a random neighbor's label follows

$$P_{ii} = \frac{a}{a + (k-1)b}, \quad P_{ij} = \frac{b}{a + (k-1)b} \quad (i \neq j).$$

This matrix has eigenvalues

$$\lambda_1 = 1, \quad \lambda_2 = \dots = \lambda_k = \frac{a - b}{a + (k-1)b},$$

where λ_2 has multiplicity $k - 1$.

Kesten–Stigum threshold The Kesten–Stigum condition is

$$d\lambda_2^2 > 1.$$

Substituting the formulas for d and λ_2 gives

$$\frac{(a - b)^2}{k(a + (k-1)b)} > 1$$

Definition 9 (Weak recovery). Given an estimator $\hat{\sigma}(G) \in [k]^n$, define the permutation-invariant overlap

$$\text{Overlap}(\hat{\sigma}, \sigma) := \max_{\pi \in S_k} \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{\hat{\sigma}_i = \pi(\sigma_i)\}.$$

We say *weak recovery* is possible if there exists $\varepsilon > 0$ and $\hat{\sigma}$ such that $\text{Overlap}(\hat{\sigma}, \sigma) \geq 1/k + \varepsilon$ with probability $\rightarrow 1$ as $n \rightarrow \infty$.

Above the Kesten–Stigum (KS) threshold, there are polynomial-time algorithms, notably spectral methods based on the non-backtracking operator and belief propagation that achieve weak recovery [BLM18, MNS15a]. Below the KS threshold, the picture depends on the number of communities. For $k = 2$, weak recovery is information-theoretically impossible: no algorithm can achieve

non-trivial correlation with the ground truth, even with exponential running time [MNS15b]. For larger k , however, KS can be only a sufficient condition: in certain regimes, weak recovery becomes information-theoretically possible strictly below KS, yet no known polynomial-time algorithm succeeds there. In particular, for sufficiently large average degree $d \geq C_0$, there are parameter ranges where recovery remains impossible for $k = 3, 4$, while for $k \geq 5$ exponential-time procedures can achieve weak recovery below KS [BMNN16, AS18]. This separation between the information-theoretic and algorithmic thresholds is often referred to as the *statistical-to-computational gap*.

Connection to broadcasting on trees A central insight in [MNS15a] is that an SBM is locally tree-like: the depth- t neighborhood of a uniform vertex converges to a Galton–Watson tree with offspring distribution $\text{Poi}(d)$. Conditional on the root label, child labels are generated by a k -state symmetric channel whose second eigenvalue is $\frac{a-b}{a+(k-1)b}$. Thus, the feasibility of recovery in an SBM is tied to *reconstruction* in this broadcast process on trees. In [EKPS00] the broadcasting problem with $k = 2$ was analyzed with a phase transition at $(a-b)^2 = 2(a+b)$. [MNS15a] turns this result into an impossibility result for an SBM below the KS threshold.

Locally-tree like structure In the sparse regime $a, b = \Theta(1)$, the stochastic block model is *locally tree-like*: for any fixed radius r , the depth- r neighborhood of a uniformly random vertex is a tree with probability $1 - o(1)$, since the breadth-first exploration reveals only $\tilde{O}(d^r)$ vertices (here $\tilde{O}$ means up to a $\log(n)$ factor), and the chance of encountering a cycle is $\tilde{O}(d^{2r}/n)$ (here $d = \frac{a+(k-1)b}{k}$ is the average degree).

More precisely, the rooted random graphs (G, v) converge in the Benjamini–Schramm (local weak) sense [Sal11] to a Galton–Watson tree with offspring distribution $\text{Poi}(d)$ (or to the corresponding multi-type Galton–Watson tree when labels are included). Conditioned on the root label X_ρ , each revealed neighbor label is generated by the k -state Markov channel with transition matrix $P_{rr} = a/(a + (k-1)b)$ and $P_{rs} = b/(a + (k-1)b)$ for $s \neq r$; hence, the label process in the neighborhood is asymptotically a broadcast process on this limiting tree. This coupling between SBM neighborhoods and broadcasting on trees underlies the appearance of the Kesten–Stigum criterion in community detection; see [MNS15a, Abb18].

Broadcast process on trees Let $k \geq 2$ and let T be an infinite rooted tree with root ρ . (One common choice is a Galton–Watson tree with offspring distribution $\text{Poi}(d)$.) We consider a label process $\{X_v \in [k]\}_{v \in V(T)}$ generated as follows.

The root label is drawn from a prior π on $[k]$, typically uniform:

$$\Pr(X_\rho = r) = \pi_r, \quad r \in [k], \quad \text{with } \pi_r = \frac{1}{k} \text{ in the symmetric case.}$$

Fix a row-stochastic matrix $P \in \mathbb{R}^{k \times k}$ (a Markov channel). Conditioned on $X_v = r$, each child u of v receives a label via

$$\Pr(X_u = s \mid X_v = r) = P_{rs}, \quad s \in [k],$$

independently across different children/edges given the parent labels. A standard example is the *symmetric k -ary channel* with noise $\eta \in [0, 1]$:

$$P_{rr} = 1 - \eta, \quad P_{rs} = \frac{\eta}{k-1} \quad (s \neq r).$$

Let $\theta := \lambda_2(P)$ denote the second-largest eigenvalue of P in magnitude (the “correlation parameter”). Let ∂T_t be the boundary at depth t . The observation is the boundary labeling

$$X_{\partial T_t} := \{X_v : v \in \partial T_t\}.$$

The *broadcasting on trees* (or *tree reconstruction*) problem asks whether information about X_ρ persists in $X_{\partial T_t}$ as $t \rightarrow \infty$. We say that *reconstruction is possible* if the Bayes-optimal estimator $\hat{X}_\rho^* = \hat{X}_\rho^*(X_{\partial T_t})$ achieves non-vanishing advantage:

$$\limsup_{t \rightarrow \infty} \left(\Pr(\hat{X}_\rho^* = X_\rho) - \frac{1}{k} \right) > 0.$$

If the advantage tends to 0 we say *non-reconstruction* holds.

If T is Galton–Watson with mean offspring d and $\theta = \lambda_2(P)$, the Kesten–Stigum condition is

$$d\theta^2 > 1.$$

Above this threshold reconstruction holds; for $k = 2$ it is also sharp [EKPS00], while for larger k it depends on the channel [Mos01, Sly09].

2.7 The non-backtracking operator

We now turn to the discussion of a spectral method that works down to the KS threshold. The key idea is to replace the adjacency matrix A by an operator that counts *non-backtracking* walks, thereby suppressing the high-variance contribution of immediate backtracks that dominates the spectrum of A in sparse graphs.

In the sparse regime, degree fluctuation implies poor concentration of A around $\mathbb{E}A$. These fluctuations create localized eigenvectors and large “noise” eigenvalues of A that can mask the community signal near the KS threshold. In contrast, by forbidding immediate backtracking, one obtains an operator with a cleaner bulk and a well-separated signal eigenvalue above the KS threshold.

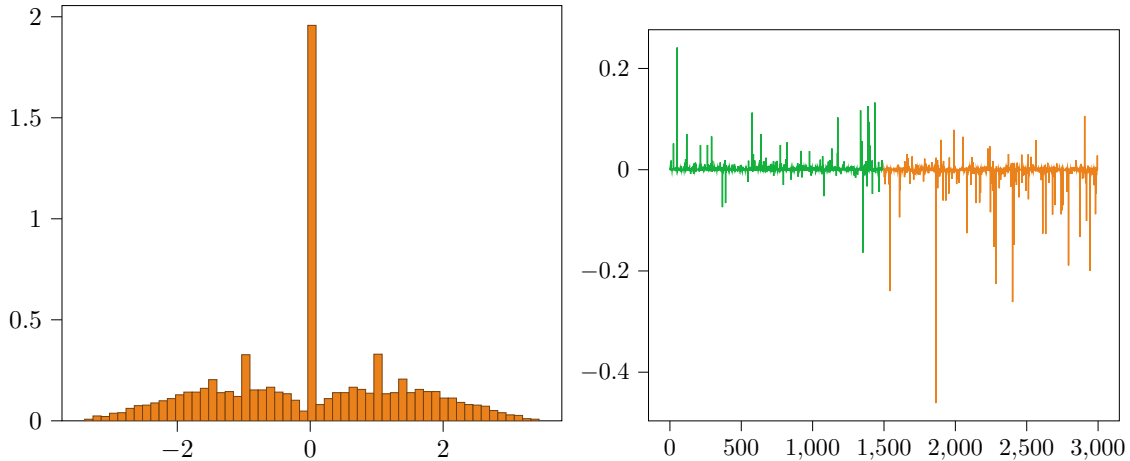


Figure 4: Eigenvalues and eigenvectors of A when $a, b = O(1)$.

Definition 10 (Non-backtracking matrix). Let $G = (V, E)$ be an undirected graph with $m = |E|$. Let $\text{vec}E$ be the set of directed edges:

$$\text{vec}E := \{(u \rightarrow v) : \{u, v\} \in E\}, \quad |\text{vec}E| = 2m.$$

The *non-backtracking matrix* B is the $(2m) \times (2m)$ matrix indexed by $\text{vec}E$ with

$$B_{(u \rightarrow v), (x \rightarrow y)} = \mathbf{1}\{v = x, y \neq u\}.$$

Thus, $(B^\ell)_{e,f}$ counts the number of length- ℓ directed walks from e to f that never immediately backtracks.

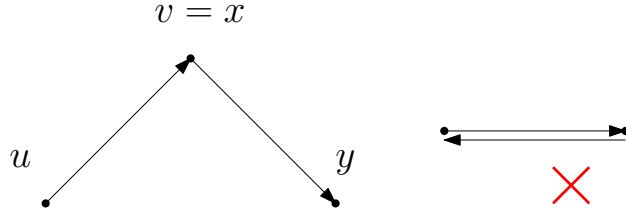


Figure 5: one step of a non-backtracking walk

An eigenvector $\psi \in \mathbb{R}^{\text{vec}E}$ of B lives on directed edges. To turn it into an n -dimensional vector, a common choice is

$$\varphi(v) := \sum_{u: \{u,v\} \in E} \psi_{(u \rightarrow v)}, \quad v \in V.$$

This allows us to construct an estimator for σ in the SBM when $k = 2$. For general k , one can use the k leading eigenvectors of B to form an embedding $v \mapsto (\varphi_2(v), \dots, \varphi_{k-1}(v)) \in \mathbb{R}^k$ and run k -means.

Ihara-Bass formula A practical issue is that B is $(2m) \times (2m)$, which is large. The *Ihara-Bass formula* reduces computations to simpler matrices:

Lemma 22 (Ihara-Bass formula). *For any graph $G = (V, E)$, and any $z \in \mathbb{C}$, the following identity holds:*

$$\det(B - zI) = (z^2 - 1)^{|E|-n} \det(z^2I - zA + D - I). \quad (23)$$

Consider the block matrix

$$\tilde{B} = \begin{bmatrix} 0 & D - I \\ -I & A \end{bmatrix} \in \mathbb{R}^{2n \times 2n}.$$

Then from (23), we have

$$\det(B - zI) = (z^2 - 1)^{|E|-n} \det(\tilde{B} - zI). \quad (24)$$

The identity (24) implies that B and $\tilde{B}$ share the same spectrum, up to the multiplicity of trivial eigenvalues ± 1 . We also call $\tilde{B}$ the *reduced non-backtracking matrix* of G .

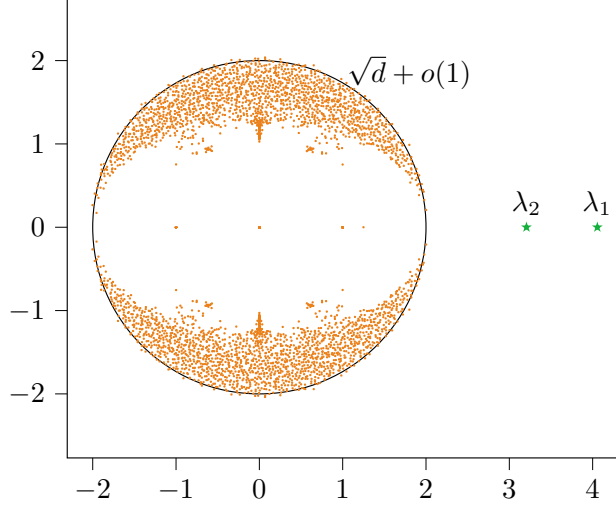


Figure 6: Spectrum of B for an SBM

Application in the SBM Above the KS threshold, there are k outliers in the spectrum of B outside a circle of radius $\sqrt{d}$, and eigenvectors associated with the outliers are informative [BLM18].

Theorem 16 (Spectrum of B). *If $(a - b)^2 > 2(a + b)$, then with high probability,*

$$\lambda_1(B) = \frac{a+b}{2} + o(1), \quad \lambda_2(B) = \frac{a-b}{2} + o(1), \quad |\lambda_3(B)| \leq \sqrt{\frac{a+b}{2}} + o(1),$$

Let $v_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix} \in \mathbb{C}^{2n}$ be an eigenvector of $\tilde{B}$ with associated eigenvalue $\lambda_i := \lambda_i(\tilde{B})$, $\|y_i\|_2 = 1$.

By solving $\tilde{B}v_i = \lambda_i v_i$, we obtain the following relation:

$$\begin{aligned} (D - I)y_i &= \lambda_i x_i, \\ -x_i + Ay_i &= \lambda_i y_i. \end{aligned}$$

Due to the degree fluctuation, x_i is noisy, but we can use y_i for spectral clustering. This gives a practical way for spectral clustering:

1. compute the spectrum of $\tilde{B}$ Find k outliers outside $\sqrt{d}$.
2. Run k -means based on the top k eigenvectors $y_1, \dots, y_k$.

The following eigenvector overlap result indicates weak recovery is possible above the KS threshold:

Theorem 17 ([SZ22]). *Let y_2 be the n -dimensional unit vector associated with In the 2-block SBM above the KS threshold, we have, with high probability,*

$$|\langle y_2, \frac{1}{\sqrt{n}}\sigma \rangle| = \sqrt{1 - \frac{2(a+b)}{(a-b)^2}} + o(1).$$

Non-backtracking spectrum for a d -regular graph To understand the spectrum of B , it is helpful to look at a different graph model where the degree is fixed. When G is a d -regular graph with n vertices, (23) can be further simplified to

$$\det(B - zI) = (z^2 - 1)^{nd/2-n} \det(z^2 I - zA + (d-1)I).$$

Lemma 23 (Spectral relation between A and $\tilde{B}$). *Let G be a d -regular graph with adjacency matrix A and non-backtracking matrix B and $d \geq 2$. Let v_i be an eigenvector of A with respect to an eigenvalue $\lambda_i, i \in [n]$. Then*

$$\mu_i, \mu'_i = \frac{\lambda_i \pm \sqrt{\lambda_i^2 - 4(d-1)}}{2}$$

are two eigenvalues of $\tilde{B}$, with two corresponding eigenvectors of the form

$$u_i = \begin{bmatrix} v_i \\ \frac{\mu_i}{d-1} v_i \end{bmatrix}, \quad u'_i = \begin{bmatrix} v_i \\ \frac{\mu'_i}{d-1} v_i \end{bmatrix}.$$

Proof. Let $\begin{bmatrix} x \\ y \end{bmatrix}, x, y \in \mathbb{C}^n$ be an eigenvector of $\tilde{B}$ with respect to an eigenvalue μ , we have

$$\mu \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & (d-1)I \\ -I & A \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

This equation implies

$$Ax = \left(\mu + \frac{d-1}{\mu} \right) x, \quad y = \frac{\mu}{d-1} x. \quad (25)$$

For each eigenvalue λ_i of A with a corresponding eigenvector $v_i \in \mathbb{R}^n$, there are two corresponding eigenvalue μ_i, μ'_i of $\tilde{B}$ satisfying the quadratic equation

$$x^2 - \lambda_i x + d-1 = 0. \quad (26)$$

In particular, $\lambda_1 = d$ gives two real eigenvalues $\mu_1 = d-1$ and $\mu'_1 = 1$.

By counting multiplicity, Equation (26) gives all the $2n$ eigenvalues of $\tilde{B}$. Thus, by Equations (25) and (26), there are n eigenvectors of $\tilde{B}$ of the form

$$u_i = \begin{bmatrix} v_i \\ \frac{\mu_i}{d-1} v_i \end{bmatrix}, \text{ where } \mu_i = \frac{\lambda_i + \sqrt{\lambda_i^2 - 4(d-1)}}{2},$$

and other n eigenvectors of the form,

$$u'_i = \begin{bmatrix} v_i \\ \frac{\mu'_i}{d-1} v_i \end{bmatrix}, \text{ where } \mu'_i = \frac{\lambda_i - \sqrt{\lambda_i^2 - 4(d-1)}}{2}.$$

□

The lemma above shows that any $|\lambda_i| > 2\sqrt{d-1}$ implies two eigenvalue of B on the real line with $|\mu_i \mu'_i| = (d-1)$. Therefore, this gives an equivalent characterization of a Ramanujan graph.

Lemma 24 (Ramanujan graph). *Let G be a d -regular graph. Then $\max\{|\lambda_2(A)|, |\lambda_n(A)|\} \leq 2\sqrt{d-1}$ if and only if $|\lambda_2(B)| \leq \sqrt{d-1}$.*

Such a characterization is used in [Bor20] for a new proof of Friedman's theorem that a random d -regular graph has second eigenvalue $2\sqrt{d-1} + o(1)$ with high probability.

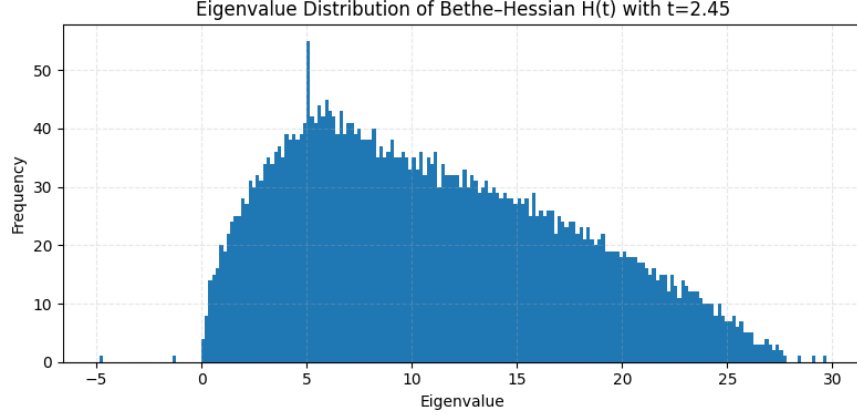


Figure 7: Bethe-Hessian of an SBM, two negative outliers

The Bethe-Hessian The *Bethe-Hessian* (also called the *deformed Laplacian*) provides a symmetric spectral method that remains effective in sparse stochastic block models (SBM), down to the Kesten-Stigum threshold [SKZ14].

Let G be a graph with adjacency matrix A and degree matrix $D = \text{diag}(d_1, \dots, d_n)$. For a scalar parameter $r \in \mathbb{R}$, define the Bethe-Hessian matrix

$$H(r) := (r^2 - 1)I - rA + D.$$

The parameter r is chosen to match the spectral radius of the non-backtracking operator; in practice one takes

$$r \approx \sqrt{\bar{d}}, \quad \bar{d} := \frac{1}{n} \sum_{i=1}^n d_i,$$

the empirical average degree. The Bethe-Hessian arises naturally from the Ihara-Bass formula, which relates the characteristic polynomial of the non-backtracking matrix B to that of $H(r)$ via

$$\det(I - rB) = (1 - r^2)^{m-n} \det(H(r)),$$

where m is the number of edges. Thus, eigenvalues of B outside the bulk correspond to values of r for which $H(r)$ becomes singular.

In sparse SBM, the adjacency spectrum is dominated by degree fluctuations and does not cleanly separate signal from noise. In contrast, when r is chosen near the bulk edge of B , the informative community directions correspond to negative (or smallest) eigenvalues of $H(r)$. Above

the Kesten–Stigum threshold, it was conjectured in [SKZ14] that $H(r)$ develops exactly $k - 1$ negative eigenvalues in the k -block model, whose eigenvectors correlate with the true community labels. See Figure 7 for a simulation.

Hence community detection can be performed by computing the eigenvectors associated with the negative eigenvalues of $H(r)$. This yields a computationally efficient symmetric spectral algorithm that matches the detection threshold.

Recently, [SZ24] determined the number of negative outliers of $H(\sqrt{d})$ above the KS-threshold when $d \geq 2$. A complete characterization of the spectral clustering algorithm above the KS threshold is still open.

3 Spiked matrix models

3.1 Spiked Wigner model and BBP transition

Fix a deterministic unit vector $u \in \mathbb{R}^n$, $\|u\| = 1$, and a spike strength $\theta \in \mathbb{R}$. Define the *spiked Wigner matrix*

$$M = W + \theta uu^\top.$$

Let $\lambda_1 \geq \dots \geq \lambda_n$ be the eigenvalues of M and let $\hat{u}$ be a top eigenvector with unit norm.

As $n \rightarrow \infty$, we may ask the following questions:

1. When does λ_1 detach from the bulk edge at 2 and form an *outlier*?
2. When is $\hat{u}$ *informative*, i.e. when does $|\langle \hat{u}, u \rangle|$ stay bounded away from 0?

The Baik–Ben Arous–Péché (BBP) transition [BBAP05] is the sharp threshold separating the regimes where the spike is spectrally invisible vs visible.

Spiked GOE model Fix $n \geq 1$. Let $W = W_n$ be a GOE matrix scaled so the limiting spectrum lives on $[-2, 2]$:

$$W = \frac{1}{\sqrt{n}}G, \quad G_{ij} = G_{ji}, \quad (G_{ij})_{i < j} \sim \mathcal{N}(0, 1), \quad G_{ii} \sim \mathcal{N}(0, 2),$$

and all these entries are independent. Fix a deterministic unit vector $u \in \mathbb{R}^n$ with $\|u\| = 1$, and a spike strength $\theta \in \mathbb{R}$. Define the spiked matrix

$$M(\theta) = W + \theta uu^\top.$$

Theorem 18 (BBP transition for spiked GOE, rank one). *As $n \rightarrow \infty$:*

1. (**Eigenvalue**) If $\theta \leq 1$, then $\lambda_1(\theta) \rightarrow 2$ in probability. If $\theta > 1$, then

$$\lambda_1(\theta) \rightarrow \theta + 1/\theta$$

in probability.

2. (**Eigenvector**) If $\theta \leq 1$, then $|\langle v_1(\theta), u \rangle|^2 \rightarrow 0$ in probability. If $\theta > 1$, then

$$|\langle v_1(\theta), u \rangle|^2 \rightarrow 1 - 1/\theta^2$$

in probability.

For the rest of the section, we prove Theorem 18.

Step 1: Reduce to $u = e_1$ by orthogonal invariance The following lemma reduces $u = e_1$ due to orthogonal invariance of W .

Lemma 25 (Orthogonal reduction). *Because GOE is orthogonally invariant, we may assume $u = e_1$ without loss of generality.*

Proof. The joint density of W satisfies

$$f(W) = Z_n^{-1} \exp\left(-\frac{n}{4}\|W\|_F^2\right),$$

with respect to Lebesgue measure on the space

$$\mathcal{S}_n = \{W \in \mathbb{R}^{n \times n} : W = W^\top\}.$$

Let O be any deterministic orthogonal matrix. We have $f(O^\top W O) = f(W)$. Therefore

$$O^\top W O \stackrel{d}{=} W.$$

Pick O such that $O^\top u = e_1$. Then

$$O^\top M(\theta) O = O^\top W O + \theta (O^\top u)(O^\top u)^\top \stackrel{d}{=} W + \theta e_1 e_1^\top.$$

Eigenvalues are invariant under conjugation, so λ_i has the same law in either representation. Also if v is an eigenvector of M , then

$$\langle v, u \rangle = \langle O^\top v, O^\top u \rangle = \langle O^\top v, e_1 \rangle,$$

where $O^\top v$ is an eigenvector of $O^\top M(\theta) O$. Thus it suffices to prove the theorem for $u = e_1$. $\square$

Define, for $z \notin \text{spec}(W)$,

$$s_n(z) := e_1^\top (zI - W)^{-1} e_1.$$

This is the $(1, 1)$ -th entry of the resolvent $(zI - W)^{-1}$.

Step 2: Rank-one resolvent equation and monotonicity

Lemma 26. *for invertible A and vectors x, y ,*

$$\det(A + xy^\top) = \det(A)(1 + y^\top A^{-1}x).$$

Proof. Factor out A :

$$A + xy^\top = A(I + A^{-1}xy^\top).$$

Taking determinants,

$$\det(A + xy^\top) = \det(A) \det(I + uv^\top),$$

where $u = A^{-1}x$ and $v = y$. Thus it suffices to show

$$\det(I + uv^\top) = 1 + v^\top u.$$

Now $uv^\top$ has rank one. Hence $I + uv^\top$ has:

- eigenvalue 1 with multiplicity $n - 1$, by finding $(n - 1)$ orthogonal eigenvectors with respect to u .
- one eigenvalue equal to $1 + v^\top u$ (since $(I + uv^\top)u = u + u(v^\top u) = (1 + v^\top u)u$).

Therefore,

$$\det(I + uv^\top) = 1^{n-1}(1 + v^\top u) = 1 + v^\top u.$$

Combining the two steps gives the result. $\square$

Lemma 27 (Rank-one characteristic equation). *For any $z \notin \text{spec}(W)$,*

$$\det(zI - M(\theta)) = \det(zI - W) \left(1 - \theta s_n(z)\right).$$

In particular, for $z \notin \text{spec}(W)$,

$$z \in \text{spec}(M(\theta)) \iff s_n(z) = \frac{1}{\theta}.$$

Proof. Use the rank-one determinant identity: for invertible A and vectors x, y ,

$$\det(A + xy^\top) = \det(A) (1 + y^\top A^{-1}x).$$

Take $A = zI - W$, $x = -\theta e_1$, $y = e_1$, giving

$$\det(zI - M) = \det((zI - W) - \theta e_1 e_1^\top) = \det(zI - W) (1 - \theta e_1^\top (zI - W)^{-1} e_1).$$

$\square$

The Weyl's inequality gives an interlacing property of eigenvalues:

Lemma 28 (Weyl's inequality). *Let $A, B \in \mathbb{C}^{n \times n}$ be Hermitian matrices and let $\lambda_1(M) \geq \lambda_2(M) \geq \dots \geq \lambda_n(M)$ denote the ordered eigenvalues of a Hermitian matrix M . Then for $1 \leq i, j \leq n$ with $i + j - 1 \leq n$,*

$$\lambda_{i+j-1}(A + B) \leq \lambda_i(A) + \lambda_j(B).$$

If $i + j - n \geq 1$, then

$$\lambda_{i+j-n}(A + B) \geq \lambda_i(A) + \lambda_j(B).$$

Lemma 29 (Monotonicity and uniqueness of the root). *Assume $\theta > 0$. Let $\mu_1 = \mu_1(W)$ be the top eigenvalue of W . Then on (μ_1, ∞) , the function $s_n(z)$ is continuous and strictly decreasing with*

$$\lim_{z \downarrow \mu_1} s_n(z) = +\infty \quad (a.s.), \quad \lim_{z \rightarrow \infty} s_n(z) = 0.$$

Hence there is a unique solution $\lambda_n^(\theta) \in (\mu_1, \infty)$ to $s_n(z) = 1/\theta$. Moreover $\lambda_n^*(\theta) = \lambda_1(\theta)$, the largest eigenvalue of $M(\theta)$.*

Proof. For $z > \mu_1$, $(zI - W)$ is positive definite so $s_n(z) > 0$. Differentiate:

$$\frac{d}{dz}s_n(z) = e_1^\top \frac{d}{dz}(zI - W)^{-1}e_1 = -e_1^\top (zI - W)^{-2}e_1 < 0,$$

so s_n is strictly decreasing and continuous. As $z \rightarrow \infty$, $(zI - W)^{-1} \sim z^{-1}I$, so $s_n(z) \sim 1/z \rightarrow 0$.

Write the spectral decomposition $W = \sum_{i=1}^n \mu_i \phi_i \phi_i^\top$ with $\mu_1 > \mu_2 > \dots$. Then

$$s_n(z) = \sum_{i=1}^n \frac{|\langle e_1, \phi_i \rangle|^2}{z - \mu_i} \geq \frac{|\langle e_1, \phi_1 \rangle|^2}{z - \mu_1}.$$

The equation $s_n(z) = 1/\theta$ has a unique solution $\lambda_n^*(\theta) \in (\mu_1, \infty)$.

Finally, by Lemma 27, zeros of $\det(zI - M)$ outside $\text{spec}(W)$ coincide with roots of $1 - \theta s_n(z) = 0$. Since $\lambda_n^*(\theta) > \mu_1$, it must be the (unique) eigenvalue of M lying above μ_1 . By Weyl's inequality (Lemma 28),

$$\lambda_2(M(\theta)) \leq \lambda_1(W) = \mu_1.$$

There is at most one eigenvalue above μ_1 , hence that eigenvalue is $\lambda_1(\theta)$. $\square$

Step 3: Use the semicircle law Diagonalize $W = Q\Lambda Q^\top$ with $\Lambda = \text{diag}(\mu_1, \dots, \mu_n)$. For GOE, Q is Haar on $O(n)$ and independent of Λ ; in particular the first row $q = (q_1, \dots, q_n)$ is uniform on the sphere and independent of (μ_i) [AGZ10]. Then for $z > \mu_1$,

$$s_n(z) = \sum_{i=1}^n \frac{q_i^2}{z - \mu_i}, \quad m_n(z) := \frac{1}{n} \text{Tr}(zI - W)^{-1} = \frac{1}{n} \sum_{i=1}^n \frac{1}{z - \mu_i}.$$

Then $s_n(z)$ is a linear combination of q_i^2 and $\{q_i\}, \{\mu_i\}$ are independent. Fix $\delta > 0$ and $z \geq 2 + \delta$. On the event $\{\mu_1(W) \leq 2 + \delta/2\}$ (which holds w.h.p. since $\mu_1(W) \rightarrow 2$), the coefficients $a_i := 1/(z - \mu_i)$ satisfy $\|a\|_\infty \leq 2/\delta$.

We need the following concentration inequality on the unit sphere to show $s_n(z)$ is concentrated around $m_n(z)$.

Lemma 30 (Exercise 5.5 in [Ver18]). *Let $X \sim \text{Unif}(S^{n-1})$. Let f be a Lipschitz function such that $|f(x) - f(y)| \leq \|f\|_{\text{Lip}} \|x - y\|_2$. For every $t \geq 0$,*

$$\mathbb{P}\{|f(X) - \mathbb{E}f(X)| \geq t\} \leq 2 \exp\left(-\frac{cnt^2}{\|f\|_{\text{Lip}}^2}\right).$$

Applying the lemma above to $F(q) = \sum a_i q_i^2$, since $|F(q) - F(q')| \leq \frac{2}{\delta} \|q - q'\|_2$, we obtain

$$s_n(z) - m_n(z) \xrightarrow{\mathbb{P}} 0 \quad \text{for each fixed } z > 2. \quad (27)$$

For each fixed $z > 2$, the semicircle law implies

$$m_n(z) \xrightarrow{\mathbb{P}} m_{\text{sc}}(z) = \frac{z - \sqrt{z^2 - 4}}{2},$$

hence by (27),

$$s_n(z) \xrightarrow{\mathbb{P}} m_{\text{sc}}(z). \quad (28)$$

Step 4: Eigenvalue transition

(1) Supercritical $\theta > 1$. Let $\rho(\theta) := \theta + 1/\theta > 2$, which satisfies $m_{\text{sc}}(\rho(\theta)) = 1/\theta$. Since both m_{sc} are continuous and strictly decreasing on $(2, \infty)$ and $s_n(\lambda_1(\theta)) = 1/\theta$, it follows that $\lambda_1(\theta) \rightarrow \rho(\theta)$ in probability.

(2) Subcritical $\theta \leq 1$. Fix $\varepsilon > 0$ and set $z = 2 + \varepsilon$. Then $m_{\text{sc}}(z) < 1 \leq 1/\theta$, so by (28), w.h.p. $s_n(z) < 1/\theta$. Since s_n is decreasing and $s_n(\lambda_1(\theta)) = 1/\theta$, we get

$$\lambda_1(\theta) \leq z = 2 + \varepsilon$$

w.h.p. Also from the semicircle law of M , with high probability,

$$\lambda_1(\theta) \geq 2 + o(1).$$

Hence $\lambda_1(\theta) \rightarrow 2$ in probability.

Step 5: Eigenvector overlap

(1) Supercritical $\theta > 1$. Assume $\theta > 1$. Let $\lambda = \lambda_1(\theta)$ and $v = v_1(\theta)$. From $(W + \theta e_1 e_1^\top)v = \lambda v$,

$$(\lambda I - W)v = \theta e_1(e_1^\top v), \quad \Rightarrow \quad v = c(\lambda I - W)^{-1}e_1$$

for $c = \theta(e_1^\top v)$. Then

$$1 = \|v\|^2 = |c|^2 e_1^\top (\lambda I - W)^{-2} e_1.$$

Using also $e_1^\top v = c e_1^\top (\lambda I - W)^{-1} e_1 = c s_n(\lambda) = c/\theta$, we obtain the exact identity

$$|\langle v, e_1 \rangle|^2 = \frac{1}{\theta^2 t_n(\lambda)}, \quad t_n(z) := e_1^\top (zI - W)^{-2} e_1. \quad (29)$$

Repeat Step 3 with coefficients $a_i = 1/(z - \mu_i)^2$ to get

$$t_n(z) = \frac{1}{n} \sum (z - \mu_i)^{-2} + o(1) = -m'_n(z) + o(1)$$

for fixed $z > 2$, where $m'_n(z)$ is the derivative. Since $\lambda \rightarrow \rho(\theta)$,

$$t_n(\lambda) \xrightarrow{\mathbb{P}} -m'_{\text{sc}}(\rho(\theta)).$$

A direct calculation gives $-m'_{\text{sc}}(\rho(\theta)) = 1/(\theta^2 - 1)$, so (29) yields

$$|\langle v_1(\theta), e_1 \rangle|^2 \xrightarrow{\mathbb{P}} \frac{1}{\theta^2} (\theta^2 - 1) = 1 - \frac{1}{\theta^2}.$$

(2) Subcritical $\theta < 1$. Define

$$f_n(\theta) = \lambda_1(W + \theta e_1 e_1^\top) = \max_{\|x\|=1} x^\top W x + \theta x_1^2.$$

Then f_n is convex (maximum of affine functions) and 1-Lipschitz in θ , and $f_n(\theta) \rightarrow 2$ for each fixed $\theta \leq 1$. One can prove $f_n(\theta)$ is differentiable when $\lambda_1(\theta)$ is simple. By convexity,

$$f'_n(\theta) \leq \frac{f_n(\theta + h) - f_n(\theta)}{h} \quad (h > 0).$$

Since $(W + \theta e_1 e_1^\top)v_1(\theta) = f_n(\theta)v_1(\theta)$, taking derivative w.r.t. θ and multiplying $v_1(\theta)^\top$ on both sides yields

$$f'_n(\theta) = |\langle v_1(\theta), e_1 \rangle|^2.$$

Since $f_n(\theta + h) - f_n(\theta) \rightarrow 0$ for fixed $\theta < 1$ and small fixed h , the right side tends to 0, hence $|\langle v_1(\theta), e_1 \rangle|^2 \rightarrow 0$ in probability when $\theta < 1$.

(3) Critical $\theta = 1$. When $\theta = 1$, we have for any fixed $h > 0$

$$f'_n(1) \leq \frac{f_n(1 + h) - f_n(1)}{h}.$$

The right-hand side converges in probability to $\frac{h}{1+h}$ by the eigenvalue BBP transition. For any $\varepsilon > 0$, choose $\frac{h}{1+h} < \varepsilon$, and one can show that

$$\mathbb{P}(|\langle v_1(\theta), e_1 \rangle|^2 > \varepsilon) \leq \mathbb{P}\left(\frac{f_n(1 + h) - f_n(1)}{h} > \varepsilon\right) \rightarrow 0.$$

Therefore, $|\langle v_1(\theta), e_1 \rangle|^2 \rightarrow 0$ in probability.

Beyond the GOE case The proof of the BBP transition in the GOE case uses the orthogonal invariance. By exploring the orthogonal invariance, the analysis can be generalized to a wide class of spiked models [BGN11]. Beyond orthogonal invariance, the work [KY13] obtained the BBP transition for general Wigner matrices by using the isotropic local semicircle law.

Generalization to multiple spikes The analysis of the rank-1 spiked model can be extended to a finite-rank perturbation of a GOE [BGN11, CDMF09].

Theorem 19 (BBP transition for finite-rank deformations of GOE). *Let*

$$X_n = W_n + A_n,$$

where W_n is an $n \times n$ GOE matrix with variance $1/n$, and

$$A_n = \sum_{j=1}^r \theta_j v_j v_j^\top$$

is a deterministic symmetric matrix of fixed rank r , where $v_1, \dots, v_r$ are orthonormal vectors and $\theta_1, \dots, \theta_r \in \mathbb{R}$ are fixed spike strengths. Let $\lambda_1(X_n) \geq \lambda_2(X_n) \geq \dots \geq \lambda_n(X_n)$ denote the eigenvalues of X_n . Then the following hold as $n \rightarrow \infty$.

1. (**Outlier eigenvalues**). For each spike θ_j satisfying $|\theta_j| > 1$, there exists an eigenvalue of X_n converging in probability to

$$\rho(\theta_j) = \theta_j + \theta_j^{-1}.$$

2. (**Subcritical spikes**). If $|\theta_j| \leq 1$,

$$\lambda_i(X_n) \rightarrow \pm 2,$$

depending on whether the spike is positive or negative.

3. (**Eigenvector alignment**). Let u_j denote the unit eigenvector associated with the outlier generated by a simple spike θ_j with $|\theta_j| > 1$. Then

$$|\langle u_j, v_j \rangle|^2 \xrightarrow{p} 1 - \theta_j^{-2}.$$

4. (**Orthogonality to other spikes**). For $i \neq j$,

$$|\langle u_j, v_i \rangle| \xrightarrow{p} 0.$$

5. (**subcritical case**). If $|\theta_j| \leq 1$,

$$|\langle u_i, v_j \rangle|^2 \xrightarrow{p} 0.$$

3.2 Spiked sample covariance model

Definition 11 (Spiked covariance model). Let

$$X = \Sigma^{1/2} Z,$$

where $Z \in \mathbb{R}^{p \times n}$ has i.i.d. $N(0, 1)$ entries, and Σ is a population covariance matrix. The sample covariance matrix is

$$S = \frac{1}{n} X X^\top.$$

We consider the *spiked model*

$$\Sigma = \text{diag}(\ell_1, \dots, \ell_r, 1, \dots, 1),$$

where r is fixed and $p - r$ eigenvalues equal 1. The parameters satisfy the high-dimensional asymptotic regime

$$\frac{p}{n} \rightarrow c \in (0, \infty).$$

When $\Sigma = I$, the eigenvalues of S follow the Marchenko–Pastur distribution.

Theorem 20 (Marchenko–Pastur). *Let $p/n \rightarrow c \in (0, \infty)$. Then the empirical spectral distribution of S converges almost surely to the Marchenko–Pastur law supported on*

$$[(1 - \sqrt{c})^2, (1 + \sqrt{c})^2].$$

The largest eigenvalue satisfies

$$\lambda_{\max}(S) \xrightarrow{p} (1 + \sqrt{c})^2.$$

Suppose Σ contains a single spike $\ell > 1$:

$$\Sigma = \text{diag}(\ell, 1, \dots, 1).$$

The sample covariance matrix can be written as

$$S = \frac{1}{n} \Sigma^{1/2} Z Z^\top \Sigma^{1/2}.$$

One can show that any outlier eigenvalue ρ must satisfy $1 = \ell m(\rho)$, where $m(z)$ is the Stieltjes transform of the Marchenko–Pastur law

$$m(z) = \int \frac{1}{x - z} \mu_{MP}(x).$$

Solving this equation yields the candidate outlier location

$$\rho(\ell) = \ell \left(1 + \frac{c}{\ell - 1} \right).$$

A similar BBP transition can be obtained [BBAP05, BGN11].

Theorem 21 (BBP transition). *Let $p/n \rightarrow c$ and suppose the population covariance matrix contains a spike $\ell > 1$. Then the largest sample eigenvalue satisfies*

$$\lambda_1(S) \rightarrow \begin{cases} (1 + \sqrt{c})^2, & \ell \leq 1 + \sqrt{c}, \\ \ell \left(1 + \frac{c}{\ell - 1} \right), & \ell > 1 + \sqrt{c}. \end{cases}$$

Theorem 22 (Eigenvector overlap). *Let u be the population spike direction and $\hat{u}$ the top sample eigenvector.*

- If $\ell > 1 + \sqrt{c}$, then

$$|\langle \hat{u}, u \rangle|^2 \rightarrow \frac{1 - \frac{c}{(\ell-1)^2}}{1 + \frac{c}{\ell-1}}.$$

- If $\ell \leq 1 + \sqrt{c}$, then

$$|\langle \hat{u}, u \rangle|^2 \rightarrow 0.$$

Information plus noise model A related spiked model is called the information plus noise model, where

$$M = W + \theta u v^\top,$$

and $W \in \mathbb{R}^{p \times n}$ has i.i.d. Gaussian entries [Cap14, BP21]. A similar BBP-type transition can be derived.

3.3 Tensor PCA

Many modern datasets contain higher-order interactions that cannot be captured by matrices alone. Examples include higher-order correlations in statistics, hypergraph data, and moment tensors in latent variable models. These naturally lead to tensor models and motivate the study of *Tensor PCA*. Let $k \geq 3$. We observe a symmetric order- k tensor

$$T = \lambda v^{\otimes k} + W,$$

where

- $v \in \mathbb{R}^n$ is an unknown unit vector ($\|v\|_2 = 1$),
- $v^{\otimes k}$ is the rank-one signal tensor,
- W is a random noise tensor.

The goal is to estimate v from the observation T .

Definition 12 (Tensor GOE). Let $k \geq 2$. An *order- k tensor GOE* is a random symmetric tensor

$$G = (G_{i_1, \dots, i_k})_{1 \leq i_1, \dots, i_k \leq n} \in \text{Sym}^k(\mathbb{R}^n)$$

such that:

1. G is symmetric, i.e.

$$G_{i_1, \dots, i_k} = G_{i_{\pi(1)}, \dots, i_{\pi(k)}} \quad \text{for every } \pi \in S_k;$$

2. the entries

$$\{ G_{i_1, \dots, i_k} : 1 \leq i_1 \leq \dots \leq i_k \leq n \}$$

are independent centered Gaussian random variables;

3. if the multi-index $(i_1, \dots, i_k)$ has multiplicities $m_1, \dots, m_p$ (so that $m_1 + \dots + m_p = k$), then

$$\text{Var}(G_{i_1, \dots, i_k}) = m_1! \cdots m_p!.$$

Tensor linear algebra We begin by recalling some basic notions from tensor linear algebra that will be used throughout the discussion of Tensor PCA.

For positive integers $n_1, \dots, n_k$, an *order- k tensor* is an array

$$T = (T_{i_1, \dots, i_k})_{1 \leq i_j \leq n_j} \in \mathbb{R}^{n_1 \times \dots \times n_k}.$$

When all dimensions are equal, we write

$$T \in (\mathbb{R}^n)^{\otimes k} \cong \mathbb{R}^{n \times \dots \times n}.$$

The case $k = 1$ corresponds to vectors, and $k = 2$ corresponds to matrices.

Given vectors $u^{(1)} \in \mathbb{R}^{n_1}, \dots, u^{(k)} \in \mathbb{R}^{n_k}$, their *outer product*

$$u^{(1)} \otimes \dots \otimes u^{(k)}$$

is the rank-one tensor defined by

$$(u^{(1)} \otimes \cdots \otimes u^{(k)})_{i_1, \dots, i_k} = u_{i_1}^{(1)} \cdots u_{i_k}^{(k)}.$$

If $u^{(1)} = \cdots = u^{(k)} = u$, we write

$$u^{\otimes k} = u \otimes \cdots \otimes u.$$

The standard inner product of two tensors $S, T \in \mathbb{R}^{n_1 \times \cdots \times n_k}$ is

$$\langle S, T \rangle = \sum_{i_1, \dots, i_k} S_{i_1, \dots, i_k} T_{i_1, \dots, i_k},$$

and the associated Frobenius norm is

$$\|T\|_F = \sqrt{\langle T, T \rangle}.$$

A tensor $T \in (\mathbb{R}^n)^{\otimes k}$ is called *symmetric* if

$$T_{i_1, \dots, i_k} = T_{i_{\pi(1)}, \dots, i_{\pi(k)}}$$

for every permutation π of $\{1, \dots, k\}$ and every index tuple $(i_1, \dots, i_k)$. Symmetric tensors are the natural higher-order analog of symmetric matrices. To a symmetric order- k tensor T , one associates the homogeneous polynomial

$$x \mapsto \langle T, x^{\otimes k} \rangle = \sum_{i_1, \dots, i_k=1}^n T_{i_1, \dots, i_k} x_{i_1} \cdots x_{i_k}.$$

The analog of matrix rank for tensors is more subtle.

Definition 13 (Tensor rank). Let $T \in \mathbb{R}^{n_1 \times \cdots \times n_k}$. The *rank* of T , also called the *CP rank*, is the smallest integer $r \geq 0$ such that

$$T = \sum_{\ell=1}^r u_\ell^{(1)} \otimes \cdots \otimes u_\ell^{(k)}$$

for some vectors $u_\ell^{(j)} \in \mathbb{R}^{n_j}$.

Thus a tensor has rank one precisely when it can be written as a single outer product. When $k = 2$, this notion reduces to the usual matrix rank. For $k \geq 3$, however, tensor rank behaves quite differently from matrix rank: it is typically much harder to compute, may fail to have a simple algebraic characterization, and is closely related to difficult optimization problems.

CP decomposition The most common low-rank tensor decomposition is the *CP decomposition*.

Definition 14 (CP decomposition). A *CP decomposition* of a tensor $T \in \mathbb{R}^{n_1 \times \cdots \times n_k}$ is a representation of the form

$$T = \sum_{\ell=1}^r \lambda_\ell u_\ell^{(1)} \otimes \cdots \otimes u_\ell^{(k)},$$

where $\lambda_\ell \in \mathbb{R}$ and $u_\ell^{(j)} \in \mathbb{R}^{n_j}$. If r is minimal, then $r = \text{rank}(T)$.

For symmetric tensors, the CP decomposition often takes the form

$$T = \sum_{\ell=1}^r \lambda_{\ell} v_{\ell}^{\otimes k}.$$

In particular, the spiked tensor model

$$T = \lambda v^{\otimes k} + W$$

is a rank-one signal plus noise model. Unlike the matrix singular value decomposition, CP decompositions may fail to be unique without additional assumptions, and their computation is generally nonconvex. Nevertheless, CP decomposition is a central notion in multilinear algebra, statistics, and machine learning.

Spectral norm The most relevant operator norm for Tensor PCA is the tensor spectral norm.

Definition 15 (Spectral norm). Let $T \in \mathbb{R}^{n_1 \times \dots \times n_k}$. Its *spectral norm* is

$$\|T\|_{\text{op}} = \sup_{\|x^{(j)}\|_2=1} \left| \left\langle T, x^{(1)} \otimes \dots \otimes x^{(k)} \right\rangle \right|,$$

where the supremum is taken over all unit vectors $x^{(j)} \in \mathbb{R}^{n_j}$.

Equivalently,

$$\|T\|_{\text{op}} = \sup_{\|x^{(j)}\|_2 \leq 1} \left| \sum_{i_1, \dots, i_k} T_{i_1, \dots, i_k} x_{i_1}^{(1)} \dots x_{i_k}^{(k)} \right|.$$

When $k = 2$, this is exactly the usual matrix operator norm. If T is symmetric, one often uses the equivalent variational formula

$$\|T\|_{\text{op}} = \sup_{\|x\|_2=1} \left| \langle T, x^{\otimes k} \rangle \right|.$$

For a rank-one tensor

$$T = u^{(1)} \otimes \dots \otimes u^{(k)},$$

one has

$$\|T\|_{\text{op}} = \|u^{(1)}\|_2 \dots \|u^{(k)}\|_2, \quad \|T\|_F = \|u^{(1)}\|_2 \dots \|u^{(k)}\|_2.$$

In particular, if $T = \lambda v^{\otimes k}$ with $\|v\|_2 = 1$, then

$$\|T\|_{\text{op}} = \|T\|_F = |\lambda|.$$

Best rank-one approximation The spectral norm is closely tied to the problem of approximating a tensor by a rank-one tensor. Indeed, maximizing

$$\langle T, x^{(1)} \otimes \dots \otimes x^{(k)} \rangle$$

over unit vectors gives the best rank-one correlation with T . In the symmetric case, this becomes

$$\sup_{\|x\|_2=1} \langle T, x^{\otimes k} \rangle.$$

For matrices, this optimization is solved by the leading singular vectors, but for tensors of order $k \geq 3$ it is generally nonconvex and computationally difficult. This distinction is one of the main reasons why Tensor PCA is much harder than ordinary matrix PCA.

Although these definitions resemble familiar notions from matrix theory, tensors behave in a much less rigid way. For $k \geq 3$:

- tensor rank is difficult to compute,
- low-rank decompositions need not be unique,
- there is no direct analog of the matrix spectral theorem,
- computing the best rank-one approximation is generally hard.

These phenomena are central to both the theory and algorithms of Tensor PCA.

Computational challenges for tensors Unlike the matrix case $k = 2$, where the leading eigenvector or singular vector can be computed efficiently by spectral methods, the tensor setting is much more delicate. For $k \geq 3$, there is no direct analog of the spectral theorem that reduces the estimation problem to a tractable linear-algebraic procedure. Indeed, even computing the best rank-one approximation of a general tensor is NP-hard [HL13]. As a result, Tensor PCA exhibits a fundamental tension between what is statistically possible and what is computationally feasible.

A natural estimator is obtained by maximizing the multilinear form

$$\langle T, x^{\otimes k} \rangle \quad \text{over } \|x\|_2 = 1.$$

If there were no noise, this objective is maximized at $x = \pm v$ (up to the natural sign ambiguity when k is odd/even). However, in the presence of noise, the optimization landscape becomes highly nonconvex and typically contains many critical points [BAMMN19]. This creates major algorithmic difficulties: local search methods may get trapped at spurious stationary points, and naive tensor power iteration may fail unless it is started from an initialization already correlated with the true spike.

Another important feature of tensor PCA is the gap between *information-theoretic* and *algorithmic* thresholds. Roughly speaking, one may ask two different questions:

1. When is it possible, in principle, to estimate v from T better than random guessing?
2. When can this be done by a polynomial-time algorithm?

For matrices, these thresholds essentially coincide, but for tensors, there exists a gap. In several tensor models, recovery is believed to be statistically possible at signal levels where all currently known efficient algorithms fail [RM14]. This makes tensor PCA a central example of a high-dimensional problem with a suspected *computational-to-statistical gap*.

These difficulties have motivated a wide range of algorithmic ideas. Simple methods include tensor unfolding or matricization [RM14], which convert the tensor into a matrix and then apply spectral techniques, though this often loses higher-order structure. More refined approaches include tensor power methods, approximate message passing, and sum-of-squares relaxations. A major theme in the subject is to understand which algorithms succeed at which signal-to-noise levels, and how this performance depends on the tensor order k and the ambient dimension n .

For this reason, Tensor PCA plays a role analogous to ordinary PCA, but in a much richer and more challenging nonlinear setting. It serves as a basic model for understanding detection, estimation, phase transitions, and computational hardness in high-dimensional inference problems involving higher-order data.

3.4 Information-theoretical threshold for tensor PCA

Maximum likelihood estimator Conditioned on (x, λ) with $\|x\| = 1$, the observation T is Gaussian with mean $\lambda x^{\otimes k}$. Hence

$$p(T \mid x, \lambda) \propto \exp\left(-\frac{1}{2\sigma^2}\|T - \lambda x^{\otimes k}\|_F^2\right).$$

Therefore, up to an additive constant independent of (x, λ) , the log-likelihood is

$$\ell(x, \lambda; T) = -\frac{1}{2\sigma^2}\|T - \lambda x^{\otimes k}\|_F^2.$$

Thus the MLE is given by

$$(\hat{x}, \hat{\lambda}) \in \arg \min_{\|x\|=1, \lambda \in \mathbb{R}} \|T - \lambda x^{\otimes k}\|_F^2.$$

Proposition 1 (MLE as best rank-one approximation). The maximum likelihood estimator in the spiked tensor model is

$$\hat{x} \in \arg \max_{\|x\|=1} |\langle T, x^{\otimes k} \rangle|, \quad \hat{\lambda} = \langle T, \hat{x}^{\otimes k} \rangle.$$

Proof. Expand the squared Frobenius norm:

$$\|T - \lambda x^{\otimes k}\|_F^2 = \|T\|_F^2 - 2\lambda \langle T, x^{\otimes k} \rangle + \lambda^2 \|x^{\otimes k}\|_F^2.$$

Since $\|x\| = 1$, we have

$$\|x^{\otimes k}\|_F^2 = 1.$$

Hence

$$\|T - \lambda x^{\otimes k}\|_F^2 = \|T\|_F^2 - 2\lambda \langle T, x^{\otimes k} \rangle + \lambda^2.$$

For fixed x , this is a quadratic function of λ , whose minimizer is

$$\hat{\lambda}(x) = \langle T, x^{\otimes k} \rangle.$$

Substituting back gives

$$\min_{\lambda \in \mathbb{R}} \|T - \lambda x^{\otimes k}\|_F^2 = \|T\|_F^2 - \langle T, x^{\otimes k} \rangle^2.$$

Since $\|T\|_F^2$ does not depend on x , minimizing over $\|x\| = 1$ is equivalent to maximizing

$$\langle T, x^{\otimes k} \rangle^2,$$

that is,

$$\hat{x} \in \arg \max_{\|x\|=1} |\langle T, x^{\otimes k} \rangle|.$$

Finally,

$$\hat{\lambda} = \langle T, \hat{x}^{\otimes k} \rangle.$$

This proves the claim. □

Therefore, the MLE is the maximizer of the spectral norm of a Gaussian symmetric tensor T . When $K = 2$, this corresponds to the largest eigenvalue and eigenvector of the spiked GOE.

Statistical phase transition Consider the rank-one spiked tensor model

$$T = \lambda v^{\otimes k} + \frac{1}{\sqrt{n}} W,$$

where

- $v \in S^{n-1}$ is an unknown unit vector,
- W is a tensor GOE noise tensor.

The statistical task is to estimate the spike v from the observation T . A natural measure of estimation accuracy is the *overlap*

$$|\langle \hat{v}, v \rangle|,$$

where $\hat{v}$ is an estimator of the spike. The maximum likelihood estimator (MLE) of v is given by

$$\hat{v}_{\text{ML}} = \arg \max_{\|x\|_2=1} \langle T, x^{\otimes k} \rangle.$$

Define

$$f_\lambda(q) = \lambda^2 q^k + \log(1 - q) + q, \quad q \in [0, 1].$$

The critical signal strength is

$$\lambda_c = \sup \left\{ \lambda \geq 0 : \sup_{q \in [0, 1]} f_\lambda(q) \leq 0 \right\}.$$

This constant depends only on the tensor order k . The main result in [JLM20] states that a sharp statistical phase transition occurs at λ_c .

Theorem 23 (Statistical phase transition). *Let $k \geq 3$ and consider the tensor PCA model above.*

1. *If $\lambda < \lambda_c$, then*

$$|\langle \hat{v}_{\text{ML}}, v \rangle| \xrightarrow{P} 0.$$

In this regime the spike is statistically undetectable, and no estimator can achieve nontrivial correlation with v .

2. *If $\lambda > \lambda_c$, then*

$$|\langle \hat{v}_{\text{ML}}, v \rangle| \xrightarrow{P} \sqrt{q^*(\lambda)},$$

where

$$q^*(\lambda) = \arg \max_{q \in [0, 1]} f_\lambda(q).$$

In particular, the estimator achieves a positive correlation with the spike.

For $k \geq 3$ the phase transition is *discontinuous*. As λ crosses λ_c , the overlap jumps from 0 to a positive constant. This behavior contrasts with the matrix case ($k = 2$), where the BBP transition is continuous, and the overlap increases smoothly above the threshold. See Figure 8.

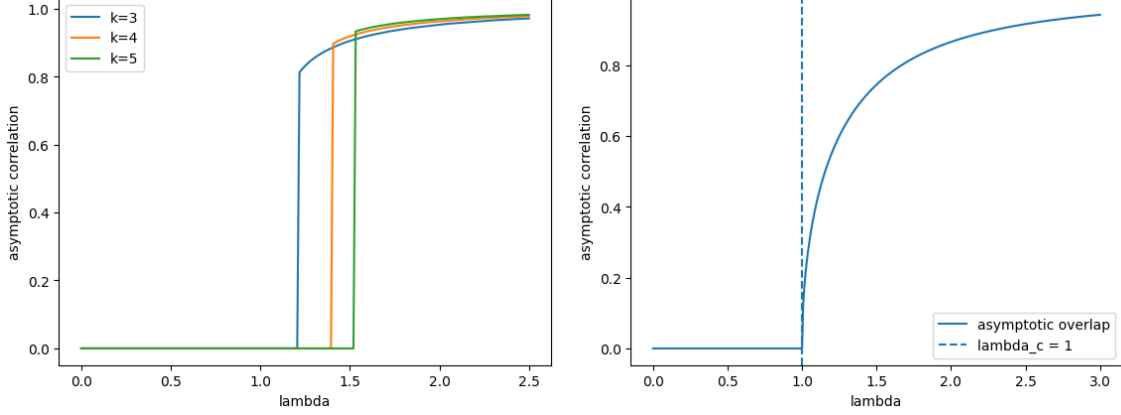


Figure 8: Asymptotic correlation as a function of λ for different $k \geq 3$ (left) and $k = 2$ (right).

3.5 Algorithmic threshold for tensor PCA

We focus on two spectral methods for tensor PCA. Other algorithms include tensor power iteration [HHYC22, WZ24b], gradient descent, approximate message passing, and sum-of-squares relaxations.

1. **Tensor unfolding:** reshape the tensor into a matrix and apply ordinary spectral methods.
2. **Kikuchi matrices:** build a higher-order spectral object from the Kikuchi Hessian (equivalently, from symmetric-difference matrices indexed by subsets), and then use its leading eigenvector.

Tensor unfolding A convenient continuous model is

$$T = \beta v_{\star}^{\otimes k} + W, \quad \|v_{\star}\|_2 = 1,$$

where W is a i.i.d. Gaussian noise tensor with $N(0, 1/n)$ entries. Fix $q = \lfloor \frac{k}{2} \rfloor$. The basic unfolding reshapes the tensor into a matrix

$$\text{Mat}_q(T) \in \mathbb{R}^{n^q \times n^{k-q}}.$$

When $k = 2q$ is even, the balanced unfolding is square:

$$\text{Mat}_{k/2}(T) \in \mathbb{R}^{n^{k/2} \times n^{k/2}}.$$

Under the spiked model,

$$\text{Mat}_q(T) = \beta \text{vec}(v_{\star}^{\otimes q}) \text{vec}(v_{\star}^{\otimes(k-q)})^{\top} + \text{Mat}_q(W).$$

Hence, unfolding turns tensor PCA into a spiked matrix problem. If the unfolded noise behaves like a random matrix of size $n^q \times n^{k-q}$ with entry size $n^{-1/2}$, its operator norm is of order $\|\text{Mat}_q(W)\| \asymp n^{(\max\{q, k-q\}-1)/2}$. Therefore the natural unfolding threshold is $\beta_{\text{unf}} \asymp n^{(\lceil k/2 \rceil - 1)/2}$. For even k this becomes

$$\beta_{\text{unf}} \asymp n^{(k-2)/4}.$$

For odd k , the naive unfolding threshold is worse by a factor $n^{1/4}$ than the exponent one would hope for from the even-order formula.

Long random matrix and tensor unfolding A recent random matrix analysis in [BAHH23] shows that tensor unfolding exhibits a sharp BBP-type threshold at the signal level

$$\beta \asymp n^{(k-2)/4},$$

and this threshold is *independent of the unfolding procedure*, namely, independent of how one reshapes the order- k tensor into an $n^q \times n^{k-q}$ matrix. A second important conclusion is that if one wants to recover the *individual* signal vectors, then it is natural to take $q = 1$, i.e. unfold into an $n \times n^{k-1}$ matrix. In tensor language, this is exactly the truncated HOSVD method [KB09].

The paper [BAHH23] works with the *asymmetric rank-one spiked tensor model*

$$X = \beta v_1 \otimes v_2 \otimes \cdots \otimes v_k + W,$$

where

- $v_1, \dots, v_k \in \mathbb{R}^n$ are unknown unit vectors,
- $\beta > 0$ is the signal-to-noise ratio,
- W has i.i.d. centered entries with variance $1/n$.

Fix an index set $I \subseteq [k]$ with $|I| = q \leq k/2$. Unfolding X along the modes indexed by I produces a matrix

$$\text{Mat}_I(X) \in \mathbb{R}^{n^q \times n^{k-q}}.$$

The unfolded signal is still rank one:

$$\text{Mat}_I(X) = \beta v_I u_I^\top + Z_I,$$

where

$$v_I \in \mathbb{R}^{n^q}, \quad u_I \in \mathbb{R}^{n^{k-q}},$$

are the vectorizations of the corresponding tensor products of the planted factors. It is convenient to normalize the unfolding by

$$\widetilde{\text{Mat}}_I(X) = \frac{\beta}{n^{(q-1)/2}} v_I u_I^\top + \widetilde{Z}_I,$$

so that the entries of $\widetilde{Z}_I$ have variance n^{-q} . The relevant aspect ratio parameter is then

$$\phi = \sqrt{\frac{n^{k-q}}{n^q}} = n^{(k-2q)/2}.$$

The core tensor result of the paper is the following.

Theorem 24 (Tensor unfolding threshold). *Fix $1 \leq q \leq k/2$, and unfold the spiked tensor into an $n^q \times n^{k-q}$ matrix. Write*

$$\beta = \lambda n^{(k-2)/4}, \quad \lambda > 0.$$

Then there is a BBP-type transition at $\lambda = 1$. More precisely, for any arbitrarily small $c > 0$ and $\delta > 0$:

1. If $\lambda > 1 + n^{-(1/3-c)q}$, then with high probability the largest singular value becomes an outlier, and the top left singular vector $\hat{v}_1$ of the normalized unfolding correlates with the planted vector v_I according to

$$|\langle \hat{v}_1, v_I \rangle| = \left(1 + O\left(\frac{n^\delta}{n^{q/2}(\lambda - 1)^{3/2}}\right)\right) \sqrt{\frac{\lambda^4 - 1}{\lambda^4 + \lambda^2/\phi}}.$$

In particular, when $\phi \rightarrow \infty$,

$$|\langle \hat{v}_1, v_I \rangle| \sim \sqrt{1 - \lambda^{-4}}.$$

2. If $\lambda \leq 1 + n^{-(1/3-c)q}$, then with high probability there is no outlier singular value, and the overlap vanishes:

$$|\langle \hat{v}_1, v_I \rangle| \leq n^{4qc} \min\left\{n^{-q/6}, \frac{1}{n^{q/2}(\lambda - 1)}\right\}.$$

Hence the spectral threshold for unfolding is $\beta_{\text{unf}} \asymp n^{(k-2)/4}$.

Let us specialize to the symmetric tensor PCA model

$$X = \beta v^{\otimes k} + W.$$

Thus, in the tensor-PCA setting, the paper identifies the sharp spectral threshold for the unfolding method:

- If $\beta \ll n^{(k-2)/4}$, tensor unfolding does not correlate with the spike.
- If $\beta \gg n^{(k-2)/4}$, the top singular vector of the unfolded matrix becomes informative.
- At the critical scaling $\beta = \lambda n^{(k-2)/4}$, the asymptotic overlap is $\sqrt{1 - \lambda^{-4}}$ for $\lambda \geq 1$.

Kikuchi matrices We now switch to the Boolean normalization

$$Y = \lambda x_\star^{\otimes p} + G, \quad x_\star \in \{\pm 1\}^n,$$

where G is a symmetric tensor with $N(0, 1)$ entries. This is the conventional setting in the Kikuchi method [WEAM25]. Assume first that p is even (when p is odd, the definition is more involved [KX26]). For an integer

$$\ell \in [p/2, n - p/2],$$

the *level- ℓ symmetric-difference matrix* is the matrix

$$M^{(\ell)} \in \mathbb{R}^{\binom{n}{\ell} \times \binom{n}{\ell}}$$

indexed by subsets $S, T \subset [n]$ of size ℓ , with entries

$$M_{S,T}^{(\ell)} = \begin{cases} Y_{S \triangle T}, & |S \triangle T| = p, \\ 0, & \text{otherwise.} \end{cases}$$

Here $S \triangle T$ is the symmetric difference.

Lemma 31 (Expectation of the Kikuchi matrix). *Let $Y = \lambda(x^*)^{\otimes p} + G$ be the spiked tensor model, where $x^* \in \{\pm 1\}^n$ and G is a centered Gaussian noise tensor. Let $M^{(\ell)}$ be the matrix indexed by subsets $S, T \subset [n]$ with $|S| = |T| = \ell$, defined by*

$$M_{S,T}^{(\ell)} = \begin{cases} Y_{S\Delta T}, & |S\Delta T| = p, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\mathbb{E}[M_{S,T}^{(\ell)}] = \lambda x_S^* x_T^* \mathbf{1}\{|S\Delta T| = p\}.$$

Proof. By definition of the model, for any p -tuple (or p -set) E ,

$$\mathbb{E}[Y_E \mid x^*] = \lambda x_E^*, \quad x_E^* := \prod_{i \in E} x_i^*.$$

Hence, by the definition of $M^{(\ell)}$,

$$\mathbb{E}[M_{S,T}^{(\ell)} \mid x^*] = \begin{cases} \lambda x_{S\Delta T}^*, & |S\Delta T| = p, \\ 0, & \text{otherwise.} \end{cases}$$

It remains to simplify $x_{S\Delta T}^*$. Since $x_i^* \in \{\pm 1\}$, we have $(x_i^*)^2 = 1$, and therefore

$$x_{S\Delta T}^* = \prod_{i \in S\Delta T} x_i^* = \left(\prod_{i \in S} x_i^* \right) \left(\prod_{i \in T} x_i^* \right) = x_S^* x_T^*.$$

Substituting this identity yields

$$\mathbb{E}[M_{S,T}^{(\ell)} \mid x^*] = \lambda x_S^* x_T^* \mathbf{1}\{|S\Delta T| = p\},$$

as claimed. $\square$

Lemma 32. *Let $u \in \mathbb{R}^{(n)}_{\ell}$ be defined by $u_S = x_S^*$. Then u is an eigenvector of $\mathbb{E}[M^{(\ell)} \mid x^*]$ with eigenvalue*

$$\lambda d_{\ell}, \quad d_{\ell} = \binom{\ell}{p/2} \binom{n-\ell}{p/2},$$

which counts the number of sets T such that $|S\Delta T| = p$.

Proof. Recall from the previous lemma that

$$\mathbb{E}[M_{S,T}^{(\ell)} \mid x^*] = \lambda x_S^* x_T^* \mathbf{1}\{|S\Delta T| = p\}, \quad |S| = |T| = \ell.$$

Let $u \in \mathbb{R}^{(n)}_{\ell}$ be defined by $u_S = x_S^*$. Then, for each S with $|S| = \ell$,

$$(\mathbb{E}[M^{(\ell)} \mid x^*]u)_S = \sum_{|T|=\ell} \mathbb{E}[M_{S,T}^{(\ell)} \mid x^*] u_T.$$

Substituting the formula above and using $u_T = x_T^*$, we obtain

$$(\mathbb{E}[M^{(\ell)} \mid x^*]u)_S = \lambda \sum_{|T|=\ell} x_S^* x_T^* \mathbf{1}\{|S \triangle T| = p\} x_T^*.$$

Since $x_i^* \in \{\pm 1\}$, we have $(x_T^*)^2 = 1$, and therefore

$$(\mathbb{E}[M^{(\ell)} \mid x^*]u)_S = \lambda x_S^* \sum_{|T|=\ell} \mathbf{1}\{|S \triangle T| = p\}.$$

It remains to count the number of sets T of size ℓ such that $|S \triangle T| = p$. Since

$$|S \triangle T| = |S \setminus T| + |T \setminus S|$$

and $|S| = |T| = \ell$, we necessarily have $|S \setminus T| = |T \setminus S| = p/2$. Thus such a set T is obtained by removing $p/2$ elements from S and adding $p/2$ elements from $[n] \setminus S$. The number of choices is therefore

$$d_\ell = \binom{\ell}{p/2} \binom{n-\ell}{p/2}.$$

Hence

$$(\mathbb{E}[M^{(\ell)} \mid x^*]u)_S = \lambda d_\ell x_S^* = \lambda d_\ell u_S.$$

Since this holds for every S , we conclude that u is an eigenvector of $\mathbb{E}[M^{(\ell)} \mid x^*]$ with eigenvalue λd_ℓ . $\square$

Remark 10. The leading eigenvector of $M^{(\ell)}$ should be thought of as an estimate of the vector of monomials

$$(x_S^*)_{|S|=\ell} \in \mathbb{R}^{\binom{n}{\ell}} \quad x_S^* := \prod_{i \in S} x_i^*.$$

A second spectral rounding step (the *voting matrix*) [WEAM25] then converts this back into an estimate of the planted vector x^* .

Kothari and Xu [KX26] proved sharp norm bounds for Kikuchi matrices, covering both even and odd tensor orders.

Theorem 25 (Smooth statistical-computational tradeoff for tensor PCA). *For every fixed tensor order $p > 2$, there exists $\delta > 0$ such that for all $\ell \leq n^\delta$, there is a spectral algorithm based on the level- ℓ Kikuchi matrix, running in time $n^{O(\ell)}$, that solves detection in the Boolean tensor PCA model whenever*

$$\lambda \gtrsim n^{-p/4} \ell^{1/2-p/4}.$$

Under the conversion to the same normalization as in the unfolding setting, this becomes

$$\beta \gtrsim n^{(p-2)/4} \ell^{1/2-p/4}.$$

Hence if $\ell = n^\delta$,

$$\beta \gtrsim n^{(1-\delta)(p-2)/4}.$$

This is the clean *smooth tradeoff*: Larger runtime buys a lower required signal in a continuous way. That is the main theoretical advantage of Kikuchi matrices over plain unfolding.

4 Nonlinear random matrices and deep learning theory

Classical random matrix theory focuses on linear objects such as Wigner matrices and sample covariance matrices, whose spectra are governed by universal laws such as the semicircle and Marchenko–Pastur distributions. In modern machine learning, however, data are processed through nonlinear transformations—most prominently, neural networks and kernel methods—which naturally give rise to *nonlinear random matrices*. These include random feature models, neural tangent kernels (NTK), and Gram matrices of nonlinear embeddings. Understanding their spectra is essential for explaining generalization, optimization, and implicit bias in deep learning. Some useful references in this area: [BMR21, Bel21, MM24, Lou, Paq].

4.1 Student–teacher model, training error, and generalization error

Student–teacher model The student–teacher framework is a canonical probabilistic model for analyzing learning algorithms in high dimensions. It assumes that the data are generated from a *teacher* model, while a *student* model is trained to approximate it.

Let $x \in \mathbb{R}^d$ be a random input drawn from a distribution $\mathcal{D}_x$ (e.g. $x \sim \mathcal{N}(0, I_d)$). The response is generated by

$$y = f^\star(x) + \varepsilon,$$

where $f^\star : \mathbb{R}^d \rightarrow \mathbb{R}$ is the unknown *teacher* function and ε is noise with $\mathbb{E}[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2$.

The *student* model is a parametric function $f(x; \theta)$ trained from data $\{(x_i, y_i)\}_{i=1}^n$, for example for a two-layer neural network can be written as

$$f(x; \theta) = \frac{1}{\sqrt{m}} \sum_{j=1}^m a_j \phi(w_j^\top x),$$

with parameters $\theta = \{(a_j, w_j)\}_{j=1}^m$. Here ϕ is the activation function, $\{w_j\}$ are the first layer weights and $\{a_j\}$ are the second layer weights.

Training error Given training data $\{(x_i, y_i)\}_{i=1}^n$, the *training error* (empirical risk) is

$$\mathcal{E}_{\text{train}}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i; \theta), y_i),$$

where $\ell(\cdot, \cdot)$ is a loss function. For squared loss,

$$\mathcal{E}_{\text{train}}(\theta) = \frac{1}{n} \sum_{i=1}^n (f(x_i; \theta) - y_i)^2.$$

Generalization error The *generalization error* (population risk) is defined as

$$\mathcal{E}_{\text{gen}}(\theta) = \mathbb{E}_{(x, y)} [\ell(f(x; \theta), y)],$$

where (x, y) is an independent test sample drawn from the same distribution. For squared loss,

$$\mathcal{E}_{\text{gen}}(\theta) = \mathbb{E}_{x, \varepsilon} \left[(f(x; \theta) - f^\star(x) - \varepsilon)^2 \right].$$

Bias–variance decomposition Expanding the squared loss yields

$$\mathcal{E}_{\text{gen}}(\theta) = \mathbb{E}[(f(x; \theta) - f^*(x))^2] + \sigma^2,$$

where

- $\mathbb{E}[(f(x; \theta) - f^*(x))^2]$ is the estimation error,
- σ^2 is the irreducible noise.

The goal of learning is to minimize the generalization error rather than just the training error. In high-dimensional and overparameterized models:

- it is often possible to achieve $\mathcal{E}_{\text{train}} \approx 0$,
- yet the excessive risk ($\mathcal{E}_{\text{gen}} - \sigma^2$) may remain small due to implicit regularization,
- the gap between training and generalization error depends on the data distribution and the model structure.

The student–teacher model provides a clean framework for studying generalization, phase transitions, and high-dimensional phenomena in modern machine learning.

4.2 Random feature regression

In the analysis of wide neural networks, a central object is the *conjugate kernel* (also called the neural network Gaussian kernel). It arises as the limiting kernel when the width of a randomly initialized network tends to infinity.

Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be an activation function and let $x, x' \in \mathbb{R}^d$. Define

$$K(x, x') = \mathbb{E}_{w \sim \mathcal{N}(0, I_d)} [\phi(w^\top x) \phi(w^\top x')].$$

This defines a positive semi-definite kernel, called the *conjugate kernel* associated with ϕ .

Given data points $x_1, \dots, x_n$, the corresponding kernel matrix is

$$K_{ij} = K(x_i, x_j) = \mathbb{E}_w [\phi(w^\top x_i) \phi(w^\top x_j)].$$

Connection to neural networks Consider a random feature model (or a two-layer neural network at initialization):

$$f(x) = \frac{1}{\sqrt{N}} \sum_{k=1}^N a_k \phi(w_k^\top x),$$

where $w_k \sim \mathcal{N}(0, I_d)$ and a_k are random coefficients. As $m \rightarrow \infty$, by the law of large numbers,

$$K_N(x_i, x_j) = \frac{1}{N} \sum_{k=1}^N \phi(w_k^\top x_i) \phi(w_k^\top x_j) \longrightarrow K(x_i, x_j).$$

Thus, the *empirical conjugate kernel* converges entrywise to the population conjugate kernel matrix.

The spectrum of K_N characterizes the neural network at random initialization and is an important ingredient to understand training and generalization of random feature regression.

Random feature regression In random feature regression, one fits a linear model on top of the random features $\phi(w_k^\top x)$ while keeping the weights w_k fixed. Given training data $\{(x_i, y_i)\}_{i=1}^n$, define the feature matrix

$$\Phi \in \mathbb{R}^{n \times N}, \quad \Phi_{ik} = \frac{1}{\sqrt{N}} \phi(w_k^\top x_i),$$

and let

$$y = (y_1, \dots, y_n)^\top \in \mathbb{R}^n.$$

For each point x , define the random feature vector

$$\Phi(x) = \frac{1}{\sqrt{N}} (\phi(w_1^\top x), \dots, \phi(w_N^\top x))^\top \in \mathbb{R}^N.$$

Consider the ridge-regularized square loss

$$L(\beta) := \|\Phi\beta - y\|_2^2 + \lambda \|\beta\|_2^2, \quad \beta \in \mathbb{R}^N,$$

where $\lambda \geq 0$ is the regularization parameter.

$$\hat{\beta} = (\Phi^\top \Phi + \lambda I_N)^{-1} \Phi^\top y.$$

$\hat{\beta}$ is the unique minimizer whenever $\Phi^\top \Phi + \lambda I_N$ is invertible. The resulting predictor is

$$\hat{f}(x) = \Phi(x)^\top \hat{\beta}.$$

We claim the following identity:

$$(\Phi^\top \Phi + \lambda I_N)^{-1} \Phi^\top = \Phi^\top (\Phi \Phi^\top + \lambda I_n)^{-1}.$$

To see this, let

$$A := \Phi^\top \Phi + \lambda I_N, \quad B := \Phi \Phi^\top + \lambda I_n.$$

Observe the key relation

$$A \Phi^\top = (\Phi^\top \Phi + \lambda I_N) \Phi^\top = \Phi^\top (\Phi \Phi^\top + \lambda I_n) = \Phi^\top B.$$

Multiplying both sides on the left by A^{-1} gives

$$\Phi^\top = A^{-1} \Phi^\top B.$$

Finally, multiplying on the right by B^{-1} yields

$$A^{-1} \Phi^\top = \Phi^\top B^{-1},$$

which is exactly the desired identity.

Then we may rewrite the predictor in dual form as

$$\hat{f}(x) = \Phi(x)^\top \Phi^\top (\Phi \Phi^\top + \lambda I_n)^{-1} y.$$

Now define the empirical conjugate kernel

$$K_N = \Phi \Phi^\top \in \mathbb{R}^{n \times n}, \quad (K_N)_{ij} = \frac{1}{N} \sum_{k=1}^N \phi(w_k^\top x_i) \phi(w_k^\top x_j),$$

and the kernel vector

$$k_N(x, X) = \Phi \Phi(x) \in \mathbb{R}^n, \quad (k_N(x, X))_i = \frac{1}{N} \sum_{k=1}^N \phi(w_k^\top x_i) \phi(w_k^\top x).$$

Then

$$\hat{f}(x) = k_N(x, X)^\top (K_N + \lambda I_n)^{-1} y.$$

Thus random feature regression can be viewed either as ridge regression in the finite-dimensional random feature space, or equivalently as kernel ridge regression with the empirical kernel K_N .

This shows that random feature regression interpolates between finite-width neural networks and kernel methods: when N is large, K_N concentrates around the population kernel K , and the predictor behaves like kernel ridge regression, while for finite N , the randomness in K_N introduces additional variance that affects both optimization and generalization. The spectral distribution of K_N (and its limit as $n, d, N \rightarrow \infty$) thus plays a central role in determining bias–variance trade-offs, training dynamics, and the emergence of phenomena such as double descent.

Spectrum of the empirical conjugate kernel matrix K_N There are two complementary types of questions one can ask about the random matrix K_N :

- **Non-asymptotic analysis:** How large must the network width N be to ensure that $\|K_N - K\|$ is small? This quantifies the width required for the finite network to approximate the behavior of its infinite-width (kernel) counterpart.
- **High-dimensional asymptotics:** When $n, d, N \rightarrow \infty$ proportionally, does the empirical spectral distribution of K_N converge to a deterministic limit μ ? How does this limiting distribution depend on the data distribution and the architecture (e.g., activation function, number of neurons N) of the neural network?

4.3 Concentration of random kernel matrices

We will first show that with Lipschitz activation functions, K_N enjoys good concentration properties [WZ24a].

Definition 16 (K -concentration property). Let X be a random vector in $\mathbb{R}^n$. We say X has the K -concentration property with constant K if for any 1-Lipschitz convex function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we have $\mathbb{E}|f(X)| < \infty$ and for any $t > 0$,

$$\mathbb{P}(|f(X) - \mathbb{E}f(X)| \geq t) \leq 2 \exp\left(-\frac{t^2}{K^2}\right).$$

Definition 17 (K -convex concentration property). Let X be a random vector in $\mathbb{R}^n$. We say X has the K -convex concentration property with constant K if for any 1-Lipschitz convex function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we have $\mathbb{E}|f(X)| < \infty$ and for any $t > 0$,

$$\mathbb{P}(|f(X) - \mathbb{E}f(X)| \geq t) \leq 2 \exp\left(-\frac{t^2}{K^2}\right).$$

We will apply the following result from [Ada15] to the nonlinear setting.

Lemma 33 (Theorem 2.5 in [Ada15]). *Let X be a mean zero random vector in $\mathbb{R}^n$. If X has the K -convex concentration property, then for any $n \times n$ matrix A and any $t > 0$,*

$$\mathbb{P}(|X^\top AX - \mathbb{E}(X^\top AX)| \geq t) \leq 2 \exp \left(-\frac{1}{C} \min \left\{ \frac{t^2}{2K^4 \|A\|_F^2}, \frac{t}{K^2 \|A\|} \right\} \right),$$

for some universal constant $C > 1$.

Theorem 26 (Quadratic form concentration). *Let $w \in \mathbb{R}^d$ be a random vector with the K -concentration property, and let $X = (x_1, \dots, x_n) \in \mathbb{R}^{d \times n}$ be a deterministic matrix. Define*

$$y = \sigma(w^\top X)^\top \in \mathbb{R}^n,$$

where σ is λ_σ -Lipschitz, and let $\mathbf{K} = \mathbb{E}yy^\top$. Let A be an $n \times n$ deterministic matrix. If $\mathbb{E}[y] = 0$, then for any $t > 0$,

$$\mathbb{P}(|y^\top Ay - \text{Tr}(A\mathbf{K})| \geq t) \leq 2 \exp \left(-\frac{1}{C} \min \left\{ \frac{t^2}{2K^4 \lambda_\sigma^4 \|X\|^4 \|A\|_F^2}, \frac{t}{K^2 \lambda_\sigma^2 \|X\|^2 \|A\|} \right\} \right),$$

where $C > 0$ is an absolute constant.

Proof. Let f be any 1-Lipschitz convex function. Since $\mathbf{y} = \sigma(w^\top X)^\top$, $f(\mathbf{y}) = f(\sigma(w^\top X)^\top)$ is a $\lambda_\sigma \|X\|$ -Lipschitz function of w . Then by the K -concentration property of w , we obtain

$$\mathbb{P}(|f(\mathbf{y}) - \mathbb{E}f(\mathbf{y})| \geq t) \leq 2 \exp \left(-\frac{t^2}{K^2 \lambda_\sigma^2 \|X\|^2} \right).$$

Therefore, $\mathbf{y}$ satisfies the $K\lambda_\sigma \|X\|$ -convex concentration property. Applying Lemma 8 finishes the proof. $\square$

We present the proof from [WZ24a] with an ε -net argument and Theorem 26.

Theorem 27 (kernel concentration with Lipschitz activation). *Assume σ is λ_σ -Lipschitz with $\mathbb{E}_{\xi \sim N(0,1)}[\sigma(\xi)] = 0$ and $\|x_i\|_2 = 1, 1 \leq i \leq n$. Then with probability at least $1 - 4e^{-2n}$,*

$$\|K_N - \mathbf{K}\| \leq C \left(\sqrt{\frac{n}{N}} + \frac{n}{N} \right) \lambda_\sigma^2 \|X\|^2.$$

where $C > 0$ is a universal constant.

Proof. Take a $1/2$ -net $\mathcal{N}$ of the sphere $\mathbb{S}^{n-1}$. Then

$$\|K_N - \mathbf{K}\| \leq 2 \sup_{v \in \mathcal{N}} \langle v, (K_N - \mathbf{K})v \rangle.$$

Let $\mathbf{y}_i^\top = \sigma(w_i^\top X)$. For each $v \in \mathcal{N}$,

$$\langle v, K_N v \rangle = \frac{1}{N} (\mathbf{y}_1, \dots, \mathbf{y}_N)^\top \begin{bmatrix} vv^\top & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & vv^\top \end{bmatrix} \begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_N \end{bmatrix} := \frac{1}{N} Y^\top A_v Y.$$

Then Y satisfies $\sqrt{2}\lambda_\sigma \|X\|$ concentration property. Moreover, $\|A_v\| = 1, \|A_v\|_F = \sqrt{N}$. Applying Theorem 3 with a union bound over $v \in \mathcal{N}$ finishes the proof. $\square$

The next concentration inequality does not require $\mathbb{E}_\xi \sigma(\xi) = 0$ but has additional log factors. The proof in [WZ23] is an application of the matrix Bernstein inequality (Theorem 10) with truncation.

Theorem 28 (Kernel concentration with matrix Bernstein). *Assume σ has a polynomial growth rate $|\sigma(x)| \leq C_\sigma(1+|x|)^{C_\sigma}$ and $\|x_i\| = 1, i \in [n]$, and $\mathbb{E}_{\xi \sim N(0,1)} \sigma^4(\xi) < \infty$. We have with probability at least $1 - N^{-2}$,*

$$\|K_N - K\| \leq C \sqrt{\frac{n}{N}} \log^{C_\sigma}(nN) \sqrt{\log n} \sqrt{\|\mathbf{K}\|} + C \log n \log^{C_\sigma}(nN) \frac{n}{N}.$$

Remark 11. The right-hand side of this concentration inequality can be improved by considering $\|K^{-1/2}(K_N - K)K^{-1/2}\|$ when K is invertible. See [WZ23, MZ22].

Proof. Denote $\tilde{\sigma}(x) := \sigma(x)\mathbf{1}_{|x| \leq B}$, where B is a parameter to be decided later. Define

$$\begin{aligned} \mathbf{K}_N &= \frac{1}{N} \sum_{i=1}^N \sigma(\mathbf{w}_i^\top \mathbf{X})^\top \sigma(\mathbf{w}_i^\top \mathbf{X}), & \mathbf{K} &= \mathbb{E}_{\mathbf{w}}[\sigma(\mathbf{w}^\top \mathbf{X})^\top \sigma(\mathbf{w}^\top \mathbf{X})], \\ \tilde{\mathbf{K}}_N &= \frac{1}{N} \sum_{i=1}^N \tilde{\sigma}(\mathbf{w}_i^\top \mathbf{X})^\top \tilde{\sigma}(\mathbf{w}_i^\top \mathbf{X}), & \tilde{\mathbf{K}} &= \mathbb{E}_{\mathbf{w}}[\tilde{\sigma}(\mathbf{w}^\top \mathbf{X})^\top \tilde{\sigma}(\mathbf{w}^\top \mathbf{X})]. \end{aligned}$$

Based on the truncated function $\tilde{\sigma}(x)$, we have that for some universal constant $c > 0$,

$$\mathbb{P}(\mathbf{K}_N \neq \tilde{\mathbf{K}}_N) \leq \mathbb{P}\left(\max_{i \in [N], k \in [n]} |\mathbf{w}_i^\top \mathbf{x}_k| > B\right) \leq Nn\mathbb{P}(|\xi| > B) \leq cNn \exp(-B^2/2),$$

where $\xi \sim \mathcal{N}(0, 1)$.

Define the event by

$$A_i := \{\mathbf{w} : |\mathbf{w}^\top \mathbf{x}_i| \leq B\}.$$

We have

$$\begin{aligned} |[\mathbf{K} - \tilde{\mathbf{K}}]_{i,j}| &= |\mathbb{E}_{\mathbf{w}}[\sigma(\mathbf{w}^\top \mathbf{x}_i) \sigma(\mathbf{w}^\top \mathbf{x}_j) \mathbf{1}_{A_j^c \cup A_i^c}]| \\ &\leq \mathbb{E}_{\mathbf{w}}[\sigma(\mathbf{w}^\top \mathbf{x}_i)^4]^{1/2} \mathbb{E}[\mathbf{1}_{A_j^c \cup A_i^c}]^{1/2} \\ &\leq \sqrt{2} \mathbb{E}[\sigma(\xi)^4]^{1/2} \mathbb{P}(A_i^c)^{1/2} \leq C_0 \exp(-B^2/4), \end{aligned}$$

for some constant $C_0 > 0$ which only depends on $\|\sigma\|_4$. Therefore,

$$\|\mathbf{K} - \tilde{\mathbf{K}}\| \leq \|\mathbf{K} - \tilde{\mathbf{K}}\|_F \leq C_0 n \exp(-B^2/4)$$

and

$$\mathbb{P}(\|(\mathbf{K}_N - \mathbf{K})\| \geq t) \leq \mathbb{P}\left(\|(\tilde{\mathbf{K}}_N - \mathbf{K})\| \geq t\right) + \mathbb{P}(\mathbf{K}_N \neq \tilde{\mathbf{K}}_N).$$

Hence, we only need to prove the concentration inequality for $\tilde{\mathbf{K}}_N - \tilde{\mathbf{K}} = \sum_{i=1}^N \mathbf{H}_i - \mathbb{E}\mathbf{H}_i$. In terms of the definition of $\tilde{\sigma}$, we know that, almost surely,

$$\sup_{\mathbf{w}_i} \|\mathbf{H}_i - \mathbb{E}\mathbf{H}_i\| \leq 2 \sup_{\mathbf{w}_i} \|\mathbf{H}_i\| \leq \frac{2}{N} \sup_{\mathbf{w}_i} \|\tilde{\sigma}(\mathbf{w}_i^\top \mathbf{X})\|^2 \leq \frac{2C_\sigma^2 n(1+B)^{2C_\sigma}}{N}.$$

Analogously, we have

$$\mathbf{H}_i^2 = \frac{1}{N^2} \tilde{\sigma}(\mathbf{w}_i^\top \mathbf{X})^\top \tilde{\sigma}(\mathbf{w}_i^\top \mathbf{X}) \tilde{\sigma}(\mathbf{w}_i^\top \mathbf{X})^\top \tilde{\sigma}(\mathbf{w}_i^\top \mathbf{X})$$

$$\mathbb{E}[(\mathbf{H}_i - \mathbb{E}[\mathbf{H}_i])^2] \preceq \mathbb{E}\mathbf{H}_i^2 \preceq \frac{C_\sigma^2 n(1+B)^{2C_\sigma}}{N^2} \tilde{\mathbf{K}}.$$

And

$$\left\| \sum_{i=1}^N \mathbb{E}[(\mathbf{H}_i - \mathbb{E}[\mathbf{H}_i])^2] \right\| \leq \frac{C_\sigma^2 n(1+B)^{2C_\sigma}}{N} \|\mathbf{K}\|.$$

Thus, applying Theorem 5.4.1 of [Ver18], we obtain

$$\mathbb{P} \left(\left\| \sum_{i=1}^N \mathbf{H}_i - \mathbb{E}\mathbf{H}_i \right\| > t \right) \leq 2n \exp \left(-\frac{t^2/2}{v + at/3} \right),$$

where

$$v \leq \frac{C_\sigma^2 n(1+B)^{2C_\sigma}}{N} \|\mathbf{K}\|, \quad a = \frac{2C_\sigma^2 n(1+B)^{2C_\sigma}}{N}.$$

Take $B = C' \sqrt{\log(nN)}$ implies with probability at least $1 - N^{-2}$,

$$\|K_N - K\| \leq C \sqrt{\frac{n}{N}} \log^{C_\sigma}(nN) \sqrt{\log n} \sqrt{\|\mathbf{K}\|} + C \log n \log^{C_\sigma}(nN) \frac{n}{N}.$$

□

Beyond the conjugate kernels, Other types of kernel random matrix concentration can be found in [DWY21, MZ22, MMM22, KRM25].

4.4 Spectrum of conjugate kernel in the proportional limit

When N, n, d are proportional, the limiting spectral distribution of K_N can be obtained in various setting depending on the data distribution in $\mathbf{X}$ and weight distribution in W [LLC18, BP21]. We present the result from [FW20] for the limiting spectrum of the conjugate kernel in a multilayer setting.

Fully connected network and conjugate kernels Consider a fully connected feedforward neural network with input dimension d_0 , hidden widths $d_1, \dots, d_L$, and scalar output

$$f_\theta(x) = w^\top \frac{1}{\sqrt{d_L}} \sigma \left(W_L \frac{1}{\sqrt{d_{L-1}}} \sigma \left(\dots \frac{1}{\sqrt{d_1}} \sigma(W_1 x) \right) \right),$$

where $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is applied entrywise, and

$$W_\ell \in \mathbb{R}^{d_\ell \times d_{\ell-1}}, \quad w \in \mathbb{R}^{d_L}.$$

Given training samples $x_1, \dots, x_n \in \mathbb{R}^{d_0}$, define

$$X_0 = X = (x_1, \dots, x_n) \in \mathbb{R}^{d_0 \times n}, \quad X_\ell = \frac{1}{\sqrt{d_\ell}} \sigma(W_\ell X_{\ell-1}) \in \mathbb{R}^{d_\ell \times n}, \quad 1 \leq \ell \leq L.$$

The conjugate kernel at layer ℓ is

$$K_\ell := X_\ell^\top X_\ell \in \mathbb{R}^{n \times n},$$

and the final conjugate kernel is

$$K^{\text{CK}} := K_L = X_L^\top X_L.$$

Asymptotic regime Fix the depth L . Let

$$n, d_0, d_1, \dots, d_L \rightarrow \infty, \quad \frac{n}{d_\ell} \rightarrow \gamma_\ell \in (0, \infty), \quad 1 \leq \ell \leq L.$$

Assume also that the empirical spectral distribution of the input Gram matrix converges:

$$\lim \text{spec}(X^\top X) = \mu_0.$$

Approximate orthogonality A matrix $X = (x_1, \dots, x_n) \in \mathbb{R}^{d \times n}$ is (ε, B) -orthonormal if

$$|\|x_\alpha\|^2 - 1| \leq \varepsilon, \quad |x_\alpha^\top x_\beta| \leq \varepsilon \quad (\alpha \neq \beta), \quad \|X\| \leq B,$$

and

$$\sum_{\alpha=1}^n (\|x_\alpha\|^2 - 1)^2 \leq B^2.$$

Assume that the input matrix X is (ε_n, B) -orthonormal with $\varepsilon_n n^{1/4} \rightarrow 0$.

Activation assumptions Assume the weights are i.i.d. $N(0, 1)$, and the activation σ is twice differentiable with bounded first and second derivatives. For $\xi \sim N(0, 1)$, assume

$$\mathbb{E}[\sigma(\xi)] = 0, \quad \mathbb{E}[\sigma(\xi)^2] = 1.$$

Define $b_\sigma := \mathbb{E}[\sigma'(\xi)]$.

Marčenko–Pastur map For a probability measure μ on $[0, \infty)$ and aspect ratio $\gamma > 0$, let $\rho_\gamma^{\text{MP}}[\mu]$ denote the Marčenko–Pastur map of μ , namely the probability law whose Stieltjes transform m solves

$$m(z) = \int \frac{1}{x(1 - \gamma - \gamma z m(z)) - z} d\mu(x).$$

This is also the free multiplicative convolution [NS06] of the two laws denoted by $\mu_{MP, \gamma} \boxtimes \mu$. This appears in the setting of the limiting spectral distribution of sample covariance matrices with non-isotropic vectors [BZ08].

Recursive CK limit Define probability measures $\mu_1, \dots, \mu_L$ recursively by

$$\mu_\ell = \rho_{\gamma_\ell}^{\text{MP}}[(1 - b_\sigma^2) + b_\sigma^2 \cdot \mu_{\ell-1}], \quad 1 \leq \ell \leq L,$$

where $(1 - b_\sigma^2) + b_\sigma^2 \cdot \mu$ denotes the pushforward of μ under the map $x \mapsto (1 - b_\sigma^2) + b_\sigma^2 x$.

Theorem 29 (Limiting spectrum for the conjugate kernel). *Under the assumptions above, for each fixed layer $\ell = 1, \dots, L$,*

$$\lim \text{spec}(K_\ell) = \mu_\ell.$$

In particular,

$$\lim \text{spec}(K^{\text{CK}}) = \mu_L.$$

Moreover, $\|K^{\text{CK}}\|$ is almost surely bounded uniformly for all large n .

- The CK spectrum evolves layer by layer by first applying an affine transformation

$$x \mapsto (1 - b_\sigma^2) + b_\sigma^2 x$$

to the previous layer spectrum, and then applying a Marčenko–Pastur map with aspect ratio γ_ℓ . The limiting distribution only depends on the first-order information of σ .

- If $b_\sigma = 0$, then

$$\mu_\ell = \rho_{\gamma_\ell}^{\text{MP}} \quad \text{for every } \ell,$$

so in this case the limiting CK spectrum no longer depends on the input spectral law μ_0 .

Proof sketch for $L = 1$

Proof sketch. Write the rows of WX_0 as $w_1^\top X_0, \dots, w_{d_1}^\top X_0$, where $w_1, \dots, w_{d_1} \stackrel{\text{i.i.d.}}{\sim} N(0, I_{d_0})$. Then the rows of X_1 are i.i.d. conditional on X_0 , and

$$K_1 = \frac{1}{d_1} \sum_{a=1}^{d_1} z_a z_a^\top, \quad z_a := \sigma(w_a^\top X_0) \in \mathbb{R}^n.$$

Thus K_1 is a sample covariance matrix built from i.i.d. rows with population second-moment matrix

$$\Phi := \mathbb{E}[z_a z_a^\top \mid X_0] = \mathbb{E}[\sigma(w^\top X_0)^\top \sigma(w^\top X_0) \mid X_0].$$

Step 1: identify the effective population matrix. The key claim is that Φ is asymptotically close to

$$\tilde{\Phi} := (1 - b_\sigma^2)I_n + b_\sigma^2 X_0^\top X_0.$$

Indeed, for the diagonal entries, the normalization $\mathbb{E}[\sigma(\xi)^2] = 1$ gives

$$\Phi_{\alpha\alpha} \approx 1.$$

For the off-diagonal entries, when $x_\alpha^\top x_\beta$ is small, Gaussian integration by parts (or a first-order Hermite expansion) gives

$$\mathbb{E}[\sigma(w^\top x_\alpha) \sigma(w^\top x_\beta)] = b_\sigma^2 x_\alpha^\top x_\beta + o(1).$$

Under the approximate pairwise orthogonality assumption on the columns of X_0 , this yields

$$\frac{1}{n} \|\Phi - \tilde{\Phi}\|_F^2 \rightarrow 0.$$

Hence Φ and $\tilde{\Phi}$ have the same limiting empirical spectral distribution, namely

$$\text{ESD}(\Phi) \Rightarrow \nu,$$

where ν is the pushforward of μ_0 by

$$x \mapsto (1 - b_\sigma^2) + b_\sigma^2 x.$$

Step 2: reduce to a sample covariance problem. Conditional on X_0 , the matrix K_1 is the sample covariance matrix of d_1 i.i.d. vectors with population covariance Φ . Therefore its limiting spectrum should be given by the Marčenko–Pastur transform of the limiting spectrum of Φ (see [BZ08]).

Step 3: resolvent equation. Let

$$m_{K_1}(z) := \frac{1}{n} \text{Tr}(K_1 - zI)^{-1}, \quad z \in \mathbb{C}^+.$$

The standard resolvent analysis for sample covariance matrices shows that any subsequential limit $m_1(z)$ must satisfy

$$m_1(z) = \int \frac{1}{x(1 - \gamma - \gamma z m_1(z)) - z} d\nu(x).$$

This is exactly the Marčenko–Pastur fixed-point equation with aspect ratio γ and population law ν .

Step 4: conclusion. Since the fixed-point equation has a unique solution in $\mathbb{C}^+$, the whole sequence converges, and

$$\text{ESD}(K_1) \Rightarrow \rho_\gamma^{\text{MP}}[\nu].$$

Substituting the definition of ν gives

$$\text{ESD}(K_1) \Rightarrow \rho_\gamma^{\text{MP}}[(1 - b_\sigma^2) + b_\sigma^2 \cdot \mu_0].$$

□

Multiple layers The case of multiple layers follows by induction on the layer index. Suppose that for some $\ell - 1 \geq 0$, the empirical spectral distribution of

$$K_{\ell-1} := X_{\ell-1}^\top X_{\ell-1}$$

converges to a probability measure $\mu_{\ell-1}$, and that the columns of $X_{\ell-1}$ remain approximately pairwise orthogonal [FW20, Lemma D.2].

Consider

$$X_\ell = \frac{1}{\sqrt{d_\ell}} \sigma(W_\ell X_{\ell-1}), \quad K_\ell = X_\ell^\top X_\ell.$$

Conditional on $X_{\ell-1}$, the rows of X_ℓ are i.i.d., so K_ℓ is again a sample covariance matrix. Its conditional population second-moment matrix is

$$\Phi_\ell := \mathbb{E} \left[\sigma(w^\top X_{\ell-1})^\top \sigma(w^\top X_{\ell-1}) \mid X_{\ell-1} \right].$$

Exactly as in the one-layer case, one shows that Φ_ℓ is asymptotically close to

$$\tilde{\Phi}_\ell = (1 - b_\sigma^2)I + b_\sigma^2 X_{\ell-1}^\top X_{\ell-1},$$

so Φ_ℓ and $\tilde{\Phi}_\ell$ have the same limiting spectral distribution. Therefore, the limiting law of Φ_ℓ is the pushforward

$$\nu_\ell = (1 - b_\sigma^2) + b_\sigma^2 \cdot \mu_{\ell-1}.$$

Applying the Marčenko–Pastur theorem with aspect ratio $\gamma_\ell = \lim n/d_\ell$, we obtain

$$\mu_\ell = \rho_{\gamma_\ell}^{\text{MP}}[\nu_\ell] = \rho_{\gamma_\ell}^{\text{MP}}[(1 - b_\sigma^2) + b_\sigma^2 \cdot \mu_{\ell-1}].$$

Thus, each layer transforms the previous spectral law by first applying the affine map

$$x \mapsto (1 - b_\sigma^2) + b_\sigma^2 x$$

and then the Marčenko–Pastur map. Iterating this recursion from $\ell = 1$ up to $\ell = L$ gives the limiting spectral distribution of the final conjugate kernel K_L .

Limiting spectral laws of conjugate kernels in other settings [LLC18, BP21]

In [LLC18] Under Lipschitz condition on σ and $\|X\| \leq C$, the limiting spectral distribution of K_1 is given by $\rho_{\gamma_1}^{\text{MP}}(\mu_\Phi)$, where $\Phi = \mathbb{E}[\sigma(w^\top X)^\top \sigma(w^\top X)]$. This can be seen as a more general result of [FW20] for the case $L = 1$, and [FW20] simplifies μ_Φ using approximate orthogonality of the data vectors.

In a different setting, [BP21] assumes W, X are both random with i.i.d. subexponential entries, and σ is an analytic function. A unified theorem for both deterministic and random data X , and for a broad class of activation functions is still open.

4.5 Benign overfitting in linear regression

A classical lesson from statistics is that exactly fitting noisy training data should lead to poor generalization. Modern overparameterized models often violate this intuition. In particular, one may see a *double-descent* risk curve: test error increases near the interpolation threshold and then decreases again as the model becomes even larger.

In linear regression, a stronger question is the following: Can an estimator that interpolates the training data still achieve nearly optimal prediction risk? When the answer is yes, this is called *benign overfitting*. The important point is that benign overfitting is stronger than merely observing a second descent. It does not just say that test risk goes down after interpolation. It says that the interpolating estimator can get close to the Bayes-optimal risk.

Setup Consider the well-specified linear model

$$y = x^\top \theta^* + \xi,$$

where

- $x \in \mathbb{R}^d$ is the covariate vector,
- $\theta^* \in \mathbb{R}^d$ is the true parameter,
- ξ is centered noise with $\mathbb{E}[\xi \mid x] = 0$ and $\text{Var}(\xi) = \sigma^2$.

Let $\Sigma = \mathbb{E}[xx^\top]$ denote the population covariance matrix. Suppose we observe n i.i.d. samples $\{(x_i, y_i)\}_{i=1}^n$. Write the design matrix as

$$X = \begin{bmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{bmatrix} \in \mathbb{R}^{n \times d}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n.$$

We focus on the overparameterized regime $d > n$, where exact interpolation (finding a θ with $X\theta = y$) is typically possible.

Minimum-norm interpolator Among all vectors θ satisfying $X\theta = y$, the most studied choice is the minimum- ℓ_2 -norm interpolator:

$$\hat{\theta}_{\min} = \arg \min_{\theta \in \mathbb{R}^d} \|\theta\|_2 \quad \text{subject to} \quad X\theta = y.$$

Assuming X has full row rank, this solution is

$$\hat{\theta}_{\min} = X^\top (XX^\top)^{-1} y.$$

This is also the ridgeless least-squares solution, i.e. the limit of ridge regression as the regularization parameter tends to 0^+ .

Prediction risk

For a fresh test point (x, y) from the same distribution, the prediction risk of an estimator $\hat{\theta}$ is

$$\mathbb{E}[(y - x^\top \hat{\theta})^2].$$

Since the model is well specified, the Bayes predictor is $x \mapsto x^\top \theta^*$ and its risk is σ^2 . Therefore the *excess prediction risk* is

$$R(\hat{\theta}) := \mathbb{E}_{(x,y)} [(x^\top (\hat{\theta} - \theta^*))^2] = (\hat{\theta} - \theta^*)^\top \Sigma (\hat{\theta} - \theta^*).$$

In this setting, benign overfitting means:

$$X\hat{\theta} = y \quad \text{and} \quad R(\hat{\theta}) \approx 0.$$

So the estimator fits the training data exactly, including noise, but still predicts well.

Assumptions Now we state the assumptions in [BLLT20]:

1. x and y are mean zero.
2. $x = \Sigma^{1/2} z$, and the coordinates of z are independent σ_x^2 -subgaussian random variables, in the sense that for every $\lambda \in \mathbb{R}^d$,

$$\mathbb{E} \exp(\lambda^\top z) \leq \exp\left(\frac{\sigma_x^2 \|\lambda\|^2}{2}\right).$$

3. The conditional noise variance is bounded below by a constant σ^2 :

$$\mathbb{E}[(y - x^\top \theta^*)^2 \mid x] \geq \sigma^2.$$

4. The noise $y - x^\top \theta^*$ is conditionally σ_y^2 -subgaussian given x , meaning that for every $\lambda \in \mathbb{R}$,

$$\mathbb{E} \left[\exp(\lambda(y - x^\top \theta^*)) \mid x \right] \leq \exp\left(\frac{\sigma_y^2 \lambda^2}{2}\right).$$

5. Almost surely, the projection of the data $X \in \mathbb{R}^{n \times d}$ onto the orthogonal complement of any eigenvector of Σ spans a space of dimension n .

The main point is that benign overfitting is characterized by the covariance spectrum, not merely by the parameter dimension. To quantify this, [BLLT20] introduces two effective-rank quantities for the covariance operator Σ , with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$:

$$r_k(\Sigma) = \frac{\sum_{i>k} \lambda_i}{\lambda_{k+1}}, \quad R_k(\Sigma) = \frac{(\sum_{i>k} \lambda_i)^2}{\sum_{i>k} \lambda_i^2}.$$

The central contribution of the paper is a nearly matching upper/lower characterization of benign overfitting for minimum-norm interpolation in linear regression, expressed in terms of the effective-rank structure of the covariance operator.

Theorem 30 (Bartlett–Long–Lugosi–Tsigler [BLLT20]). *For any σ_x , there exist constants $b, c, c_1 > 1$ such that the following holds. Define*

$$k^* = \min\{k \geq 0 : r_k(\Sigma) \geq bn\},$$

with the convention that the minimum of the empty set is ∞ . Suppose $\delta < 1$ and $\log(1/\delta) < n/c$.

1. *If $k^* \geq n/c_1$, then*

$$\mathbb{E}R(\hat{\theta}) \geq \frac{\sigma^2}{c}.$$

2. *Otherwise, with probability at least $1 - \delta$,*

$$R(\hat{\theta}) \leq c\|\theta^*\|^2\|\Sigma\| \max\left\{\sqrt{\frac{r_0(\Sigma)}{n}}, \frac{r_0(\Sigma)}{n}, \sqrt{\frac{\log(1/\delta)}{n}}\right\} + c\log(1/\delta)\sigma_y^2\left(\frac{k^*}{n} + \frac{n}{R_{k^*}(\Sigma)}\right).$$

Also,

$$\mathbb{E}R(\hat{\theta}) \geq \frac{\sigma^2}{c}\left(\frac{k^*}{n} + \frac{n}{R_{k^*}(\Sigma)}\right).$$

3. *Moreover, there exist universal constants a_1, a_2, n_0 such that for all $n \geq n_0$, for all Σ , and for all $t \geq 0$, there exists θ^* with $\|\theta^*\| = t$ such that for $x \sim N(0, \Sigma)$ and*

$$y \mid x \sim N(x^\top \theta^*, \|\theta^*\|^2 \|\Sigma\|),$$

with probability at least $1/4$,

$$R(\hat{\theta}) \geq \frac{1}{a_1} \|\theta^*\|^2 \|\Sigma\| \mathbf{1}\left\{\frac{r_0(\Sigma)}{n \log(1 + r_0(\Sigma))} \geq a_2\right\}.$$

Main finite-sample result Let

$$k^* = \min\{k \geq 0 : r_k(\Sigma) \geq bn\},$$

for a universal constant $b > 0$. If k^* is too large (specifically, $k^* \gtrsim n$), then the expected excess risk of the minimum-norm interpolator is bounded below by a constant multiple of the noise variance, so benign overfitting cannot occur. Otherwise, the excess risk is bounded above by two terms: a “signal” term depending on $r_0(\Sigma)$, and a “noise-fitting” term of order

$$\frac{k^*}{n} + \frac{n}{R_{k^*}(\Sigma)}.$$

Moreover, the lower bound shows that this second term is essentially sharp. Hence benign overfitting requires

$$\frac{r_0(\Sigma)}{n} \rightarrow 0, \quad \frac{k^*}{n} \rightarrow 0, \quad \frac{n}{R_{k^*}(\Sigma)} \rightarrow 0.$$

In words, one needs substantial overparameterization together with many low-variance directions where the label noise can be spread out harmlessly.

The result therefore shows that overparameterization is necessary but not sufficient. What matters is a heavy enough spectral tail: the covariance must have many directions that are individually unimportant for prediction but collectively numerous enough to absorb the training noise.

Example 1 (A sharp example in infinite dimension, Theorem 2 in [BLLT20]). Two examples are given for benign overfitting:

- If

$$\mu_k(\Sigma) = k^{-\alpha} \log^{-\beta}(k+1),$$

then benign overfitting occurs if and only if $\alpha = 1, \beta > 1$. Thus, in a fixed infinite-dimensional problem, benign overfitting is only possible in a narrow spectral regime.

- For a growing finite-dimensional model with eigenvalues

$$\mu_k(\Sigma_n) = \begin{cases} \gamma_k + \epsilon_n, & \text{if } k \leq p_n, \\ 0, & \text{otherwise,} \end{cases}$$

and $\gamma_k = \Theta(\exp(-k/\tau))$, then Σ_n with $\|\Sigma_n\| = 1$ is benign if and only if

$$p_n = \omega(n) \quad \text{and} \quad ne^{-o(n)} = \epsilon_n p_n = o(n).$$

Furthermore, for $p_n = \Omega(n)$ and $\epsilon_n p_n = ne^{-o(n)}$,

$$R(\hat{\theta}) = O\left(\frac{\epsilon_n p_n + 1}{n} + \frac{\ln(n/(\epsilon_n p_n))}{n} + \max\left\{\frac{1}{n}, \frac{n}{p_n}\right\}\right).$$

So benign overfitting is much easier to realize in a large finite-dimensional model with a broad low-energy tail than in a fixed infinite-dimensional model.

Proof of the upper bound

Step 1: Bias and variance decomposition

Lemma 34 (exact risk decomposition). *The excess risk of the minimum-norm estimator satisfies*

$$R(\hat{\theta}) = \mathbb{E}_x(x^\top(\theta^* - \hat{\theta}))^2 \leq 2\theta^{*\top}B\theta^* + 2\varepsilon^\top C\varepsilon,$$

where

$$B = (I - X^\top(XX^\top)^{-1}X)\Sigma(I - X^\top(XX^\top)^{-1}X),$$

$$C = (XX^\top)^{-1}X\Sigma X^\top(XX^\top)^{-1}.$$

Then, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the noise vector ε (conditional on X),

$$R(\hat{\theta}) \leq 2\theta^{*\top}B\theta^* + c\sigma_y^2 \log\left(\frac{1}{\delta}\right) \text{tr}(C).$$

Proof. Let

$$\hat{\theta} = X^\top(XX^\top)^{-1}y, \quad \varepsilon := y - X\theta^*,$$

and define

$$P_X := X^\top(XX^\top)^{-1}X,$$

$$B = (I - P_X)\Sigma(I - P_X), \quad C = (XX^\top)^{-1}X\Sigma X^\top(XX^\top)^{-1}.$$

Write

$$y = X\theta^* + \varepsilon,$$

so

$$\hat{\theta} = X^\top(XX^\top)^{-1}(X\theta^* + \varepsilon) = P_X\theta^* + X^\top(XX^\top)^{-1}\varepsilon.$$

Hence

$$\hat{\theta} - \theta^* = -(I - P_X)\theta^* + X^\top(XX^\top)^{-1}\varepsilon.$$

Let

$$a := (I - P_X)\theta^*, \quad b := X^\top(XX^\top)^{-1}\varepsilon.$$

Then

$$R(\hat{\theta}) = (a - b)^\top \Sigma(a - b) \leq 2a^\top \Sigma a + 2b^\top \Sigma b.$$

Note that

$$a^\top \Sigma a = \theta^{*\top}(I - P_X)\Sigma(I - P_X)\theta^* = \theta^{*\top}B\theta^*,$$

and

$$b^\top \Sigma b = \varepsilon^\top(XX^\top)^{-1}X\Sigma X^\top(XX^\top)^{-1}\varepsilon = \varepsilon^\top C\varepsilon.$$

Therefore

$$R(\hat{\theta}) \leq 2\theta^{*\top}B\theta^* + 2\varepsilon^\top C\varepsilon.$$

Now condition on X . Then $C \succeq 0$ is deterministic and, by the sub-gaussian assumption on ε , the Hanson-Wright inequality [RV13] implies that for any $t > 0$, with probability at least $1 - e^{-t}$,

$$\varepsilon^\top C\varepsilon \leq \sigma_y^2 \text{tr}(C) + 2\sigma_y^2 \|C\|t + 2\sigma_y^2 \sqrt{\|C\|^2 t^2 + \text{tr}(C^2)t}.$$

Since $C \succeq 0$,

$$\|C\| \leq \text{tr}(C), \quad \text{tr}(C^2) \leq \text{tr}(C)^2,$$

and hence

$$\varepsilon^\top C \varepsilon \leq c \sigma_y^2 t \text{tr}(C)$$

for a universal constant c (assuming $t \geq 1$). Taking $t = \log(1/\delta)$ yields, with probability at least $1 - \delta$,

$$\varepsilon^\top C \varepsilon \leq c \sigma_y^2 \log\left(\frac{1}{\delta}\right) \text{tr}(C).$$

□

Step 2: Control the signal-distortion term $\theta^{*\top} B \theta^*$.

Lemma 35 (upper bound on the B -term). *There exists a constant c , depending only on σ_x , such that for any $1 < t < n$, with probability at least $1 - e^{-t}$,*

$$\theta^{*\top} B \theta^* \leq c \|\theta^*\|^2 \|\Sigma\| \max\left\{\sqrt{\frac{r_0(\Sigma)}{n}}, \frac{r_0(\Sigma)}{n}, \sqrt{\frac{t}{n}}\right\}.$$

Proof. Recall

$$P_X = X^\top (X X^\top)^{-1} X, \quad B = (I - P_X) \Sigma (I - P_X).$$

First observe that

$$(I - P_X) X^\top = 0, \quad (I - P_X) X^\top X (I - P_X) = 0.$$

We may write

$$B = (I - P_X) \left(\Sigma - \frac{1}{n} X^\top X \right) (I - P_X).$$

Let $u := (I - P_X) \theta^*$. We have

$$\theta^{*\top} B \theta^* = u^\top \left(\Sigma - \frac{1}{n} X^\top X \right) u.$$

Taking absolute values and using the operator norm,

$$\theta^{*\top} B \theta^* \leq \left\| \Sigma - \frac{1}{n} X^\top X \right\| \cdot \|u\|^2 \leq \left\| \Sigma - \frac{1}{n} X^\top X \right\| \cdot \|\theta^*\|^2. \quad (30)$$

Thus it suffices to control the deviation

$$\left\| \frac{1}{n} X^\top X - \Sigma \right\|.$$

By assumption, $x = \Sigma^{1/2} z$, where z has independent σ_x^2 -subgaussian coordinates. Applying [KL17], we obtain with probability $1 - e^{-t}$ for $1 < t < n$,

$$\left\| \frac{1}{n} X^\top X - \Sigma \right\| \leq c \|\Sigma\| \max\left\{\sqrt{\frac{r_0(\Sigma)}{n}}, \frac{r_0(\Sigma)}{n}, \sqrt{\frac{t}{n}}\right\}.$$

Substituting this into (30) yields

$$\theta^{*\top} B \theta^* \leq c \|\theta^*\|^2 \|\Sigma\| \max\left\{\sqrt{\frac{r_0(\Sigma)}{n}}, \frac{r_0(\Sigma)}{n}, \sqrt{\frac{t}{n}}\right\},$$

which completes the proof. □

Step 3: Control the noise term

Lemma 36 (representation of $\text{tr}(C)$ using independent subgaussian vectors). *Assume $\lambda_i = \mu_i(\Sigma)$ and $\lambda_n > 0$. Write the spectral decomposition*

$$\Sigma = \sum_j \lambda_j v_j v_j^\top,$$

where the v_j are orthonormal eigenvectors. For each i with $\lambda_i > 0$, define $z_i := \frac{Xv_i}{\sqrt{\lambda_i}}$. Then

$$\text{tr}(C) = \sum_i \left[\lambda_i^2 z_i^\top \left(\sum_j \lambda_j z_j z_j^\top \right)^{-2} z_i \right],$$

and the vectors $z_i \in \mathbb{R}^n$ are independent and σ_x^2 -subgaussian. Moreover, for every i with $\lambda_i > 0$,

$$\lambda_i^2 z_i^\top \left(\sum_j \lambda_j z_j z_j^\top \right)^{-2} z_i = \frac{\lambda_i^2 z_i^\top A_{-i}^{-2} z_i}{(1 + \lambda_i z_i^\top A_{-i}^{-1} z_i)^2},$$

where

$$A_{-i} := \sum_{j \neq i} \lambda_j z_j z_j^\top.$$

Proof. Recall that

$$C = (XX^\top)^{-1} X \Sigma X^\top (XX^\top)^{-1}.$$

Since

$$\Sigma = \sum_j \lambda_j v_j v_j^\top,$$

we have

$$X \Sigma X^\top = \sum_j \lambda_j X v_j v_j^\top X^\top.$$

Using the definition $Xv_j = \sqrt{\lambda_j} z_j$, it follows that

$$X \Sigma X^\top = \sum_j \lambda_j (\sqrt{\lambda_j} z_j) (\sqrt{\lambda_j} z_j)^\top = \sum_j \lambda_j^2 z_j z_j^\top.$$

Likewise, since $I = \sum_j v_j v_j^\top$,

$$XX^\top = X I X^\top = \sum_j X v_j v_j^\top X^\top = \sum_j (\sqrt{\lambda_j} z_j) (\sqrt{\lambda_j} z_j)^\top = \sum_j \lambda_j z_j z_j^\top.$$

Let

$$A := XX^\top = \sum_j \lambda_j z_j z_j^\top.$$

Then

$$C = A^{-1} (X \Sigma X^\top) A^{-1}.$$

And

$$C = A^{-1} \left(\sum_j \lambda_j^2 z_j z_j^\top \right) A^{-1} = \sum_j \lambda_j^2 A^{-1} z_j z_j^\top A^{-1}.$$

Taking traces, we obtain

$$\text{tr}(C) = \sum_j \lambda_j^2 \text{tr}(A^{-1} z_j z_j^\top A^{-1}) = \sum_j \lambda_j^2 z_j^\top A^{-2} z_j.$$

Since $A = \sum_j \lambda_j z_j z_j^\top$, this is exactly

$$\text{tr}(C) = \sum_j \left[\lambda_j^2 z_j^\top \left(\sum_\ell \lambda_\ell z_\ell z_\ell^\top \right)^{-2} z_j \right].$$

Write the m -th sample point as x_m , so the m -th coordinate of z_i is

$$(z_i)_m = \frac{\langle x_m, v_i \rangle}{\sqrt{\lambda_i}}.$$

By the model assumption,

$$x_m = V \Lambda^{1/2} \xi_m,$$

where the coordinates of ξ_m in the eigenbasis $\{v_j\}$ are independent σ_x^2 -subgaussian random variables. Therefore

$$\langle x_m, v_i \rangle = \sqrt{\lambda_i} (\xi_m)_i, \quad \text{so} \quad (z_i)_m = (\xi_m)_i.$$

Hence z_i is exactly the vector of the i -th coordinates of the n i.i.d. latent vectors $\xi_1, \dots, \xi_n$. Since the coordinates $(\xi_m)_i$ are independent across both m and i , the vectors $z_i \in \mathbb{R}^n$ are independent.

Fix i and write

$$A = A_{-i} + \lambda_i z_i z_i^\top, \quad A_{-i} := \sum_{j \neq i} \lambda_j z_j z_j^\top.$$

Since $\lambda_n > 0$, the matrix A_{-i} is invertible almost surely under the Assumption 5. Apply the Sherman–Morrison formula (Lemma 6):

$$(A_{-i} + \lambda_i z_i z_i^\top)^{-1} = A_{-i}^{-1} - \frac{\lambda_i A_{-i}^{-1} z_i z_i^\top A_{-i}^{-1}}{1 + \lambda_i z_i^\top A_{-i}^{-1} z_i}.$$

Multiplying this identity on the left by $z_i^\top$ and on the right by z_i yields

$$z_i^\top A^{-1} z_i = z_i^\top A_{-i}^{-1} z_i - \frac{\lambda_i (z_i^\top A_{-i}^{-1} z_i)^2}{1 + \lambda_i z_i^\top A_{-i}^{-1} z_i} = \frac{z_i^\top A_{-i}^{-1} z_i}{1 + \lambda_i z_i^\top A_{-i}^{-1} z_i}.$$

Now use the rank-one update identity in the form

$$A^{-1} z_i = \frac{A_{-i}^{-1} z_i}{1 + \lambda_i z_i^\top A_{-i}^{-1} z_i},$$

which follows immediately from the Sherman–Morrison formula above. Therefore

$$z_i^\top A^{-2} z_i = (A^{-1} z_i)^\top (A^{-1} z_i) = \frac{z_i^\top A_{-i}^{-2} z_i}{(1 + \lambda_i z_i^\top A_{-i}^{-1} z_i)^2}.$$

Multiplying by λ_i^2 gives

$$\lambda_i^2 z_i^\top A^{-2} z_i = \frac{\lambda_i^2 z_i^\top A_{-i}^{-2} z_i}{(1 + \lambda_i z_i^\top A_{-i}^{-1} z_i)^2},$$

which is the desired identity. $\square$

The lemma above gives the identity

$$\boxed{\text{tr}(C) = \sum_i \lambda_i^2 z_i^\top A^{-2} z_i = \sum_i \frac{\lambda_i^2 z_i^\top A_{-i}^{-2} z_i}{(1 + \lambda_i z_i^\top A_{-i}^{-1} z_i)^2}.$$

Fix an integer l . Splitting the sum at l gives

$$\text{tr}(C) = \sum_{i=1}^l \frac{\lambda_i^2 z_i^\top A_{-i}^{-2} z_i}{(1 + \lambda_i z_i^\top A_{-i}^{-1} z_i)^2} + \sum_{i>l} \lambda_i^2 z_i^\top A^{-2} z_i. \quad (3)$$

The first sum is the *head* contribution; the second is the *tail* contribution³. We will take $\ell = k^*$.

Step 4: Some spectral bounds

Lemma 37 (Spectral bounds for A_k). *Let $A_k := \sum_{j>k} \lambda_j z_j z_j^\top$. There exists a constant $c \geq 1$, depending only on σ_x , such that for any $k \geq 0$, with probability at least $1 - 2e^{-n/c}$,*

$$\frac{1}{c} \sum_{j>k} \lambda_j - c \lambda_{k+1} n \leq \mu_n(A_k) \leq \mu_1(A_k) \leq c \left(\sum_{j>k} \lambda_j + \lambda_{k+1} n \right),$$

where $\mu_1(\cdot)$, $\mu_n(\cdot)$ denote the largest and smallest eigenvalues, respectively.

Proof. This can be proved by a standard ε -net argument for bounding $\|A_k - (\sum_{j>k} \lambda_j) I\|$. $\square$

Lemma 38 (spectral control of A , A_{-i} , and A_k). *There are constants $b, c \geq 1$ such that for any $k \geq 0$, with probability at least $1 - 2e^{-n/c}$,*

1. for all $i \geq 1$,

$$\mu_{k+1}(A_{-i}) \leq \mu_{k+1}(A) \leq \mu_1(A_k) \leq c \left(\sum_{j>k} \lambda_j + \lambda_{k+1} n \right);$$

2. for all $1 \leq i \leq k$,

$$\mu_n(A) \geq \mu_n(A_{-i}) \geq \mu_n(A_k) \geq \frac{1}{c} \sum_{j>k} \lambda_j - c \lambda_{k+1} n;$$

³The idea of the head-tail decomposition of the covariance matrix was also used in [LWK⁺24, PXZ26].

3. if $r_k(\Sigma) \geq bn$, then

$$\frac{1}{c}\lambda_{k+1}r_k(\Sigma) \leq \mu_n(A_k) \leq \mu_1(A_k) \leq c\lambda_{k+1}r_k(\Sigma).$$

Proof. We work on this event of Lemma 37 throughout.

(1) Since

$$A - A_k = \sum_{j=1}^k \lambda_j z_j z_j^\top,$$

the matrix $A - A_k$ is a sum of at most k rank-one matrices, hence

$$\text{rank}(A - A_k) \leq k.$$

Therefore there exists a subspace $L \subseteq \mathbb{R}^n$ with

$$\dim(L) \geq n - k$$

such that

$$(A - A_k)v = 0 \quad \text{for all } v \in L.$$

Hence for every $v \in L$,

$$v^\top A v = v^\top A_k v \leq \mu_1(A_k) \|v\|^2.$$

By the Courant–Fischer variational characterization of eigenvalues,

$$\mu_{k+1}(A) = \min_{\dim(L) \geq n-k} \max_{v \in L \setminus \{0\}} \frac{v^\top A v}{\|v\|^2} \leq \mu_1(A_k).$$

Next, for every $i \geq 1$,

$$A = A_{-i} + \lambda_i z_i z_i^\top,$$

with $\lambda_i z_i z_i^\top \succeq 0$. Hence $A_{-i} \preceq A$. By monotonicity of eigenvalues under the Loewner order,

$$\mu_{k+1}(A_{-i}) \leq \mu_{k+1}(A).$$

This proves the first claim.

(2) Similarly, the second claim holds due to Lemma 37.

(3) Assume now that

$$r_k(\Sigma) = \frac{\sum_{j>k} \lambda_j}{\lambda_{k+1}} \geq bn.$$

Then

$$\sum_{j>k} \lambda_j = \lambda_{k+1} r_k(\Sigma).$$

Substituting this into the upper bound in (1) gives

$$\mu_1(A_k) \leq c_1(\lambda_{k+1} r_k(\Sigma) + \lambda_{k+1} n) = c_1 \lambda_{k+1} r_k(\Sigma) \left(1 + \frac{n}{r_k(\Sigma)}\right).$$

Since $r_k(\Sigma) \geq bn$,

$$\frac{n}{r_k(\Sigma)} \leq \frac{1}{b},$$

and therefore

$$\mu_1(A_k) \leq c_1 \left(1 + \frac{1}{b}\right) \lambda_{k+1} r_k(\Sigma). \quad (35)$$

Similarly, the lower bound in Claim 2 becomes

$$\mu_n(A_k) \geq \frac{1}{c_1} \lambda_{k+1} r_k(\Sigma) - c_1 \lambda_{k+1} n = \lambda_{k+1} r_k(\Sigma) \left(\frac{1}{c_1} - c_1 \frac{n}{r_k(\Sigma)} \right).$$

Again using $r_k(\Sigma) \geq bn$, we get

$$\mu_n(A_k) \geq \lambda_{k+1} r_k(\Sigma) \left(\frac{1}{c_1} - \frac{c_1}{b} \right). \quad (36)$$

Choose $b > c_1^2$. Then, after enlarging the constant c if necessary, (35) and (36) imply

$$\frac{1}{c} \lambda_{k+1} r_k(\Sigma) \leq \mu_n(A_k) \leq \mu_1(A_k) \leq c \lambda_{k+1} r_k(\Sigma).$$

This is exactly Claim 3. $\square$

Lemma 39 (norm and projected norm of a subgaussian vector). *Suppose $z \in \mathbb{R}^n$ is a centered random vector with independent σ^2 -subgaussian coordinates and unit variances, $\mathcal{L}$ is a random subspace of $\mathbb{R}^n$ of codimension k , and $\mathcal{L}$ is independent of z . Then there exists a universal constant a such that for every $t > 0$, with probability at least $1 - 3e^{-t}$,*

$$\|z\|^2 \leq n + a\sigma^2(t + \sqrt{nt}),$$

$$\|\Pi_{\mathcal{L}} z\|^2 \geq n - a\sigma^2(k + t + \sqrt{nt}),$$

where $\Pi_{\mathcal{L}}$ is the orthogonal projection onto $\mathcal{L}$.

Proof. This is an application of Bernstein's inequality and Hanson-Wright inequality [Ver18]. $\square$

With the spectral bounds above, we are ready to analyze the head-tail decomposition.

Step 5: Analyzing the head and tail parts

Bound the head part in (3) For each $1 \leq i \leq l$, let $\mathcal{L}_i$ be the span of the $n - l$ eigenvectors of A_{-i} corresponding to its smallest $n - l$ eigenvalues. By Lemma 38,

$$z_i^\top A_{-i}^{-2} z_i \leq c \frac{\|z_i\|^2}{(\lambda_{l+1} r_l(\Sigma))^2}, \quad z_i^\top A_{-i}^{-1} z_i \geq \frac{1}{\mu_{l+1}(A_{-i})} \|\Pi_{\mathcal{L}_i} z_i\|^2 \geq c^{-1} \frac{\|\Pi_{\mathcal{L}_i} z_i\|^2}{\lambda_{l+1} r_l(\Sigma)}.$$

Hence

$$\frac{\lambda_i^2 z_i^\top A_{-i}^{-2} z_i}{(1 + \lambda_i z_i^\top A_{-i}^{-1} z_i)^2} \leq c \frac{\|z_i\|^2}{\|\Pi_{\mathcal{L}_i} z_i\|^4}. \quad (4)$$

Now apply Lemma 39 to each pair $(z_i, \mathcal{L}_i)$ and use a union bound. Because $\mathcal{L}_i$ depends only on $\{z_j : j \neq i\}$, it is independent of z_i . Thus, with high probability,

$$\|z_i\|^2 \lesssim n, \quad \|\Pi_{\mathcal{L}_i} z_i\|^2 \gtrsim n, \quad 1 \leq i \leq l.$$

Substituting into (4) and summing over $i \leq l$ yields

$$\sum_{i=1}^l \frac{\lambda_i^2 z_i^\top A_{-i}^{-2} z_i}{(1 + \lambda_i z_i^\top A_{-i}^{-1} z_i)^2} \leq c \frac{l}{n}.$$

Bound the tail part in (3). Again by Lemma 38,

$$\mu_n(A) \gtrsim \lambda_{l+1} r_l(\Sigma),$$

so

$$\sum_{i>l} \lambda_i^2 z_i^\top A^{-2} z_i \leq c \frac{\sum_{i>l} \lambda_i^2 \|z_i\|^2}{(\lambda_{l+1} r_l(\Sigma))^2}. \quad (6)$$

Now use Lemma 39 on the weighted sum $\sum_{i>l} \lambda_i^2 (\|z_i\|^2 - n)$. This gives, with high probability,

$$\sum_{i>l} \lambda_i^2 \|z_i\|^2 \leq c n \sum_{i>l} \lambda_i^2.$$

Plugging this into (6) gives

$$\sum_{i>l} \lambda_i^2 z_i^\top A^{-2} z_i \leq c n \frac{\sum_{i>l} \lambda_i^2}{(\lambda_{l+1} r_l(\Sigma))^2} = \frac{cn}{R_l(\Sigma)}.$$

Step 6: Finishing the proof The bias variance decomposition and head-tail decomposition with $l = k^*$ imply that with probability at least $1 - \delta$,

$$R(\hat{\theta}) \leq c \|\theta^*\|^2 \|\Sigma\| \max \left\{ \sqrt{\frac{r_0(\Sigma)}{n}}, \frac{r_0(\Sigma)}{n}, \sqrt{\frac{\log(1/\delta)}{n}} \right\} + c \log\left(\frac{1}{\delta}\right) \sigma_y^2 \left(\frac{k^*}{n} + \frac{n}{R_{k^*}(\Sigma)} \right).$$

This is exactly the upper bound in Theorem 30.

Remark 12. Very roughly, the excess risk can be understood as the sum of:

- a *signal distortion* term, related to how well the sample captures the important directions,
- a *noise fitting* term, related to whether the spectral tail is large and diffuse enough.

Overparameterization helps when the extra directions are mostly low-variance directions that matter little for prediction but provide room to harmlessly store noise.

Uniform convergence and benign overfitting [KZSS21] gives a different proof of benign overfitting under the condition that each x_i is i.i.d. $N(0, \Sigma)$. Their proof explores the convex Gaussian min-max theorem [TOH15] and provides a bound for near-minimal-norm interpolators.

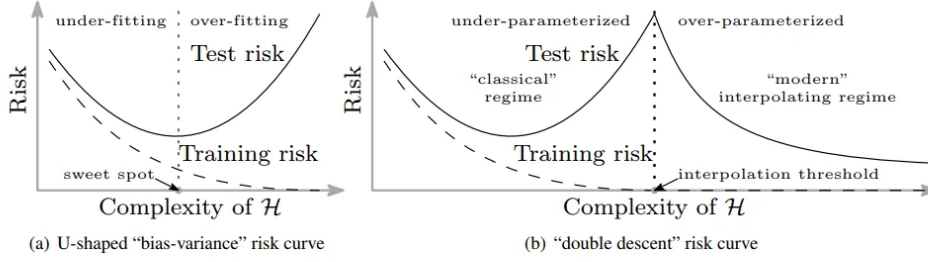


Figure 1: Curves for training risk (dashed line) and test risk (solid line). (a) The classical *U-shaped risk curve* arising from the bias-variance trade-off. (b) The *double descent risk curve*, which incorporates the U-shaped risk curve (i.e., the “classical” regime) together with the observed behavior from using high complexity function classes (i.e., the “modern” interpolating regime), separated by the interpolation threshold. The predictors to the right of the interpolation threshold have zero training risk.

Figure 9: Classical “bias-variance tradeoff” vs modern “double descent” from [BHMM19]

4.6 Double descent

The analysis of linear regression can be viewed as the simplest instance of the student–teacher framework introduced in Section 4.1, in which both the student and the teacher are linear functions. Introducing nonlinearity leads to more expressive models; among these, random feature regression stands out as a tractable and analytically solvable extension.

The double descent phenomenon. In classical statistical learning theory, the generalization error is often understood through the *bias–variance tradeoff*: as model complexity increases, the bias decreases while the variance increases, leading to a U-shaped test error curve. In particular, the optimal performance is expected to occur at an intermediate level of complexity.

However, modern overparameterized models exhibit a striking deviation from this classical picture. As the model complexity continues to increase beyond the interpolation threshold—where the model can fit the training data perfectly—the test error does not necessarily deteriorate. Instead, it often decreases again, giving rise to a *double descent* curve: the test error first decreases, then increases near the interpolation threshold, and finally decreases once more in the highly overparameterized regime. See Figure 9.

Understanding this phenomenon has become a central question in modern high-dimensional statistics and machine learning. Random feature regression [MM19] provides one of the simplest settings in which the double descent behavior can be analyzed precisely, allowing for exact asymptotic characterizations of the generalization error.

Generalization error in random feature regression

Throughout,

$$d, n, N \rightarrow \infty, \quad \frac{N}{d} \rightarrow \psi_1 \in (0, \infty), \quad \frac{n}{d} \rightarrow \psi_2 \in (0, \infty).$$

Consider spherical covariates and spherical random features:

$$x_i \stackrel{\text{iid}}{\sim} \text{Unif}(\mathbb{S}^{d-1}(\sqrt{d})), \quad \theta_i \stackrel{\text{iid}}{\sim} \text{Unif}(\mathbb{S}^{d-1}(\sqrt{d})),$$

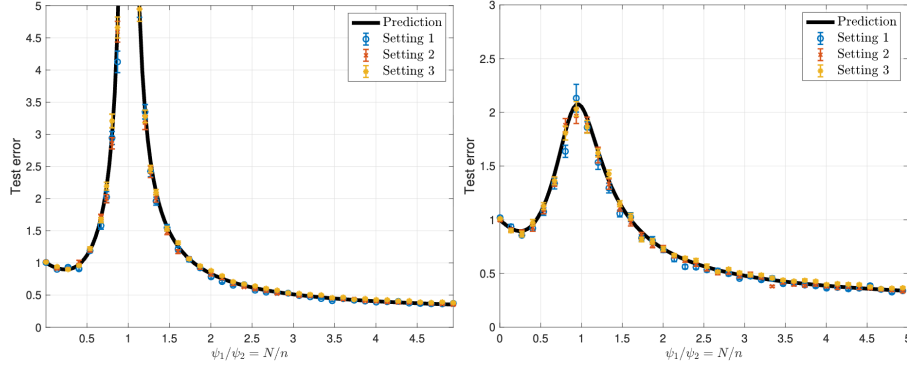


FIGURE 5.1. Random features regression with ReLU activation ($\sigma = \max\{x, 0\}$). Data are generated according to one of three settings: (1) $f_d(\mathbf{x}) = x_1$ and $\mathbb{E}[\varepsilon^2] = 0.5$; (2) $f_d(\mathbf{x}) = x_1 + (x_1^2 - 1)/2$ and $\mathbb{E}[\varepsilon^2] = 0$; (3) $f_d(\mathbf{x}) = x_1 + x_1 x_2 / \sqrt{2}$ and $\mathbb{E}[\varepsilon^2] = 0$. Within any of these settings, the total power of nonlinearity and noise is $F_\star^2 + \tau^2 = 0.5$, while the power of the linear part is $F_1^2 = 1$. Left frame: $\lambda = 10^{-8}$. Right frame: $\lambda = 10^{-3}$. Here $n = 300$, $d = 100$. The continuous black line is our theoretical prediction, and the colored symbols are numerical results. Symbols are averages over 20 instances and the error bars report the standard error of the means over these 20 instances.

Figure 10: Simulation from [MM19]

independently. The random-features predictor is

$$\hat{f}(x) = \sum_{j=1}^N \hat{a}_j \sigma\left(\frac{\langle \theta_j, x \rangle}{\sqrt{d}}\right),$$

where $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is a fixed activation and the coefficients are chosen by ridge regression:

$$\hat{a}_\lambda = \arg \min_{a \in \mathbb{R}^N} \left\{ \frac{1}{n} \sum_{i=1}^n \left(y_i - \sum_{j=1}^N a_j \sigma\left(\frac{\langle \theta_j, x_j \rangle}{\sqrt{d}}\right) \right)^2 + \frac{N\lambda}{d} \|a\|_2^2 \right\}. \quad (37)$$

The test risk is

$$R_{\text{RF}}(f_d, X, \Theta, \lambda) = \mathbb{E}_x \left[(f_d(x) - \hat{f}(x))^2 \mid X, \Theta \right],$$

where the expectation is taken over a fresh sample from $\text{Unif}(\mathbb{S}^{d-1}(\sqrt{d}))$. This is a random quantity depends on the training data X and random weight Θ .

The *teacher model* is decomposed into constant, linear, and (random) nonlinear parts. The data are

$$y_i = f_d(x_i) + \varepsilon_i$$

with $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \tau^2)$, where

$$f_d(x) = \beta_{d,0} + \langle \beta_{d,1}, x \rangle + f_d^{\text{NL}}(x),$$

and $f_d^{\text{NL}}(x)$ is an isotropic Gaussian process indexed by $x \in \text{Unif}(\mathbb{S}^{d-1}(\sqrt{d}))$. This term is to quantify how much the non-linear component of a teacher model can be learned.

Define

$$F_1^2 = \lim_{d \rightarrow \infty} \|\beta_{d,1}\|_2^2, \quad F_*^2 = \lim_{d \rightarrow \infty} \mathbb{E}_x(f_d^{\text{NL}}(x))^2$$

The following parameters for the activation function are needed:

Definition 18 (Activation parameters). Let $G \sim N(0, 1)$ and define

$$\mu_0 = \mathbb{E}[\sigma(G)], \quad \mu_1 = \mathbb{E}[G\sigma(G)], \quad \mu_*^2 = \mathbb{E}[\sigma(G)^2] - \mu_0^2 - \mu_1^2,$$

and

$$\zeta = \frac{\mu_1}{\mu_*}, \quad \bar{\lambda} = \frac{\lambda}{\mu_*^2}.$$

Here μ_*^2 is the sum of squares for all higher-order (≥ 1) Hermite coefficients of σ .

The main result is the following:

Theorem 31 (Main asymptotic risk formula). Fix $\lambda > 0$. For $\xi \in \mathbb{C}_+ = \{z \in \mathbb{C} : \text{Im } z > 0\}$, let $\nu_1(\xi), \nu_2(\xi) \in \mathbb{C}_+$ be the unique solution of

$$\nu_1 = \psi_1 \left(-\xi - \nu_2 - \frac{\zeta^2 \nu_2}{1 - \zeta^2 \nu_1 \nu_2} \right)^{-1}, \quad \nu_2 = \psi_2 \left(-\xi - \nu_1 - \frac{\zeta^2 \nu_1}{1 - \zeta^2 \nu_1 \nu_2} \right)^{-1}.$$

Define

$$\chi = \nu_1(i\sqrt{\psi_1 \psi_2 \bar{\lambda}}) \nu_2(i\sqrt{\psi_1 \psi_2 \bar{\lambda}}).$$

Then

$$\mathbb{E}[R_{\text{RF}}(f_d, X, \Theta, \lambda)] = F_1^2 B(\zeta, \psi_1, \psi_2, \bar{\lambda}) + (F_*^2 + \tau^2) V(\zeta, \psi_1, \psi_2, \bar{\lambda}) + F_*^2 + o_d(1), \quad (38)$$

where B and V are explicit rational functions of χ and the parameters $(\zeta, \psi_1, \psi_2, \bar{\lambda})$.

Equation (38) shows:

- the linear signal term F_1^2 is multiplied by a *bias factor* B ;
- the combination $F_*^2 + \tau^2$ is multiplied by a *variance factor* V ;
- an irreducible term F_*^2 remains.
- The constant power F_0^2 does not appear in the final asymptotic formula. Heuristically, the nonzero mean μ_0 creates a rank-one feature direction aligned with constants, so the constant component is learned essentially perfectly.

Exact ridge solution and risk decomposition Let

$$\tilde{Z}_{ij} = \sigma\left(\frac{\langle x_i, \theta_j \rangle}{\sqrt{d}}\right), \quad Z = \frac{1}{\sqrt{d}} \tilde{Z} \in \mathbb{R}^{n \times N}.$$

Then the normal equations for (37) become

$$(Z^\top Z + \lambda \psi_1 \psi_2 \text{Id}_N) \hat{a}_\lambda = \frac{1}{\sqrt{d}} Z^\top y.$$

Define

$$\Xi = (Z^\top Z + \lambda \psi_1 \psi_2 \text{Id}_N)^{-1}, \quad \Pi = (ZZ^\top + \lambda \psi_1 \psi_2 \text{Id}_n)^{-1}. \quad (39)$$

Then the ridge solution is

$$\hat{a}_\lambda = \frac{1}{\sqrt{d}} \Xi Z^\top y. \quad (40)$$

For a fresh test point x , define the unnormalized feature vector

$$\sigma_\Theta(x) = \left(\sigma\left(\frac{\langle \theta_1, x \rangle}{\sqrt{d}}\right), \dots, \sigma\left(\frac{\langle \theta_N, x \rangle}{\sqrt{d}}\right) \right)^\top \in \mathbb{R}^N.$$

Also define

$$V = \mathbb{E}_x[f_d(x)\sigma_\Theta(x)] \in \mathbb{R}^N, \quad U = \mathbb{E}_x[\sigma_\Theta(x)\sigma_\Theta(x)^\top] \in \mathbb{R}^{N \times N}. \quad (41)$$

Lemma 40 (Exact risk identity). *We have the following identity for the risk:*

$$R_{\text{RF}}(f_d, X, \Theta, \lambda) = \mathbb{E}_x[f_d(x)^2] - \frac{2}{\sqrt{d}} y^\top Z \Xi V + \frac{1}{d} y^\top Z \Xi U \Xi Z^\top y.$$

Proof. Insert (40) into

$$\hat{f}(x) = \sigma_\Theta(x)^\top \hat{a}_\lambda = \frac{1}{\sqrt{d}} \sigma_\Theta(x)^\top \Xi Z^\top y.$$

Expand $(f_d(x) - \hat{f}(x))^2$, average over the fresh test point x , and use (41). $\square$

The rest of the proof is to analyze the asymptotics of quantities related to matrix resolvents in (39).

Lemma 41. *The following holds:*

$$\mathbb{E}[R_{\text{RF}}(f_d, X, \Theta, \lambda)] = F_1^2(1 - 2\Psi_1 + \Psi_2) + (F_\star^2 + \tau^2)\Psi_3 + F_\star^2 + o_d(1).$$

The quantities Ψ_1, Ψ_2, Ψ_3 are defined by:

$$Q = \frac{\Theta\Theta^\top}{d}, \quad H = \frac{XX^\top}{d}, \quad Z_1 = \frac{\mu_1}{d} X\Theta^\top,$$

and

$$\Psi_1 = \frac{1}{d} \text{tr}(Z_1^\top Z \Xi), \quad (42)$$

$$\Psi_2 = \frac{1}{d} \text{tr}\left[\Xi(\mu_1^2 Q + \mu_\star^2 \text{Id}_N) \Xi Z^\top H Z\right],$$

$$\Psi_3 = \frac{1}{d} \text{tr}\left[\Xi(\mu_1^2 Q + \mu_\star^2 \text{Id}_N) \Xi Z^\top Z\right]. \quad (43)$$

Remark 13. Because σ has only a degree-0 part, a degree-1 part, and an orthogonal residual, in the proportional regime, the kernel matrix U from (41) is asymptotically

$$U \approx \mu_0^2 \mathbf{1}\mathbf{1}^\top + \mu_1^2 Q + \mu_\star^2 \text{Id}_N.$$

The constant term explains why degree 0 is learned perfectly; the degree-1 part is detected through Q and H ; and all higher degrees see only the isotropic remainder $\mu_\star^2 \text{Id}_N$. That is why the entire nonlinear target collapses to the single scalar observable Ψ_3 . The paper makes this precise with operator-norm control of the harmonic kernels.

The next step is to compute Ψ_1, Ψ_2, Ψ_3 . The key trick is to place all of them inside one block matrix and differentiate a log-determinant. Define the parameter tuple

$$q = (s_1, s_2, t_1, t_2, p),$$

and set

$$A(q) = \begin{bmatrix} s_1 \text{Id}_N + s_2 Q & Z^\top + p Z_1^\top \\ Z + p Z_1 & t_1 \text{Id}_n + t_2 H \end{bmatrix}.$$

For $\xi \in \mathbb{C}_+$ let

$$G_d(\xi; q) = \frac{1}{d} \log \det(A(q) - \xi \text{Id}_{N+n}). \quad (44)$$

At $q = 0$, if we choose

$$\xi = i\sqrt{\psi_1 \psi_2 \lambda},$$

then

$$A(0) - \xi \text{Id} = \begin{bmatrix} -\xi \text{Id}_N & Z^\top \\ Z & -\xi \text{Id}_n \end{bmatrix}.$$

A direct Schur-complement computation gives

$$(A(0) - \xi \text{Id})^{-1} = \begin{bmatrix} \xi \Xi & \Xi Z^\top \\ Z \Xi & \xi \Pi \end{bmatrix},$$

with Ξ, Π from (39). (Here we use that $-\xi^2 = \psi_1 \psi_2 \lambda$.) Now differentiate (44). Since

$$\partial_p A(q)|_{q=0} = \begin{bmatrix} 0 & Z_1^\top \\ Z_1 & 0 \end{bmatrix},$$

the resolvent identity yields

$$\begin{aligned} \partial_p G_d(\xi; 0) &= \frac{1}{d} \text{tr} \left[(A(0) - \xi \text{Id})^{-1} \partial_p A(0) \right] \\ &= \frac{1}{d} \text{tr}(\Xi Z^\top Z_1) + \frac{1}{d} \text{tr}(\Pi Z Z_1^\top) \\ &= \frac{2}{d} \text{tr}(Z_1^\top Z \Xi) = 2\Psi_1, \end{aligned}$$

because $Z \Xi = \Pi Z$. The mixed second derivatives similarly produce the other trace terms:

$$\begin{aligned} \Psi_1 &= \frac{1}{2} \partial_p G_d(\xi; 0), \\ \Psi_2 &= -\mu_\star^2 \partial_{s_1 t_2} G_d(\xi; 0) - \mu_1^2 \partial_{s_2 t_2} G_d(\xi; 0), \\ \Psi_3 &= -\mu_\star^2 \partial_{s_1 t_1} G_d(\xi; 0) - \mu_1^2 \partial_{s_2 t_1} G_d(\xi; 0). \end{aligned}$$

The observables (42)–(43) look unrelated. The matrix $A(q)$ packages them into a single spectral object. Once we know the limit of $G_d(\xi; q)$, the asymptotic risk follows by differentiation.

Double descent in linear regression Other models with rigorous results on double descent and related phenomena can be found in [BHX20, HMRT22, HL22a, AP20].

We mention here a simpler result for linear regression from [HMRT22]. Consider the following setting:

- Training dataset: $(\mathbf{x}_i, y_i)_{i=1}^n$, $\mathbf{x}_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$.
- A linear model with noise: $y_i = \langle \boldsymbol{\beta}_0, \mathbf{x}_i \rangle + \varepsilon_i$, $\boldsymbol{\beta}_0 \sim N(0, \frac{1}{d}I_d)$.
- $\mathbf{x}_i$ are independent random vectors with distribution $N(0, I_d)$.
- $\varepsilon_i \sim N(0, \sigma^2)$ is the noise: an unobserved random variable that adds "noise" to the linear relationship.
- Training step: find the best $\hat{\boldsymbol{\beta}}$ that minimize the training error

$$L(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n (y_i - \langle \boldsymbol{\beta}, \mathbf{x}_i \rangle)^2 = \frac{1}{n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$$

$$\mathbf{y} = (y_1, \dots, y_n)^\top. \quad \mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{bmatrix} \text{ is an } n \times d \text{ random matrix, each entry is a Gaussian random}$$

variable $N(0, 1)$. Almost surely $\text{rank}(\mathbf{X}) = \min\{n, d\}$.

Definition 19 (pseudo-inverse). For a symmetric matrix $\mathbf{A}$ of rank r , write $\mathbf{A} = \sum_{i=1}^r \lambda_i u_i u_i^\top$. Its pseudo-inverse is defined by

$$\mathbf{A}^\dagger = \sum_{i=1}^r \frac{1}{\lambda_i} u_i u_i^\top.$$

In all cases for d, n , the *minimal norm* solution that minimize $L(\boldsymbol{\beta})$ satisfies

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^\dagger \mathbf{X}^\top \mathbf{y}.$$

The test error can be written as

$$\begin{aligned} \mathcal{R} &= \mathbb{E}_{\mathbf{x}, \boldsymbol{\beta}_0, \varepsilon} [(\underbrace{\langle \mathbf{x}, \hat{\boldsymbol{\beta}} \rangle}_{\text{prediction}} - \underbrace{\langle \mathbf{x}, \boldsymbol{\beta}_0 \rangle}_{\text{truth}})^2] \\ &= \mathbb{E} \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_2^2 \\ &= \underbrace{\frac{1}{d} \text{Tr}(\text{Id} - (\mathbf{X}^\top \mathbf{X})^\dagger \mathbf{X}^\top \mathbf{X})}_{\text{bias: } B} + \underbrace{\sigma^2 \text{Tr}([\mathbf{X}^\top \mathbf{X}]^\dagger)^2 \mathbf{X}^\top \mathbf{X})}_{\text{variance: } V}. \end{aligned}$$

If we consider $n \rightarrow \infty$, $d \rightarrow \infty$ and $\frac{d}{n} \rightarrow \gamma > 0$, explicit calculation on Gaussian random matrices in [HMRT22] implies

$$\lim_{n \rightarrow \infty} B = \begin{cases} 0 & \gamma \in (0, 1) \\ 1 - \frac{1}{\gamma} & \gamma > 1 \end{cases}, \quad \lim_{n \rightarrow \infty} V = \begin{cases} \frac{\sigma^2 \gamma}{1 - \gamma} & \gamma \in (0, 1) \\ \frac{\sigma^2}{\gamma - 1} & \gamma > 1 \end{cases}$$

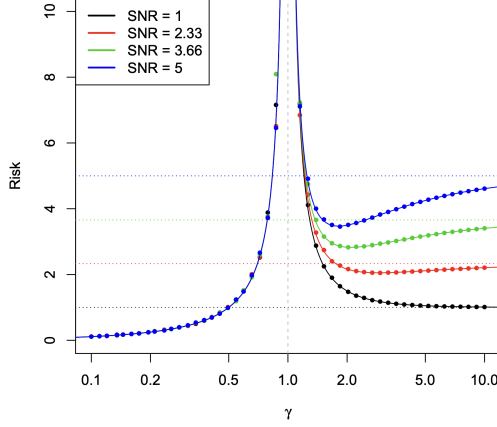


Figure 11: Double descent in linear regression

Therefore $\mathcal{R} = B + V$ has a limit

$$\lim_{n \rightarrow \infty} \mathcal{R} = \begin{cases} \frac{\sigma^2 \gamma}{1-\gamma} & \gamma \in (0, 1) \\ (1 - \frac{1}{\gamma}) + \frac{\sigma^2}{\gamma-1} & \gamma > 1 \end{cases}.$$

See Figure 11 for a double descent curve. In this model, the first phase has no descent: the test error is monotonically increasing to the interpolation point $\gamma = 1$.

4.7 Neural tangent kernel

The *Neural Tangent Kernel* (NTK) provides a powerful framework for analyzing the training dynamics of neural networks in the overparameterized regime. A key insight is that, under suitable scaling and initialization, gradient descent on a neural network can be accurately approximated by a linear model whose behavior is governed by a kernel matrix.

Let $f(x; \theta), \theta \in \mathbb{R}^p$, be a neural network, and let $\{(x_i, y_i)\}_{i=1}^n$ be the training data. We consider gradient flow under the squared loss

$$\mathcal{L}(\theta) = \frac{1}{2} \sum_{i=1}^n (f(x_i; \theta) - y_i)^2.$$

Let θ_0 denote the initialization. The starting point of the NTK analysis is the first-order Taylor expansion around θ_0 :

$$f(x; \theta) \approx f(x; \theta_0) + \nabla_{\theta} f(x; \theta_0)^{\top} (\theta - \theta_0).$$

Define the (random) feature map

$$\phi(x) := \nabla_{\theta} f(x; \theta_0) \in \mathbb{R}^p.$$

Neural Tangent Kernel The *neural tangent kernel* is defined by

$$K(x, x') = \langle \phi(x), \phi(x') \rangle = \nabla_{\theta} f(x; \theta_0)^{\top} \nabla_{\theta} f(x'; \theta_0).$$

On the training data, define the Jacobian matrix

$$J \in \mathbb{R}^{n \times p}, \quad J_{i,:} = \nabla_{\theta} f(x_i; \theta_0),$$

so that the empirical NTK matrix is

$$K = JJ^{\top} \in \mathbb{R}^{n \times n}.$$

Training Dynamics Consider gradient flow:

$$\frac{d}{dt}\theta_t = -\nabla_{\theta}\mathcal{L}(\theta_t) = -\nabla_{\theta}f(X; \theta_t)^{\top}(f(X; \theta_t) - y),$$

where $f(X; \theta_t) = (f(x_1; \theta_t), \dots, f(x_n; \theta_t))$. Consider the evolution of the output. We find

$$\frac{d}{dt}f_t(X; \theta_t) = -\nabla_{\theta}\mathcal{L}(\theta_t) = -\nabla_{\theta}f(X; \theta_t)\nabla_{\theta}f(X; \theta_t)^{\top}(f(X; \theta_t) - y).$$

Therefore, *without any linearization*, the training dynamics are governed by a time-evolving neural tangent kernel

$$K_t = \nabla_{\theta}f(X; \theta_t)\nabla_{\theta}f(X; \theta_t)^{\top}.$$

In the infinite-width limit, or with a small learning rate, the NTK remains constant during training [JGH18, DZPS19, LBN⁺18, LXS⁺19], which allows us to replace K_t with K at initialization.

Under the linearization (i.e., replacing $\nabla_{\theta}f(X; \theta_t)$ by $\nabla_{\theta}f(X; \theta_0)$), the dynamics of the predictions satisfy the closed ODE

$$\frac{d}{dt}f_t(X) = -K(f_t(X) - y).$$

This linear ODE has the explicit solution

$$f_t(X) = y + e^{-Kt}(f_0(X) - y).$$

Thus, the entire training dynamics are governed by the spectrum of the kernel matrix K .

Spectral Decomposition Let

$$K = \sum_{i=1}^n \lambda_i u_i u_i^{\top}$$

be the eigendecomposition of K , where $\lambda_i \geq 0$. Then

$$f_t(X) - y = \sum_{i=1}^n e^{-\lambda_i t} \langle f_0(X) - y, u_i \rangle u_i.$$

Each eigenmode decays at rate λ_i , so directions corresponding to small eigenvalues converge slowly.

Convergence Rate Let $\lambda_{\min}(K)$ denote the smallest eigenvalue of K . Since $K \succeq 0$,

$$\|e^{-Kt}\|_{\text{op}} = \max_i e^{-\lambda_i t} = e^{-\lambda_{\min}(K)t}.$$

Therefore,

$$\|f_t(X) - y\| \leq e^{-\lambda_{\min}(K)t} \|f_0(X) - y\|.$$

Hence, $\lambda_{\min}(K)$ controls the worst-case convergence rate.

This bound highlights that a small $\lambda_{\min}(K)$ leads to large parameter movement, which in turn may invalidate the linearization approximation. The least singular value of NTK was estimated in [BAM22].

Theorem 32 (Explicit NTK formula for a two-layer network). *Let*

$$f(x; \theta) = \frac{1}{\sqrt{N}} \sum_{r=1}^N a_r \sigma(w_r^\top x), \quad x \in \mathbb{R}^d,$$

with parameters

$$\theta = (a_1, \dots, a_m, w_1, \dots, w_N) \in \mathbb{R}^{N+Nd}.$$

Assume σ is differentiable. Then the neural tangent kernel is

$$K_\theta(x, x') = \frac{1}{N} \sum_{r=1}^N \left[\sigma(w_r^\top x) \sigma(w_r^\top x') + a_r^2 \sigma'(w_r^\top x) \sigma'(w_r^\top x') \langle x, x' \rangle \right].$$

Proof. We compute the gradient with respect to all parameters.

$$\frac{\partial f(x; \theta)}{\partial a_r} = \frac{1}{\sqrt{N}} \sigma(w_r^\top x).$$

$$\nabla_{w_r} f(x; \theta) = \frac{1}{\sqrt{N}} a_r \sigma'(w_r^\top x) x.$$

Splitting contributions from a_r and w_r coordinates,

$$K_\theta(x, x') = \sum_{r=1}^N \frac{\partial f(x; \theta)}{\partial a_r} \frac{\partial f(x'; \theta)}{\partial a_r} + \sum_{r=1}^N \langle \nabla_{w_r} f(x; \theta), \nabla_{w_r} f(x'; \theta) \rangle.$$

Substituting yields the desired formula. □

Decomposition The NTK naturally decomposes as

$$K_\theta = K_\theta^{(a)} + K_\theta^{(w)},$$

where

$$K_\theta^{(a)}(x, x') = \frac{1}{N} \sum_{r=1}^N \sigma(w_r^\top x) \sigma(w_r^\top x'),$$

and

$$K_\theta^{(w)}(x, x') = \frac{\langle x, x' \rangle}{N} \sum_{r=1}^N a_r^2 \sigma'(w_r^\top x) \sigma'(w_r^\top x').$$

We can write

$$K_\theta^{(w)} = \frac{1}{N} (X^\top X) \odot \sigma'(WX)^\top D^2 \sigma'(WX),$$

where $D = \text{diag}(a_1, \dots, a_N)$.

Using the tensorization trick, since

$$\langle x_1, y_1 \rangle \langle x_2, y_2 \rangle = \langle x_1 \otimes y_1, x_2 \otimes y_2 \rangle,$$

we can write

$$\begin{aligned} K_\theta^{(w)}(x, x') &= \frac{1}{N} \langle x, x' \rangle \langle D\sigma'(Wx), D\sigma'(Wx') \rangle \\ &= \frac{1}{N} \langle x \otimes D\sigma'(Wx), x' \otimes D\sigma'(Wx') \rangle. \end{aligned}$$

Therefore $K_\theta^{(w)}$ can be seen as $\Phi^\top \Phi$ with feature vectors given by $\phi(x) = x \otimes D\sigma'(Wx) \in \mathbb{R}^{dN}$. A high-dimensional limit or a Wishart-type matrix behavior is expected when $Nd \asymp n$. Under a quadratic scaling, $n \propto dN$, $N \propto d$, the limiting spectral distribution of $K_\theta^{(w)}$ was derived in [BP25]. When $Nd \gg n$, kernel regression with $K_\theta^{(w)}$ was studied in [MZ22].

Extension to deep fully connected networks We now extend the previous computation from a two-layer network to a fully connected network with multiple hidden layers. The main point is that the NTK is still obtained by differentiating with respect to all parameters and summing the corresponding gradient inner products. For deep networks, the resulting formula has a clean recursive structure.

Fix a depth $L \geq 1$. For an input $x \in \mathbb{R}^d$, define

$$h^{(0)}(x) = x.$$

For $1 \leq \ell \leq L$, let

$$z^{(\ell)}(x) = W^{(\ell)} h^{(\ell-1)}(x) \in \mathbb{R}^{m_\ell}, \quad h^{(\ell)}(x) = \sigma(z^{(\ell)}(x)) \in \mathbb{R}^{m_\ell},$$

where

$$W^{(\ell)} \in \mathbb{R}^{m_\ell \times m_{\ell-1}}, \quad m_0 = d.$$

The scalar output is

$$f(x; \theta) = \frac{1}{\sqrt{m_L}} a^\top h^{(L)}(x), \quad a \in \mathbb{R}^{m_L}.$$

The full parameter vector is

$$\theta = (a, W^{(1)}, \dots, W^{(L)}).$$

For convenience, let

$$D^{(\ell)}(x) := \text{diag}(\sigma'(z_1^{(\ell)}(x)), \dots, \sigma'(z_{m_\ell}^{(\ell)}(x))).$$

To compute the NTK explicitly, it is convenient to introduce the backpropagated vectors. For $x \in \mathbb{R}^d$, define

$$\delta^{(L)}(x) := \frac{1}{\sqrt{m_L}} D^{(L)}(x) a \in \mathbb{R}^{m_L},$$

and for $1 \leq \ell \leq L-1$,

$$\delta^{(\ell)}(x) := D^{(\ell)}(x) (W^{(\ell+1)})^\top \delta^{(\ell+1)}(x) \in \mathbb{R}^{m_\ell}.$$

These are exactly the gradients of the output with respect to the pre-activations:

$$\delta^{(\ell)}(x) = \nabla_{z^{(\ell)}(x)} f(x; \theta).$$

Lemma 42 (Gradient formulas). *For each $x \in \mathbb{R}^d$, the gradients of $f(x; \theta)$ with respect to the parameters are*

$$\nabla_a f(x; \theta) = \frac{1}{\sqrt{m_L}} h^{(L)}(x),$$

and for each $1 \leq \ell \leq L$,

$$\nabla_{W^{(\ell)}} f(x; \theta) = \delta^{(\ell)}(x) (h^{(\ell-1)}(x))^\top.$$

Proof. The derivative with respect to a is immediate from

$$f(x; \theta) = \frac{1}{\sqrt{m_L}} a^\top h^{(L)}(x).$$

Thus

$$\nabla_a f(x; \theta) = \frac{1}{\sqrt{m_L}} h^{(L)}(x).$$

Now fix $1 \leq \ell \leq L$. By the chain rule, the gradient with respect to the matrix $W^{(\ell)}$ is obtained from the dependence of f on

$$z^{(\ell)}(x) = W^{(\ell)} h^{(\ell-1)}(x).$$

Since

$$\frac{\partial z_i^{(\ell)}(x)}{\partial W_{ij}^{(\ell)}} = h_j^{(\ell-1)}(x),$$

we get

$$\frac{\partial f(x; \theta)}{\partial W_{ij}^{(\ell)}} = \frac{\partial f(x; \theta)}{\partial z_i^{(\ell)}(x)} h_j^{(\ell-1)}(x) = \delta_i^{(\ell)}(x) h_j^{(\ell-1)}(x).$$

Thus, in matrix form,

$$\nabla_{W^{(\ell)}} f(x; \theta) = \delta^{(\ell)}(x) (h^{(\ell-1)}(x))^\top.$$

□

Theorem 33 (Explicit NTK formula for deep fully connected networks). *For the network above,*

$$K_\theta(x, x') = K_\theta^{(a)}(x, x') + \sum_{\ell=1}^L K_\theta^{(\ell)}(x, x'),$$

where

$$K_\theta^{(a)}(x, x') = \frac{1}{m_L} \langle h^{(L)}(x), h^{(L)}(x') \rangle,$$

and

$$K_\theta^{(\ell)}(x, x') = \langle \delta^{(\ell)}(x), \delta^{(\ell)}(x') \rangle \langle h^{(\ell-1)}(x), h^{(\ell-1)}(x') \rangle, \quad 1 \leq \ell \leq L.$$

Equivalently,

$$K_\theta(x, x') = \frac{1}{m_L} \langle h^{(L)}(x), h^{(L)}(x') \rangle + \sum_{\ell=1}^L \langle \delta^{(\ell)}(x), \delta^{(\ell)}(x') \rangle \langle h^{(\ell-1)}(x), h^{(\ell-1)}(x') \rangle.$$

Proof. By definition,

$$K_\theta(x, x') = \langle \nabla_\theta f(x; \theta), \nabla_\theta f(x'; \theta) \rangle.$$

Since the parameter vector is block-decomposed as

$$\theta = (a, W^{(1)}, \dots, W^{(L)}),$$

the Euclidean inner product splits blockwise:

$$K_\theta(x, x') = \langle \nabla_a f(x; \theta), \nabla_a f(x'; \theta) \rangle + \sum_{\ell=1}^L \langle \nabla_{W^{(\ell)}} f(x; \theta), \nabla_{W^{(\ell)}} f(x'; \theta) \rangle_F,$$

where $\langle \cdot, \cdot \rangle_F$ denotes the Frobenius inner product of matrices. For the output layer,

$$\langle \nabla_a f(x; \theta), \nabla_a f(x'; \theta) \rangle = \left\langle \frac{1}{\sqrt{m_L}} h^{(L)}(x), \frac{1}{\sqrt{m_L}} h^{(L)}(x') \right\rangle = \frac{1}{m_L} \langle h^{(L)}(x), h^{(L)}(x') \rangle.$$

For a hidden layer $1 \leq \ell \leq L$, using the previous lemma,

$$\nabla_{W^{(\ell)}} f(x; \theta) = \delta^{(\ell)}(x) (h^{(\ell-1)}(x))^\top,$$

and similarly for x' . Hence

$$\langle \nabla_{W^{(\ell)}} f(x; \theta), \nabla_{W^{(\ell)}} f(x'; \theta) \rangle_F = \left\langle \delta^{(\ell)}(x) (h^{(\ell-1)}(x))^\top, \delta^{(\ell)}(x') (h^{(\ell-1)}(x'))^\top \right\rangle_F.$$

Using the identity

$$\langle uv^\top, \tilde{u}\tilde{v}^\top \rangle_F = \langle u, \tilde{u} \rangle \langle v, \tilde{v} \rangle,$$

we obtain

$$\langle \nabla_{W^{(\ell)}} f(x; \theta), \nabla_{W^{(\ell)}} f(x'; \theta) \rangle_F = \langle \delta^{(\ell)}(x), \delta^{(\ell)}(x') \rangle \langle h^{(\ell-1)}(x), h^{(\ell-1)}(x') \rangle.$$

Summing over all blocks yields the formula. $\square$

5 High-dimensional learning dynamics

The preceding sections studied spectra of random matrices that arise from data, kernels, and neural networks. We now turn to the dynamics of learning algorithms. The main example is stochastic gradient descent (SGD), and the main message is that in high dimensions, SGD has a nontrivial scaling limit [Paq].

5.1 Streaming least squares with isotropic data

We begin with the student-teacher linear model. A fresh sample is

$$(a, b) \sim \mathcal{D}, \quad b = \langle a, \beta \rangle + \eta w,$$

where

$$\mathbb{E}a = 0, \quad \mathbb{E}aa^\top = K,$$

w is centered with variance one, and w is independent of a . The population risk (test error) for x is

$$P(x) := \frac{1}{2} \mathbb{E}_{(a,b)} (\langle a, x \rangle - b)^2 = \frac{\eta^2}{2} + \frac{1}{2} \langle K, (x - \beta) \otimes (x - \beta) \rangle.$$

Excess risk is:

$$R(x) = \frac{1}{2} \langle K, (x - \beta) \otimes (x - \beta) \rangle.$$

The empirical loss is

$$L(x) = \frac{1}{2} \sum_{i=1}^n (a_i^\top x - b_i)^2 := \sum_{i=1}^n f_i(x).$$

Streaming SGD with step size α_d is

$$x_{k+1} = x_k - \alpha_d \nabla_x f_{k+1}(x_k) = x_k - \alpha_d (\langle a_{k+1}, x_k \rangle - b_{k+1}) a_{k+1}.$$

Equivalently, for the error vector $v_k := x_k - \beta$,

$$v_{k+1} = v_k - \alpha_d (\langle a_{k+1}, v_k \rangle - \eta w_{k+1}) a_{k+1}.$$

We start with the simplest setting when $K = I_d$.

Theorem 34 (Risk scaling limit for isotropic one-pass SGD). *Assume*

$$a_k \stackrel{\text{i.i.d.}}{\sim} N(0, I_d), \quad b_k = \langle a_k, \beta \rangle + \eta w_k, \quad w_k \stackrel{\text{i.i.d.}}{\sim} N(0, 1),$$

with all (a_k, w_k) independent. Consider one-pass SGD

$$x_{k+1} = x_k - \frac{\gamma}{d} (\langle a_{k+1}, x_k \rangle - b_{k+1}) a_{k+1}.$$

Let

$$v_k := x_k - \beta, \quad R_k := R(x_k) := \frac{1}{2} \|v_k\|^2.$$

Assume

$$\sup_d R_0 < \infty.$$

Define ρ as the solution of

$$\dot{\rho}(t) = (-2\gamma + \gamma^2) \rho(t) + \frac{\gamma^2 \eta^2}{2}, \quad \rho(0) = R_0. \quad (45)$$

Then for every fixed $T > 0$, there exists a constant

$$C_T = C_T(\gamma, \eta, T, \sup_d R_0) < \infty$$

such that, for every $d \geq 1$,

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |R_{\lfloor td \rfloor} - \rho(t)|^2 \right] \leq \frac{C_T}{d}. \quad (46)$$

Consequently, for every $T > 0$ and every $\delta \in (0, 1)$,

$$\mathbb{P} \left(\sup_{0 \leq t \leq T} |R_{\lfloor td \rfloor} - \rho(t)| \leq \sqrt{\frac{C_T}{d\delta}} \right) \geq 1 - \delta.$$

Proof. We write

$$\mathcal{F}_k := \sigma\{(a_j, w_j) : 1 \leq j \leq k\}.$$

Since

$$b_{k+1} = \langle a_{k+1}, \beta \rangle + \eta w_{k+1},$$

the SGD recursion becomes

$$v_{k+1} = v_k - \frac{\gamma}{d} (\langle a_{k+1}, v_k \rangle - \eta w_{k+1}) a_{k+1}.$$

For notational simplicity, throughout one step we write

$$a := a_{k+1}, \quad w := w_{k+1}, \quad Z := \langle a, v_k \rangle.$$

Then

$$v_{k+1} = v_k - \frac{\gamma}{d} (Z - \eta w) a.$$

We split the proof into several steps.

Step 1: Exact one-step expansion.

Using

$$R_k = \frac{1}{2} \|v_k\|^2,$$

we have

$$\begin{aligned} R_{k+1} - R_k &= \frac{1}{2} \|v_{k+1}\|^2 - \frac{1}{2} \|v_k\|^2 \\ &= -\frac{\gamma}{d} (Z - \eta w) Z + \frac{\gamma^2}{2d^2} (Z - \eta w)^2 \|a\|^2. \end{aligned} \quad (47)$$

Since v_k is $\mathcal{F}_k$ -measurable and (a, w) is independent of $\mathcal{F}_k$, we may compute conditional expectations treating v_k as deterministic. First,

$$Z = \langle a, v_k \rangle \sim N(0, \|v_k\|^2) = N(0, 2R_k).$$

Therefore

$$\mathbb{E}[Z^2 \mid \mathcal{F}_k] = \|v_k\|^2 = 2R_k.$$

Since w is centered and independent of a ,

$$\mathbb{E}[wZ \mid \mathcal{F}_k] = 0.$$

Thus

$$\mathbb{E}[(Z - \eta w)Z \mid \mathcal{F}_k] = 2R_k. \quad (48)$$

Next, by rotational invariance of the Gaussian distribution,

$$\mathbb{E}[Z^2 \|a\|^2 \mid \mathcal{F}_k] = \mathbb{E}[\langle a, \|v_k\|e_1 \rangle^2 \|a\|^2] = (d+2)\|v_k\|^2 = 2(d+2)R_k.$$

Also,

$$\mathbb{E}[wZ \|a\|^2 \mid \mathcal{F}_k] = 0,$$

and

$$\mathbb{E}[w^2 \|a\|^2] = \mathbb{E}[w^2] \mathbb{E}[\|a\|^2] = d.$$

Hence

$$\mathbb{E}[(Z - \eta w)^2 \|a\|^2 \mid \mathcal{F}_k] = 2(d+2)R_k + d\eta^2. \quad (49)$$

Combining (47), (48), and (49), we obtain the exact conditional drift

$$\begin{aligned} \mathbb{E}[R_{k+1} - R_k \mid \mathcal{F}_k] &= -\frac{2\gamma}{d}R_k + \frac{\gamma^2}{2d^2} (2(d+2)R_k + d\eta^2) \\ &= \frac{1}{d} \left[(-2\gamma + \gamma^2)R_k + \frac{\gamma^2\eta^2}{2} \right] + \frac{2\gamma^2}{d^2}R_k. \end{aligned} \quad (50)$$

Define

$$f(r) := (-2\gamma + \gamma^2)r + \frac{\gamma^2\eta^2}{2}$$

and

$$\epsilon_{k,d} := \frac{2\gamma^2}{d^2}R_k.$$

Then

$$\mathbb{E}[R_{k+1} - R_k \mid \mathcal{F}_k] = \frac{1}{d}f(R_k) + \epsilon_{k,d}.$$

Define the martingale difference

$$\xi_{k+1} := R_{k+1} - R_k - \mathbb{E}[R_{k+1} - R_k \mid \mathcal{F}_k].$$

Then

$$\mathbb{E}[\xi_{k+1} \mid \mathcal{F}_k] = 0,$$

and the exact recursion becomes

$$R_{k+1} = R_k + \frac{1}{d}f(R_k) + \epsilon_{k,d} + \xi_{k+1}. \quad (51)$$

Step 2: Conditional second-moment bound.

We claim that there exists a constant $C = C(\gamma, \eta) < \infty$ such that

$$\mathbb{E}[|R_{k+1} - R_k|^2 \mid \mathcal{F}_k] \leq \frac{C}{d^2}(1 + R_k^2). \quad (52)$$

Indeed, from (47),

$$|R_{k+1} - R_k| \leq \frac{\gamma}{d}|Z - \eta w||Z| + \frac{\gamma^2}{2d^2}(Z - \eta w)^2 \|a\|^2.$$

Therefore

$$|R_{k+1} - R_k|^2 \leq \frac{2\gamma^2}{d^2} (Z - \eta w)^2 Z^2 + \frac{\gamma^4}{2d^4} (Z - \eta w)^4 \|a\|^4.$$

We estimate the two conditional expectations separately.

First, since $Z \sim N(0, 2R_k)$ and $w \sim N(0, 1)$ are independent, Gaussian moment bounds imply

$$\mathbb{E}[(Z - \eta w)^2 Z^2 \mid \mathcal{F}_k] \leq C(1 + R_k^2).$$

Second, using

$$(u - v)^4 \leq 8(u^4 + v^4),$$

we get

$$(Z - \eta w)^4 \leq C(Z^4 + w^4).$$

By rotational invariance, we may write

$$Z = \|v_k\|g_1, \quad g_1 \sim N(0, 1),$$

and

$$\|a\|^2 = g_1^2 + \dots + g_d^2.$$

Since

$$\mathbb{E}[g_1^4 \|a\|^4] \leq Cd^2$$

and

$$\mathbb{E}[w^4] \mathbb{E}[\|a\|^4] \leq Cd^2,$$

we have

$$\mathbb{E}[(Z - \eta w)^4 \|a\|^4 \mid \mathcal{F}_k] \leq Cd^2(1 + R_k^2).$$

Substituting these bounds gives (52).

Since ξ_{k+1} is obtained from $R_{k+1} - R_k$ by subtracting its conditional mean,

$$\mathbb{E}[\xi_{k+1}^2 \mid \mathcal{F}_k] \leq \mathbb{E}[|R_{k+1} - R_k|^2 \mid \mathcal{F}_k],$$

and therefore

$$\mathbb{E}[\xi_{k+1}^2 \mid \mathcal{F}_k] \leq \frac{C}{d^2} (1 + R_k^2). \quad (53)$$

Step 3: Uniform second-moment bound for R_k .

We prove that for every fixed $T > 0$,

$$\sup_{0 \leq k \leq Td} \mathbb{E}[R_k^2] \leq C_T. \quad (54)$$

Let

$$\Delta_k := R_{k+1} - R_k.$$

Then

$$R_{k+1}^2 = R_k^2 + 2R_k\Delta_k + \Delta_k^2.$$

Taking conditional expectation gives

$$\mathbb{E}[R_{k+1}^2 \mid \mathcal{F}_k] = R_k^2 + 2R_k\mathbb{E}[\Delta_k \mid \mathcal{F}_k] + \mathbb{E}[\Delta_k^2 \mid \mathcal{F}_k].$$

Using (50) and (52),

$$\mathbb{E}[R_{k+1}^2 \mid \mathcal{F}_k] \leq R_k^2 + \frac{C}{d}(R_k + R_k^2) + \frac{C}{d^2}(1 + R_k^2).$$

Since

$$R_k \leq 1 + R_k^2,$$

we obtain

$$\mathbb{E}[R_{k+1}^2 \mid \mathcal{F}_k] \leq \left(1 + \frac{C}{d}\right) R_k^2 + \frac{C}{d}.$$

Taking expectations,

$$\mathbb{E}[R_{k+1}^2] \leq \left(1 + \frac{C}{d}\right) \mathbb{E}[R_k^2] + \frac{C}{d}.$$

By the discrete Gronwall inequality, for all $0 \leq k \leq Td$,

$$\mathbb{E}[R_k^2] \leq C_T.$$

This proves (54).

Step 4: Martingale maximal bound.

Define the martingale

$$M_n := \sum_{j=1}^n \xi_j.$$

By Doob's L^2 maximal inequality,

$$\mathbb{E} \left[\max_{0 \leq n \leq N} |M_n|^2 \right] \leq 4\mathbb{E}[|M_N|^2].$$

Because the increments are martingale differences,

$$\mathbb{E}[|M_N|^2] = \sum_{j=1}^N \mathbb{E}[\xi_j^2].$$

Using (53) and (54), for $N = \lfloor Td \rfloor$,

$$\mathbb{E} \left[\max_{0 \leq n \leq \lfloor Td \rfloor} |M_n|^2 \right] \leq 4 \sum_{j=1}^{\lfloor Td \rfloor} \frac{C}{d^2} (1 + \mathbb{E}[R_{j-1}^2]) \leq \frac{C_T}{d}. \quad (55)$$

Step 5: Comparison with the Euler scheme.

Let $\bar{\rho}_k$ be the explicit Euler scheme for (45) with step size $1/d$:

$$\bar{\rho}_0 = R_0, \quad \bar{\rho}_{k+1} = \bar{\rho}_k + \frac{1}{d}f(\bar{\rho}_k).$$

Since f is affine, hence globally Lipschitz, the standard Euler error estimate gives

$$\max_{0 \leq k \leq Td} |\bar{\rho}_k - \rho(k/d)| \leq \frac{C_T}{d}. \quad (56)$$

Define the error

$$e_k := R_k - \bar{\rho}_k.$$

Subtracting the Euler recursion from (51), we get

$$e_{k+1} = e_k + \frac{1}{d} (f(R_k) - f(\bar{\rho}_k)) + \epsilon_{k,d} + \xi_{k+1}.$$

Since f is affine with Lipschitz constant

$$L := |-2\gamma + \gamma^2|,$$

we have

$$|f(R_k) - f(\bar{\rho}_k)| \leq L|e_k|.$$

Thus for every $n \leq Td$,

$$|e_n| \leq \frac{L}{d} \sum_{j=0}^{n-1} |e_j| + \sum_{j=0}^{n-1} |\epsilon_{j,d}| + \max_{0 \leq m \leq n} |M_m|.$$

Let

$$E_n := \max_{0 \leq m \leq n} |e_m|.$$

Then

$$E_n \leq \frac{L}{d} \sum_{j=0}^{n-1} E_j + \sum_{j=0}^{n-1} |\epsilon_{j,d}| + \max_{0 \leq m \leq n} |M_m|.$$

By discrete Gronwall,

$$E_n \leq e^{LT} \left(\sum_{j=0}^{n-1} |\epsilon_{j,d}| + \max_{0 \leq m \leq n} |M_m| \right). \quad (57)$$

We now bound the deterministic finite- d correction. Since

$$\epsilon_{j,d} = \frac{2\gamma^2}{d^2} R_j,$$

Cauchy–Schwarz gives

$$\begin{aligned} \mathbb{E} \left[\left(\sum_{j=0}^{\lfloor Td \rfloor - 1} |\epsilon_{j,d}| \right)^2 \right] &= \frac{4\gamma^4}{d^4} \mathbb{E} \left[\left(\sum_{j=0}^{\lfloor Td \rfloor - 1} R_j \right)^2 \right] \\ &\leq \frac{4\gamma^4}{d^4} \lfloor Td \rfloor \sum_{j=0}^{\lfloor Td \rfloor - 1} \mathbb{E}[R_j^2] \\ &\leq \frac{C_T}{d^2}. \end{aligned} \quad (58)$$

Combining (57), (55), and (58), we obtain

$$\mathbb{E} \left[\max_{0 \leq k \leq \lfloor Td \rfloor} |R_k - \bar{\rho}_k|^2 \right] \leq \frac{C_T}{d}. \quad (59)$$

Step 6: Passing from the Euler scheme to the ODE.

For $t \in [0, T]$, let

$$k = \lfloor td \rfloor.$$

Then

$$|R_{\lfloor td \rfloor} - \rho(t)| \leq |R_k - \bar{\rho}_k| + |\bar{\rho}_k - \rho(k/d)| + |\rho(k/d) - \rho(t)|.$$

The first term is controlled by (59). The second is controlled by (56). For the third, since ρ solves a linear ODE on a compact time interval, it is Lipschitz:

$$|\rho(k/d) - \rho(t)| \leq \frac{C_T}{d}.$$

Therefore

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |R_{\lfloor td \rfloor} - \rho(t)|^2 \right] \leq \frac{C_T}{d}.$$

This proves (46). A Markov inequality implies the probability bound. $\square$

Remark 14 (Stability and stationary risk). For constant γ , the ODE (45) is stable if and only if

$$2\gamma - \gamma^2 > 0, \quad \text{i.e.} \quad 0 < \gamma < 2.$$

In that case,

$$\rho(t) = e^{-(2\gamma - \gamma^2)t} \rho(0) + \frac{\gamma^2 \eta^2 / 2}{2\gamma - \gamma^2} \left(1 - e^{-(2\gamma - \gamma^2)t} \right),$$

and therefore

$$\rho(\infty) = \frac{\gamma^2 \eta^2}{4\gamma - 2\gamma^2}.$$

The limiting risk is positive when $\eta > 0$, even though the population risk is strongly convex. This is the high-dimensional cost of keeping a constant effective step size.

5.2 Anisotropic least squares and homogenized SGD

We keep the least-squares model

$$b = \langle a, \beta \rangle + \eta w, \quad \mathbb{E} a a^\top = K,$$

with $\|K\|$ bounded uniformly in d and $\text{Tr } K \asymp d$. We also assume that quadratic forms in a concentrate: for deterministic B ,

$$a^\top B a \approx \text{Tr}(BK)$$

with high probability. This is guaranteed, for example, by a Hanson–Wright inequality for sub-Gaussian random vectors.

The SGD recursion with critical scaling is

$$v_{k+1} = v_k - \frac{\gamma}{d} (\langle a_{k+1}, v_k \rangle - \eta w_{k+1}) a_{k+1}, \quad v_k = x_k - \beta. \quad (60)$$

Failure of closure In contrast to the isotropic case $K = I_d$, the recursion (60) does not admit a closed description in terms of a finite number of scalar observables. Indeed, consider the evolution of the risk

$$R_k := \frac{1}{2} \|v_k\|^2.$$

Expanding one step and taking conditional expectation, we obtain terms of the form

$$\mathbb{E}[\langle a_{k+1}, v_k \rangle^2 \mid v_k] = v_k^\top K v_k,$$

which already shows that the dynamics of R_k depend not only on $\|v_k\|^2$ but on the quadratic form $v_k^\top K v_k$. This is in stark contrast with the isotropic setting, where rotational symmetry forces all such quantities to collapse to functions of $\|v_k\|^2$ alone. The lack of closure reflects the fact that the covariance K introduces a nontrivial geometry, and different spectral directions evolve at different rates. Consequently, to obtain a tractable limit, one must instead pass to suitably averaged observables (e.g., resolvent quantities), which admit a closed deterministic description in high dimension.

The homogenized SGD process The correct continuous-time approximation keeps the deterministic population drift but replaces the high-dimensional stochastic gradient noise by Brownian noise with the same leading covariance.

Recall the population risk (test error) for x is

$$P(x) := \frac{1}{2} \mathbb{E}_{(a,b)} (\langle a, x \rangle - b)^2 = \frac{\eta^2}{2} + \frac{1}{2} \langle K, (x - \beta) \otimes (x - \beta) \rangle.$$

Definition 20 (Homogenized SGD for correlated least squares). Let B_t be a standard Brownian motion in $\mathbb{R}^d$. Homogenized SGD is the process X_t solving

$$dX_t = -\gamma \nabla P(X_t) dt + \gamma \sqrt{\frac{2P(X_t)}{d}} K^{1/2} dB_t, \quad X_0 = x_0. \quad (61)$$

Since $\nabla P(x) = K(x - \beta)$, this is equivalently

$$d(X_t - \beta) = -\gamma K(X_t - \beta) dt + \gamma \sqrt{\frac{2P(X_t)}{d}} K^{1/2} dB_t.$$

Definition 21 (C^2 norm for quadratic statistics). For a quadratic function $q : \mathbb{R}^d \rightarrow \mathbb{R}$, define

$$\|q\|_{C^2} := \sup_x \|\nabla^2 q(x)\| + \|\nabla q(0)\| + |q(0)|.$$

For quadratic functions this is finite and measures the size of the statistic independently of the ambient dimension.

Theorem 35 (Homogenized SGD comparison for quadratic statistics). *Assume the correlated least-squares data satisfy the concentration assumptions above, and suppose $\|x_0\|$ and $\|\beta\|$ are bounded by a fixed power of d . Let x_k solve (60), and let X_t solve (61) with the same deterministic initial condition. Then for every fixed $T < \infty$ and every quadratic q with $\|q\|_{C^2} \leq 1$,*

$$\sup_{0 \leq k \leq Td} |q(x_k) - q(X_{k/d})| \leq d^{-1/2+o(1)}$$

with probability $1 - \exp(-\omega(\log d))$.

Remark 15 (A stochastic process but deterministic risks). The process X_t is still d -dimensional and stochastic. Nevertheless, when tested against quadratic statistics of bounded C^2 norm, its fluctuations are typically $d^{-1/2+o(1)}$. Thus risk curves concentrate around deterministic limits, even though the parameter vector itself does not converge to a deterministic path in Euclidean norm.

Resolvents, Volterra equations, and step-size criticality The homogenized process gives the right continuous-time approximation, but it is still d -dimensional. To obtain explicit risk curves, we use resolvents to encode the infinitely many quadratic statistics

$$\langle K^r, (X_t - \beta) \otimes (X_t - \beta) \rangle, \quad r = 0, 1, 2, \dots$$

This is the same idea behind the Stieltjes-transform method in random matrix theory.

Resolvent order parameters Fix a positively oriented contour $\Gamma \subset \mathbb{C}$ enclosing the spectrum of K . For $z \in \mathbb{C} \setminus \text{spec}(K)$, write

$$R(z; K) := (K - zI)^{-1}.$$

Define the resolvent observable

$$Q_t(z) := \frac{1}{2} \langle R(z; K), (X_t - \beta) \otimes (X_t - \beta) \rangle.$$

By holomorphic functional calculus,

$$\frac{1}{2} v_t^\top K v_t = P(X_t) = \frac{\eta^2}{2} - \frac{1}{2\pi i} \oint_{\Gamma} z Q_t(z) dz.$$

Thus the population risk can be recovered from the resolvent curve $z \mapsto Q_t(z)$.

Lemma 43 (Gradient flow for least squares). *Let Y_t solve gradient flow for P :*

$$\dot{Y}_t = -\nabla P(Y_t), \quad Y_0 = x_0.$$

Then

$$Y_t - \beta = e^{-tK}(x_0 - \beta).$$

Consequently,

$$F_\gamma(t) := P(Y_{\gamma t}) = \frac{\eta^2}{2} + \frac{1}{2} \langle K e^{-2\gamma t K}, (x_0 - \beta) \otimes (x_0 - \beta) \rangle. \quad (62)$$

Proof. Since $\nabla P(x) = K(x - \beta)$,

$$\frac{d}{dt}(Y_t - \beta) = -K(Y_t - \beta).$$

Solving this linear ODE gives the claimed formula. Substituting into P gives (62). $\square$

Definition 22 (Finite-dimensional Volterra risk model). Let

$$\mathcal{K}_\gamma(t) := \frac{\gamma^2}{d} \text{Tr}(K^2 e^{-2\gamma t K}), \quad t \geq 0.$$

The finite-dimensional Volterra risk curve is the solution of

$$\Psi(t) = F_\gamma(t) + \int_0^t \mathcal{K}_\gamma(t-s) \Psi(s) ds,$$

where F_γ is the gradient-flow risk (62).

Theorem 36 (Risk curve of homogenized SGD). *For every fixed $T < \infty$,*

$$\sup_{0 \leq t \leq T} |P(X_t) - \Psi(t)| \leq d^{-1/2+o(1)}$$

with overwhelming probability. Consequently, by Theorem 35, the same Volterra curve describes the risk of streaming SGD at times $k = td$:

$$P(x_{\lfloor td \rfloor}) = \Psi(t) + o_{\mathbb{P}}(1).$$

uniformly for t in compact intervals.

The Volterra equation separates the deterministic gradient-flow part from the additional variance created by SGD.

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